An Integrated NLP-Driven Decision Support Framework for Automated Judicial Judgment Summarization and Context-Aware Legal Precedent Recommendation

A Thesis Project

Presented to the

School of Information Technology

Mapúa University

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Computer Science

by

Sophia Erin Nicole D. Castro

Juan Paolo Angelo T. Dellomas

Maria Junella May E. Oalican

[February, 2026]

CAPSTONE / THESIS PROPOSAL APPROVAL FORM	Document No: FM-SIT-08-01 Effective Date: Aug. 22, 2017

This capstone / thesis proposal paper entitled :

An Integrated NLP-Driven Decision Support Framework for Automated Judicial Judgement Summarization and Context-Aware Legal Precedent Recommendation

prepared and presented by the following students on February 27, 2026

Sophia Erin Nicole D. Castro

Juan Paolo Angelo T. Dellomas

Maria Junella May E. Oalican

was supervised and endorsed by the advisor/s

Dr. Lysa V. Comia
Name & Signature

Name & Signature

and hereby approved and recommended to be accepted as fulfillment to the requirement of the course Thesis 1 by:

Mr. Polycarpio V. Cabalag
Name & Signature

Mr. Jefferson A. Costales
Name & Signature

Mrs. Renilda S. Layno
Name & Signature

Noted by:

Mr. John Paul Q. Tomas
Name & Signature of Course Coordinator

March 16, 2026
Date

THESIS/CAPSTONE CONFORMITY OF REVISIONS

Date : March 11, 2026

TITLE : An Integrated NLP-Driven Decision Support Framework for Automated Judicial Judgment Summarization and Context-Aware Legal Precedent Recommendation

Name of Student	Student No.	Program of Study
Sophia Erin Nicole D. Castro	2023103031	Bachelor of Science in Computer Science
Juan Paolo Angelo T. Dellomas	2021130466	Bachelor of Science in Computer Science
Maria Junella May E. Oalican	2023100787	Bachelor of Science in Computer Science

IMPLEMENTATION OF THE RECOMMENDED REVISIONS

Nature of Revision	Actions Made	Page Reflected Manuscript
The consistency across the documents.	Added the missing discussion for one figure (evaluation workflow) and one table (domain expert task).	page 54-55, and 65.
Data privacy and ethical considerations of the proposed framework.	Added a discussion on data privacy and ethical considerations in using the proposed framework. Added in scope and limitations, and a new section in chapter 3 which is 3.5.4.	page 14, and 68-69.
Clarity of the preprocessing methodology.	Added the pre-processing architecture and specified the algorithms used in each process. Modify Figure 4, Data Collection and Annotation, specified the tools used.	page 37, and 41-42.
Justification for the selection of models and tools.	Provided the basis for choosing the NLM models and other MC tools, including a table of comparison. Added table 1-4 in chapter 2.	page 18, 20, 23, and 24.
Validation of legal summarizer and precedent recommender.	Added a 5-point Likert scale and discussion for the legal summarizer and precedent recommender, which is used for expert evaluation.	page 65-68.

THESIS/CAPSTONE CONFORMITY OF REVISIONS

Nature of Revision	Actions Made	Page Reflected Manuscript
Interpretation of evaluation metric scores.	Added the interpretation of score ranges for each evaluation metric used in the study to clarify the meaning of the obtained results and their implications in the evaluation of the generated summaries.	page 56-60.
Description of the sample dataset and sampling method.	Added a discussion of the sample dataset obtained from the survey of legal experts, including the relevant survey details and the sampling method used.	page 68.

Sophia Erin Nicole D. Castro
Student

Mr. Polycarpio V. Cabalag
Panel Member

Juan Paolo Angelo T. Dellomas
Student

Mr. Jefferson A. Costales
Panel Member

Maria Junella May E. Oalican
Student

Mrs. Renilda S. Layno
Panel Member

Dr. Lysa V. Comia
Adviser

MAPÚA

REVISIONS LIST
FOR CAPSTONE / THESIS ORAL EXAMINATION

Document No: FM-SIT-07-01
Effective Date: Aug. 22, 2017

TITLE: An NLP-Driven Decision Support Framework for Automated Judicial Judgement Summarization and Context-Aware

DATE / TIME: 02/27/2026 8:00 am

TECHNCIAL ADVISER: Dr. Lysa V. Comia

☑ Thesis
☐ Capstone

☑ Proposal Defense
☐ Final Defense

LIST OF REVISIONS

DOCUMENT	SOFTWARE / SYSTEM
1. Discussions for Figure 11. Evaluation Workflow.	
2. Table 3. Domain Expert Task	
3. Include discussion on data privacy and ethical considerations in using the proposed framework	
4. Provide pre-processing architecture w/ algorithms per process	
5. Provide a basis for choosing NLM models and other ML tools (w/ table of comparison).	7. Discuss sample dataset obtained from survey (legal experts). – develop survey questionnaires. – type of sampling method (purposive) –
6. ~~to~~ Provide Legal Summarization Benchmarks. (5-point Likert Scale)	

(Use additional sheet if necessary)

STUDENTS	PANEL MEMBERS
Maria Junella Mae E. Dalican Name & Signature	POLYCARPIO CABALAN Name & Signature
Juan Paolo Angelo T. Dellomas Name & Signature	JEFFERSON A. COSTALES Name & Signature
Sophia Trina Nicole D. Castro Name & Signature	RENICIA SO LAYNO Name & Signature

Check Appropriate Box:

☐ Approved with no revision
☑ Approved with revision
☐ Re-defense
☐ Not accepted

TABLE OF CONTENTS

TITLE PAGE....................i

APPROVAL FORM/SHEET.................... ii

CONFORMITY OF REVISIONS....................iii

REVISIONS LIST....................vi

TABLE OF CONTENTS.................... vii

LIST OF FIGURES....................xiii

LIST OF TABLES....................xiv

Chapter 1....................1

1.1. Context of the Study.................... 2

1.2. The Problem, Gap or Opportunity.................... 5

1.3. Research Questions....................8

1.4. Research Objectives....................9

1.5. Significance of the Study....................10

1.6. Scope and Limitations (or Delimitations)....................12

Chapter 2....................15

2.1. Foundations of Legal NLP for AI-Assisted Legal Decision Support....................15

2.1.1. Growth of Legal Text and Motivation for NLP Support.................... 15

2.1.2. Core Tasks in Legal NLP....................16

2.1.3. Legal Language and Long-Document Constraints.................... 16

2.1.4. Responsible AI, Explainability, and Human Oversight in Legal Contexts..17

2.1.5. System Context: From Document Platforms to Domain-Adapted LLMs... 17

2.2. Legal Case Retrieval and Context-Aware Precedent Identification......................19

2.2.1. Lexical Retrieval and Hybrid Retrieval Pipelines...................................... 19

2.2.2. Transformer-Based Semantic and Structure-Aware Retrieval..................... 19

2.2.3. LLM-Assisted Relevance Judgment, Query Refinement, and Explanation 20

2.2.4. Retrieval as Legal-Element Generation.. 21

2.2.5. Implications for Decision Support... 21

2.3. Legal Text Summarization for Judgment Understanding.................................... 21

2.3.1. Role of Summarization in Legal Review..21

2.3.2. Hybrid Extractive–Abstractive Summarization for Long Judgments..........21

2.3.3. Structure-Guided and Knowledge-Guided Summarization......................... 25

2.3.4. Reasoning-Guided Summarization Using Judicial Logic...........................25

2.4. Benchmarking and Evaluation of Legal NLP Systems..26

2.4.1. Why Legal NLP Evaluation Is Harder Than General NLP Evaluation....... 26

2.4.2. Evaluation of Legal Retrieval and Precedent Recommendation................. 27

2.4.3. Evaluation of Legal Summarization and the Limits of Overlap Metrics.....27

2.4.4. Benchmarks and Shared Tasks as Evaluation Infrastructure....................... 29

2.4.5. Toward Practitioner-Aligned and Human-Centered Evaluation.................. 30

2.5. Research Gaps and Synthesis...30

2.5.1. Fragmentation of Judgment Understanding and Precedent Identification...31

2.5.2. Long-Document Constraints and Cost–Quality Trade-offs Remain

Unresolved... 31
2.5.3. Evaluation Gaps: Legal Faithfulness, Stability, and Reproducibility.......... 32
2.5.4. Practitioner Alignment Gap: Systems vs. Workflows................................ 33
2.5.5. Consolidated Research Gap and Relevance to the Proposed Study........... 34
Chapter 3...37
3.1. Data Gathering...37
3.1.1 Data Sources... 37
3.1.2. Case Categories and Dataset Size...38
3.1.3. Inclusion Criteria.. 39
3.1.4. Exclusion Criteria and Quality Screening...39
3.1.5. Data Collection Procedure...40
3.1.6. Dataset Documentation and Metadata Fields.. 43
3.1.7. Data Handling and Storage..43
3.2. Data Pre-processing.. 44
3.2.1. Preprocessing Environment and Tools...46
3.2.2. Cleaning, Segmentation, and Legal Section Structuring............................ 46
3.2.3. Section Completeness and Case Validity Rules..47
3.2.4. Dataset Splitting for Training and Evaluation.. 47
3.2.5. Output of Preprocessing Stage...48
3.3. NLP Model Training and Fine-tuning... 48
3.3.1. Semantic Representation via Sentence-BERT..48

3.3.2. Retrieve-then-Read Judicial Judgment Summarization (BERTSum-PH to Command-A)........ 50
3.3.3. Precedent Recommendation via Sentence-BERT........ 53
3.3.4. Objective Functions and Loss Formulations........ 55
3.3.5. Implementation Settings and Training Documentation........ 57
3.4. Testing and Evaluation Procedures........ 58
3.4.1. Summarization Evaluation........ 58
3.4.2. Precedent Recommendation Evaluation........ 58
3.4.3. Computational Efficiency Measures........ 59
3.4.4. Evaluation Metrics and Similarity Scoring Formulas........ 60
3.5. Human-in-the-Loop Validation........ 70
3.5.1 Expert Task and Review Procedure........ 71
3.5.2 Likert-Scale Expert Evaluation........ 73
3.5.3 Survey Dataset Description........ 75
3.5.4 Data Privacy and Ethical Considerations........ 76
Chapter 4........ 78
4.1. Experimental Configuration........ 78
4.2. Experimental Results........ 81
4.2.1. PEGASUS Baseline Results........ 81
4.2.2. BART with Dual Regex Extractive Results........ 82
4.2.3. Gemini 2.5 as Model with Revised Evaluation Metrics Results........ 84
4.2.4. Qwen 2.5 70B using Hybrid Extractive Result........ 85

4.2.5. Qwen 2.5 70B using Original Pipeline with Revised Metrics Result.......... 87
4.2.6. Llama 3.3 (70B) as a Model using Hybrid Extractive Result...................... 89
4.2.7. Llama 3.3 (70B) as a Model with Revised Metrics Result.......................... 91
4.2.8. Command A as a Model with the Hybrid Extractive Result........................ 92
4.2.9. Command A as a Model with G-Eval (Command R Plus via API as a Judge) Result.. 94
4.2.10. Command A as a Model with G-Eval (Llama 3.3 70B via API as a Judge) Result.. 95
4.2.11. Command A as a Model with G-Eval (Qwen 2.5 7B) Result.................... 97
4.2.12. Command A as a Model with G-Eval (Qwen 2.5 7B) Tweaked Result.... 98
4.2.13. Command A as a Model with G-Eval (Qwen 2.5 14B) Result.................. 99
4.2.14. Command A as a Model with G-Eval (Llama 3.1 70B) Result............... 101
4.2.15. Command A as a Model with G-Eval (Llama 3.1 70B) (Retrieved-then-Read Pipeline) Result.. 102
4.3. Final Model Selection... 104
4.4. Pipeline Summarization Metrics... 105
4.5. Web Deployment.. 107
4.5.1 Summarization Engine.. 108
4.5.2 Precedent Retrieval... 108
4.6. Human-in the-Loop Validation.. 108
4.6.1 Summarization Evaluation.. 109
4.6.2. Precedent Recommendation.. 116
4.7. Precision@K and Recall@K.. 123

Chapter 5 126

5.1. Conclusion 126

5.2. Summary of Findings 127

5.2.1. Research Question 1: Development of the NLP-Driven Legal Decision-Support Framework 128

5.2.2. Research Question 2: Effect of the Integrated Summarization and Precedent Recommendation Pipeline 128

5.2.3. Research Question 3: Automatic and Human-Centered Evaluation of the Framework 129

5.3. Contribution of Knowledge 130

5.4. Practical and Legal Decision-Support Implications 131

5.5. Limitations of the Study 132

5.6. Recommendations for Future Work 133

5.7. Concluding Statement 134

APPENDICES 136

REFERENCES 141

LIST OF FIGURES

Figure 1. Number of Legal NLP Papers over Time..16

Figure 2. A Comprehensive Survey on Legal Summarization………………………..…28

Figure 3. Performance Comparison of Legislative Summarization Models……………..29

Figure 4. Conceptual Framework Diagram...35

Figure 5. Dataset Distribution...38

Figure 6. Data Collection and Annotation...40

Figure 7. Preprocessing Architecture……………………………………………………45

Figure 8. Preprocessing and Section Structuring Pipeline...47

Figure 9. Sentence-BERT Vector Generation and Embedding Export Workflow 49

Figure 10. Retrieve-then-Read Summarization Workflow (BERTSum to Command-A)...52

Figure 11. Precedent Retrieval and Ranking Workflow 54

Figure 12. Evaluation Workflow...59

Figure 13. Human Review Loop..71

Figure 14. Normalized Model Footprint Comparison…………………………………105

Figure 15. Philippine Legal Summarization: Comprehensive Model Averages………..106

Figure 16. Summarization Engine……………………………………………………...109

Figure 17. Precedent Retrieval…………………………………………………………110

Figure 18. Overall Precedent Retrieval Performance…………………………………..125

LIST OF TABLES

Table 1. Adapted benchmark evidence supporting LEGAL-BERT for legal-domain language representation……………………………………………………………….18

Table 2. Adapted benchmark evidence supporting Sentence-BERT for efficient semantic similarity retrieval……………………………………………………………………..20

Table 3. Adapted literature support for the BERTSum → Command A sequential hybrid summarization design ……………………………...………………………………23

Table 4. Document-wide ROUGE and BERTScore results on the IN-Ext dataset…………………………………………………………………………...…….24

Table 5. Objective Functions Used in Each NLP Component …………………....…….55

Table 6. Evaluation Metrics and Their Purpose…………………………………...…….60

Table 7. Interpretation of Summarization Metric Score Ranges……………….......…….61

Table 8. Interpretation of Precedent Recommendation Metric Score Ranges …....…….64

Table 9. Domain Expert Task……………………………………………………...…….71

Table 10. Legal Summarization Evaluation Criteria (5-Point Likert Scale)......................73

Table 11. Legal Precedent Recommendation Evaluation Criteria (5-Point Likert Scale).74

Table 12. Survey Dataset Characteristics for Expert Evaluation……………….....…….75

Table 13. Experimental Configurations Used in the Study………………………………78

Table 14. BART with Dual Regex Extractive Results…………………………………...83

Table 15. Gemini 2.5 as Model with Revised Evaluation Metrics Results………………85

Table 16. Qwen 2.5 70B using Hybrid Extractive Result………………………………..87

Table 17. Qwen 2.5 70B using Original Pipeline with Revised Metrics Result…………88

Table 18. Llama 3.3 (70B) as a Model using Hybrid Extractive Result…………………90

Table 19. Llama 3.3 (70B) as a Model with Revised Metrics Result……………………92

Table 20. Command A as a Model with the Hybrid Extractive Result…………………93

Table 21. Command A as a Model with G-Eval (Command R Plus via API as a Judge) Result……………………………………………………………………………95

Table 22. Command A as a Model with G-Eval (Llama 3.3 70B via API as a Judge) Result……………………………………………………………………………96

Table 23. Command A as a Model with G-Eval (Qwen 2.5 7B) Result…………………98

Table 24. Command A as a Model with G-Eval (Qwen 2.5 7B) Tweaked Result………99

Table 25. Command A as a Model with G-Eval (Qwen 2.5 14B) Result………………101

Table 26. Command A as a Model with G-Eval (Llama 3.1 70B) Result……………...102

Table 27. Command A as a Model with G-Eval (Llama 3.1 70B) (Retrieved-then-Read Pipeline) Result…………………………………………………………………104

Table 28. Extractive Summarization Evaluation Results (BERTSum)...........................107

Table 29. Abstractive Summarization Evaluation Results (Command A).....................108

Table 30. Accuracy of Legal Content)...111

Table 31. Completeness of Legal Elements……………………………………………112

Table 32. Clarity and Readability………………………………………………………113

Table 33. Preservation of Judicial Reasoning…………………………………………..115

Table 34. Usefulness for Legal Research………………………………………………116

Table 35. Relevance to the Query Case………………………………………………...118

Table 36. Alignment of Legal Issues…………………………………………………...119

Table 37. Alignment of Judicial Reasoning……………………………………………120

Table 38. Defensibility as Precedent Support…………………………………………..122

Table 39. Usefulness for Legal Research……………………………………………….123

Table 40. Precedent Retrieval Performance Metrics by Offense Category…………….126

Table 41. Dataset Metadata Schema……………………………………..…....…….140

Table 42. Domain Expert Profile A……………………………………......………..141

Table 43. Domain Expert Profile B……………………………………………………..142

Chapter 1

INTRODUCTION

Justice systems operate through documented processes. In the Philippines, pleadings, statutes, and judicial decisions create an official record that supports accountability and consistency in applying law to facts. The 1987 Constitution requires that court decisions clearly and distinctly state the facts and the law, and it sets periods for resolving cases, reflecting the need for reasoned and timely adjudication [46]. The Civil Code also recognizes that judicial decisions applying or interpreting the Constitution or laws form part of the Philippine legal system, reinforcing the role of jurisprudence in precedent-based reasoning [47]. Strengthening institutions and improving access to justice are likewise national priorities aligned with Sustainable Development Goal 16 [1].

With the continued publication and digitization of jurisprudence, legal work increasingly depends on efficient access to large collections of decisions. The Supreme Court publishes decisions and signed resolutions online and points users to the Supreme Court E-Library for older decisions and deeper research; however, the Decisions portal notes that only case titles are directly searchable, which can still make review difficult at scale [48], [49]. The Court also provides digital services such as eCourt PH, a unified electronic filing and case management system accessed through the Philippine Judiciary Platform [50]. These conditions highlight the need for decision-support tools that reduce reading burden while keeping legal judgment under human oversight.

The volume of court decisions continues to grow, and many decisions are long, dense, and difficult to search, so even simple tasks such as identifying what the judge ruled and locating the correct precedent can become costly and time-intensive [2], [3]. Keyword-based search is helpful, but it often fails to capture what legal relevance depends on: the issue, the rationale, and the context that can make two cases legally similar even when they do not use the same words [11], [12], [13]. Current legal NLP research suggests that language models trained on legal text and evaluated properly can support tasks such as judgment summarization and case retrieval [4], [5]. However, long judgments remain difficult to process, and summaries can be risky when they appear correct but omit critical facts or reasoning [6]–[8]. In practice, summarization and precedent retrieval are also commonly treated as separate tasks, even though they must work together in legal work: first understand the decision, then ground it using relevant precedent [14], [21]. Because the legal domain is high-stakes, researchers emphasize that these tools must be explainable and remain under human oversight, not treated as replacements for legal judgment [16], [17], [42]. This study therefore focuses on developing an integrated decision-support framework that combines judgment summarization with context-sensitive precedent recommendation and includes human validation to keep outputs defensible in real legal practice [3], [15], [43].

1.1. Context of the Study

The administration of justice is anchored on the consistent and accountable interpretation of legal rules, factual narratives, and precedent-based reasoning. Strengthening institutional capacity and improving access to justice are widely recognized development priorities, including the directions reflected in Sustainable

Development Goal 16 on peace, justice, and strong institutions [1]. Within this setting, there is a practical need for computational support that helps legal actors manage legal information at scale while still preserving professional discretion and accountability.

The legal domain is characterized by the continuous production and use of text-heavy documents such as judgments, pleadings, and precedent cases. Surveys in legal NLP emphasize that legal documents are typically long, structurally complex, and written in technical language, which makes manual review demanding and time-consuming. These surveys also highlight recurring constraints in legal NLP, including extensive document length and limited openly available datasets for research and evaluation [2]. In parallel, legal information retrieval research notes that digitization has produced large volumes of legal information that are often unstructured, increasing the need for retrieval methods that go beyond simple browsing and keyword search [3].

Recent developments in NLP, particularly transformer-based language models, have improved the ability to represent contextual meaning in text. Domain-adapted pretraining is repeatedly shown to be beneficial for legal NLP tasks, since models trained or further pre-trained on legal corpora better capture legal language patterns than general-domain baselines [4]. Benchmark efforts such as LexGLUE further support this direction by providing standardized tasks and evaluation settings that help measure progress in legal language understanding and domain adaptation [5]. However, despite these advances, long-document constraints remain a persistent technical barrier in legal NLP, which motivates methods that can process long sequences more effectively [6].

Judicial judgment summarization is increasingly studied because it can condense long legal decisions into shorter representations that support review. At the same time, legal summarization surveys emphasize that legal summaries must remain faithful to the source, preserve key legal elements, and avoid distortions that may misrepresent facts, issues, or holdings. These works highlight risks such as faithfulness errors and evaluation limitations when applying generic summarization approaches to legal texts [7]. Studies comparing extractive and abstractive methods further note that legal case summarization must balance brevity with preservation of essential reasoning, and that evaluation must account for legal structure and content fidelity, not only surface similarity [8]. Recent AI and Law studies also explore improving judgment summarization by explicitly incorporating structural and semantic information from legal documents, reflecting the importance of legal document structure in summary quality [17], and by using clustering-based approaches to better capture sentence relationships in legal summaries [18].

Legal case retrieval and precedent recommendation are also core tasks in legal AI. Surveys of legal case retrieval report that traditional lexical matching approaches remain common, but they may miss deeper semantic relevance, motivating neural and embedding-based retrieval methods [9]. Empirical work on precedent discovery using textual similarity likewise supports the need for stronger semantic representations and careful similarity modeling when the case volume is large and manual precedent identification becomes a bottleneck [10]. In addition, retrieval research has shown that legal case structure can be used to improve retrieval quality. Approaches that incorporate structural information in case retrieval report gains over structure-agnostic baselines, indicating that legal document organization contains useful signals for relevance [11].

Structure-aware pretraining approaches, such as SAILER, further demonstrate that explicitly modeling legal structure can improve legal case retrieval performance [12].

Although summarization and retrieval are often studied as separate tasks, some work suggests that they can be combined in a way that reflects practical legal workflows. For example, COLIEE retrieval systems have explored dense retrieval with summarization-based re-ranking, which directly links retrieval stages with summarization components [13]. Related work also investigates the interaction of summarization and modern language models for cost-efficient retrieval, indicating that structured condensation can affect retrieval effectiveness and system practicality [14]. In high-stakes domains like law, these capabilities also intersect with the need for transparency. Surveys on explainable AI in law emphasize that legal decision-support tools must support accountability and provide explanations that legal stakeholders can understand [15]. Similarly, studies on explainable decision-support tools for the European Court of Human Rights demonstrate that explanations grounded in legal reasoning can be designed and evaluated as part of trustworthy legal AI tool development [16].

1.2. The Problem, Gap or Opportunity

The administration of justice in the Republic of the Philippines depends inextricably on the consistent, accurate, and timely interpretation of historical jurisprudence. As the digitization of judicial archives accelerates, legal professionals face an unprecedented volume of complex, unstructured textual data. While the transition from physical law libraries to digital repositories has successfully resolved the logistical issue of document accessibility, it has inadvertently generated a severe cognitive bottleneck for practitioners. Judges, litigation attorneys, and legal researchers are now inundated with hundreds of pages of jurisprudence on a daily basis. These legal professionals are forced to expend disproportionate amounts of cognitive energy and temporal resources manually parsing lengthy, highly technical texts merely to extract the core factual premises, the specific legal issues at hand, the underlying judicial logic, and the final dispositive rulings.

The fundamental problem lies in the structural and functional limitations of current digital legal workflows, which fail to accommodate the semantic complexity of Philippine Supreme Court decisions. Existing electronic platforms predominantly function as static document repositories reliant on basic lexical keyword matching and Boolean search parameters. These traditional retrieval architectures are inherently flawed when applied to the legal domain because jurisprudential relevance is rarely dictated by exact word overlap. Legal doctrines are frequently discussed using varied terminologies, and cases that share profound underlying judicial logic may utilize entirely disparate vocabularies depending on the specific factual matrices of the disputes. Consequently, lexical search systems generate massive amounts of irrelevant noise while frequently missing critical, controlling precedents, thereby compromising the integrity of legal research.

Compounding this functional inadequacy is the severe architectural fragmentation of legal text processing. In the current operational paradigm, the distinct processes of understanding a dense legal text and locating structurally analogous precedents are treated as completely isolated, disconnected manual tasks. A legal practitioner must first independently read and mentally condense the primary text to understand its controlling principles. Subsequently, the practitioner must initiate a disjointed, iterative search process on a separate platform to validate newly established legal logic against historical cases. There is no fluid continuity between document comprehension and precedent verification.

Therefore, the primary research gap addressed by this investigation is the critical absence of a unified, natively integrated artificial intelligence framework that seamlessly bridges document comprehension and semantic precedent retrieval within the highly specific jurisdictional context of the Philippine Supreme Court. While isolated, experimental algorithmic solutions currently exist for either text summarization or semantic search independently, there is a pronounced lack of end-to-end computational architectures that perform both functions simultaneously. In a legal context, however, existing general-purpose natural language processing tools fail to preserve strict structural integrity, specifically the separation of facts, issues, reasoning, and rulings, which legal professionals absolutely require to maintain unbroken traceability and factual fidelity.

This void presents a definitive, high-impact opportunity to architect an integrated, domain-adapted natural language processing decision-support framework. By leveraging advanced machine learning models trained specifically on the linguistic patterns of Philippine jurisprudence, an opportunity exists to automatically transform lengthy, unstructured criminal decisions into highly structured, ruling-centered abstracts. By simultaneously generating an abstractive narrative and extracting traceable, sentence-level highlights from the original text, the system can preserve the exact phrasing required for legal verification. Furthermore, this processed semantic data presents an opportunity to power a context-aware recommendation engine that retrieves topically and logically analogous precedents based on deep vector embeddings rather than surface-level keywords. The ultimate opportunity lies in unifying these dual capabilities—summarization and semantic retrieval—into a singular, cohesive digital workflow. This integration will directly mitigate the cognitive overload currently paralyzing legal research, surface latent semantic connections between complex cases, and preserve the absolute primacy of human judicial interpretation through transparent, traceable, and highly structured algorithmic outputs.

1.3. Research Questions

The aforementioned objectives were formulated to address the following research questions, which guide the study in examining how an integrated NLP-based decision support framework can support judgment understanding and precedent retrieval while ensuring that automated outputs remain aligned with practitioner needs and the primacy of human judgment:

(1) What NLP-based approach can an integrated decision support framework use to generate structured, concise, and coherent judicial judgment summaries that preserve key legal reasoning, issues, and context?

(2) Compared with standalone summarization or standalone retrieval, to what extent does an integrated summarization plus context-aware precedent recommendation pipeline affect summary fidelity, retrieval relevance, and practitioner usefulness?

(3) What combination of automatic metrics and human-centered evaluation methods (expert review and task-based studies) most effectively validates that the framework supports legal analysis while mitigating overreliance and preserving human judgment?

1.4. Research Objectives

To examine and evaluate an integrated NLP-based decision support framework for automated judicial judgment summarization and context-aware legal precedent recommendation that enhances legal analysis while preserving the primacy of human judgment. Specifically, this study's attempts to:

(1) Construct a structured, ruling-centered judicial judgment summarization pipeline specifically calibrated for Philippine Supreme Court criminal decisions, utilizing a dual-stage architecture that generates coherent abstractive summaries via Command A (cohere) alongside highly traceable, algorithmically selected extractive highlights via BERTSum, thereby preserving critical facts, legal issues, and judicial logic under strict long-document constraints.

(2) Formulate a context-aware legal precedent recommendation engine that transforms section-structured judicial documents (combining both Facts and Reasoning) into dense semantic embeddings using domain-adapted transformer models (specifically Sentence-BERT architectures like MPNet). This enables the algorithmic ranking of candidate cases via mathematically derived cosine similarity to output a highly relevant, top-10 list of historically analogous precedents to support legal verification.

(3) Evaluate the operational efficacy, computational stability, and analytical reliability of the integrated decision-support pipeline by measuring summarization fidelity through G-Evaluation (LLM as a Judge utilizing a Llama 3.1 70B model) metrics (Faithfulness, Answer Relevance, Answer

Correctness and BertScore F1), measuring semantic retrieval accuracy through ranking metrics (Precision@10, Recall@10), and validating the framework's practical utility and mitigation of automation bias through structured, human-in-the-loop expert appraisals conducted by qualified legal practitioners.

1.5. Significance of the Study

This study introduces a highly specialized, algorithmic artifact designed to optimize the intersection of artificial intelligence and complex legal jurisprudence. By departing from fragmented, single-function tools and proposing a unified, end-to-end computational framework for both document comprehension and semantic retrieval, the study delivers profound operational, pedagogical, and theoretical value across multiple sectors. The significance of this research is distinctly observed across the following stakeholder domains:

For legal practitioners and judicial researchers, the framework directly mitigates the severe cognitive overload and temporal exhaustion associated with parsing voluminous criminal decisions. By deploying a dual-stage summarization model that provides an immediate, highly readable abstractive overview anchored by exact, traceable extractive highlights, trial judges, appellate justices, and litigation attorneys are spared countless hours of manual reading. The system immediately surfaces the controlling legal doctrine of a case without omitting the factual context. Furthermore, the transition from antiquated lexical keyword searching to context-aware semantic retrieval ensures that practitioners discover highly relevant, logically analogous precedents that traditional search engines routinely obscure. This optimization accelerates case preparation, expedites the drafting of judicial resolutions, and allows legal professionals to redirect their finite cognitive resources toward complex analytical judgment and strategic argumentation, rather than basic textual processing.

For law students and legal educators, the ability of the framework to systematically deconstruct dense judicial texts into distinct, structured components provides a revolutionary pedagogical instrument for academic legal institutions. Law students frequently struggle with case digestion—the process of isolating the controlling legal logic from the surrounding contextual commentary. The system's structured output serves as an interactive, algorithmic learning aid, explicitly demonstrating how complex factual narratives map to specific judicial issues and ultimate rulings. This visual and structural separation accelerates the development of analytical legal reading skills, allowing students to comprehend the architecture of Supreme Court decisions with greater clarity and speed.

For the computer science and AI research community, this study provides a critical, empirical contribution to the localized application of domain-specific large language models and transformer architectures. By fine-tuning models such as LEGAL-BERT, BERTSum, Command A (Cohere), and Sentence-BERT exclusively on the highly specific linguistic and structural patterns of Philippine Supreme Court criminal decisions, the study expands the boundaries of computational linguistics in non-Western, highly specialized legal contexts. Additionally, by treating summarization and retrieval not as isolated algorithms but as a continuously integrated computational pipeline, the research offers a novel architectural blueprint. It provides empirical data on the efficiency trade-offs, vector space behaviors, and loss function optimizations required when handling extreme long-document constraints in high-stakes domain applications.

For civil society and institutional transparency, At a macro-societal level, delays in the justice system are frequently tied to severe docket congestion and the excruciatingly slow pace of case resolution. By radically accelerating the rate at which judicial actors can accurately research, review, and draft legal decisions, this technological framework indirectly supports the constitutional mandate for speedy trial and efficient case disposition. When judges and clerks are empowered by advanced decision-support tools, the entire judicial machinery operates with reduced friction. Ultimately, providing the judiciary with transparent, traceable, and highly accurate algorithmic assistance strengthens institutional capacity, reduces case backlogs, and fosters broader societal trust in the efficiency, fairness, and modernization of the justice system.

1.6. Scope and Limitations (or Delimitations)

This study focuses exclusively on the technical construction, algorithmic fine-tuning, and empirical evaluation of an integrated, natural language processing-based decision-support framework encompassing judicial text summarization and context-aware legal precedent recommendation. To ensure deep technical viability, rigorous experimental control, and computational feasibility within the allotted research timeframe, the investigation is strictly bounded by specific programmatic, data-related, and operational parameters. The computational models are trained, validated, and evaluated strictly on a highly curated dataset of English-language criminal decisions promulgated exclusively by the Supreme Court of the Republic of the Philippines. The textual corpus is explicitly restricted to five predefined criminal case categories: murder, homicide, illegal drugs, rape, and estafa. To maintain dataset balance and computational efficiency, the data volume is strictly bounded to a target extraction of 600 to 1,900 complete, valid decisions per category, yielding a total analytical corpus of 6,431 documents. Decisions rendered in regional languages, trial court judgments, Court of Appeals decisions, civil cases, corporate disputes, administrative disciplinary cases, and scanned HTML documents that fail automated text extraction protocols are definitively excluded from the data pipeline and the scope of this study.

In terms of algorithmic and architectural delimitations, the summarization component of the framework is structurally constrained to a retrieve-then-read dual-stage architecture, using BERTSumfor extractive sentence selection followed by Command A for abstractive synthesis grounded on the extracted legal highlights, Facts, and Ruling/Decision sections. The semantic recommendation engine is strictly bound to dense vector embeddings generated through Sentence-BERT, with cosine similarity used for Top-10 precedent ranking. The mathematical ranking of precedent cases relies entirely on cosine similarity formulations operating over these high-dimensional embeddings. The system processes textual structures based exclusively on four predefined, extracted legal elements: Facts, Issues, Judicial Reasoning, and Ruling. The research does not attempt to engineer entirely new neural network architectures from scratch, nor does it explore multilingual modeling, cross-jurisdictional transfer learning, or generative pre-trained transformers outside of the specified BERT and Command A (cohere) architectures.

Finally, the quantitative assessment of the summarization outputs is delimited to G-Evaluation (LLM as a Judge) metrics, specifically measuring Faithfulness, Answer Relevance, Answer Correctness, and BertScore F1. The performance of the semantic precedent recommendation module is quantitatively constrained to Top-10 algorithmic ranking metrics, specifically computing Precision@10 and Recall@10 based on a predefined evaluation pool. Qualitative, human-in-the-loop validation is operationally constrained to a highly specialized expert panel comprising exactly two legal professionals: both an active litigation lawyer. This targeted sample size is sufficient for deep structural prototype validation, but is explicitly not intended to represent a statistically significant user acceptance survey. Operationally, the developed software artifact functions strictly as an experimental, offline decision-support prototype engineered to assist human legal review rather than execute autonomous legal logic or predict case outcomes. It is a localized computational research prototype and does not feature the robust cybersecurity architecture, cloud scalability, or institutional API integration required for live deployment within the production servers of the Philippine Judiciary. This study is further limited to the use of publicly accessible Supreme Court decisions obtained from LawPhil, and does not involve confidential court records or privately submitted legal documents. Although this reduces privacy risk at the dataset level, broader governance and compliance safeguards for large-scale public web deployment are beyond the scope of the present study.

Chapter 2

REVIEW OF RELATED LITERATURE & STUDIES

This chapter reviews the literature and studies relevant to AI-assisted legal analysis, with emphasis on legal Natural Language Processing (NLP) for judicial judgment understanding and precedent identification. Section 2.1 introduces core legal NLP tasks, long-document constraints, and responsible use requirements. Section 2.2 reviews retrieval methods from lexical baselines to transformer-based, structure-aware, and LLM-assisted approaches. Section 2.3 examines legal summarization methods, emphasizing hybrid and structure-guided strategies for long judgments. Section 2.4 summarizes evaluation practices and their limitations in high-stakes legal settings. Section 2.5 synthesizes the consolidated research gap motivating the proposed integrated framework.

2.1. Foundations of Legal NLP for AI-Assisted Legal Decision Support

2.1.1. Growth of Legal Text and Motivation for NLP Support

Legal systems produce large volumes of text (decisions, pleadings, briefs, and statutes). This growth increases the time required for review and verification, motivating legal NLP as a support layer that improves access and efficiency while preserving professional accountability [1], [3], [38].

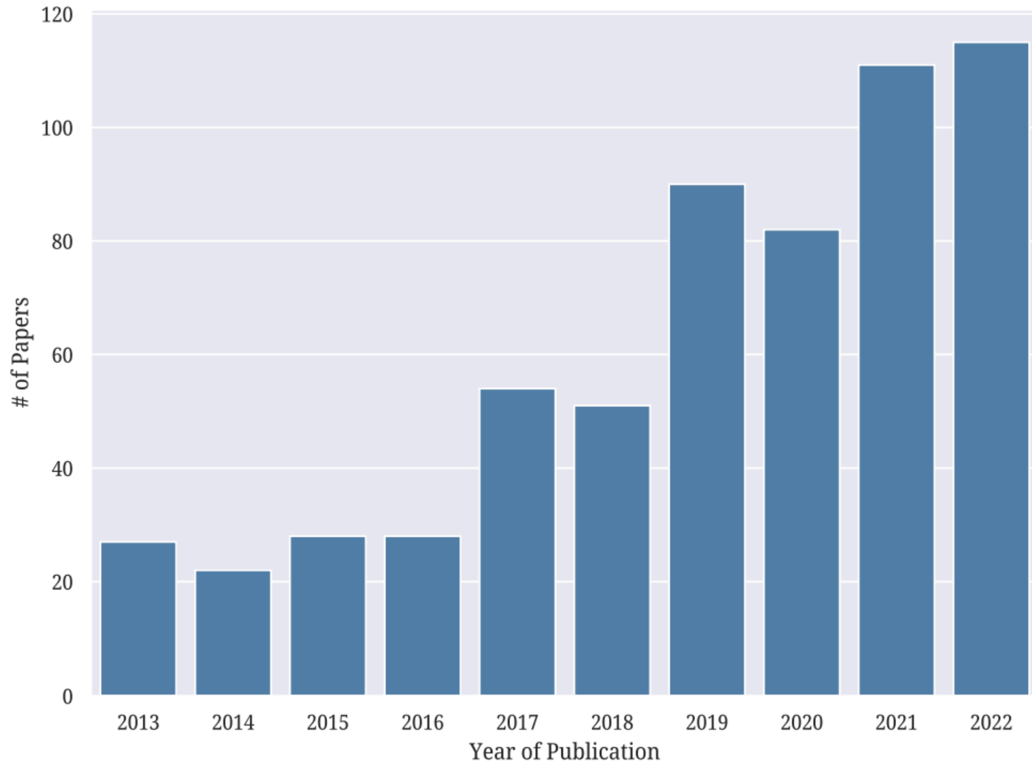

Figure 1. Number of Legal NLP Papers over Time

2.1.2. Core Tasks in Legal NLP

Survey literature frames legal NLP as a collection of tasks that map to legal work, including classification, information extraction, retrieval, entailment, summarization, and related decision-support functions [3], [38], [39]. In practice, these tasks are rarely independent: retrieval supports precedent research, summarization supports fast judgment understanding, and entailment supports reasoning about support or contradiction between texts [3], [39].

2.1.3. Legal Language and Long-Document Constraints

A repeated theme in legal NLP is that legal text differs from general text. It contains specialized terminology, formal structures, and reasoning patterns that must remain intact for the output to be legally meaningful [3, 7, 38]. Another persistent constraint is length. Judgments and briefs can exceed typical model limits, creating truncation risks and loss of important context [6, 7]. As a result, literature often emphasizes structure-aware processing, careful input management, and evaluation for faithfulness, since fluent outputs can still be legally incorrect if the reasoning chain is distorted [7, 39, 40].

2.1.4. Responsible AI, Explainability, and Human Oversight in Legal Contexts

Legal decision-making is high-stakes, so literature on AI and law emphasizes transparency, accountability, and human oversight. AI is generally framed as an assistive tool rather than a substitute for human legal judgment [42]. This position aligns with explainability-focused work that treats justification, traceability, and evidence linking as necessary for trust in legal AI outputs [16, 17]. In practical terms, this supports the view that systems should help users review and verify, not simply produce final answers.

2.1.5. System Context: From Document Platforms to Domain-Adapted LLMs

Platform-oriented work shows that many legal technologies prioritize storage and document management but provide limited support for workflow-level judgment understanding and precedent verification [44]. In parallel, domain adaptation studies report that legal-domain pretraining and fine-tuning improve performance by capturing legal language patterns [4], [5], [45]. These findings suggest that usefulness depends on both model capability and workflow integration [43], [44].

Benchmark evidence further supports the use of legal-domain pretraining in legal NLP. The LexGLUE benchmark shows that legal-oriented pretrained models outperform generic transformer baselines in aggregated legal language understanding performance, with LEGAL-BERT remaining one of the strongest overall models [5]. This indicates that domain-adapted pretraining captures legal terminology, stylistic conventions, and task-specific patterns more effectively than general-domain encoders. For the present study, this strengthens the choice of LEGAL-BERT as the domain-adapted backbone for the extractive summarization module, ensuring that the algorithmic selection of key sentences accurately captures Philippine Supreme Court reasoning.

Table 1. Adapted benchmark evidence supporting LEGAL-BERT for legal-domain language representation

Model / Source	**Reported pattern from the studies**	**Relevance to the proposed study**
Generic BERT-family baselines [5]	General-domain transformer baselines perform competitively, but not as strongly as legal-oriented models in aggregated legal benchmark performance	Shows that generic pretraining alone is not optimal for legal semantic representation
LEGAL-BERT [4]	One of the strongest overall models in aggregated LexGLUE legal benchmark results	Supports using LEGAL-BERT as the main legal-domain representation model
CaseLaw-BERT [4]	Also performs strongly, confirming the value of legal-domain pretraining	Reinforces that legal-domain pretraining is more suitable than generic pretraining
Main synthesis	Legal-oriented pretrained models perform better overall on legal language understanding tasks	Justifies domain adaptation before downstream extractive summarization and legal feature extraction.

2.2. Legal Case Retrieval and Context-Aware Precedent Identification

2.2.1. Lexical Retrieval and Hybrid Retrieval Pipelines

Traditional legal search relies on lexical matching (keywords, citations, and Boolean queries). However, legally relevant cases may use different terminology, and relevance often depends on issues and reasoning rather than word overlap [2], [11]–[13]. Hybrid pipelines address this by using fast lexical retrieval to collect candidates and stronger neural models to re-rank them [29].

2.2.2. Transformer-Based Semantic and Structure-Aware Retrieval

Transformer-based retrieval improves semantic matching by representing context beyond exact terms. Recent work also shows gains from leveraging legal structure (e.g., matching facts, issues, reasoning, and rulings instead of treating the document as uniform text) [11]–[13]. Benchmark-oriented studies (e.g., COLIEE) demonstrate that transformer-based approaches can improve retrieval and entailment performance, but at higher computational cost than classical baselines [26].

Beyond general transformer retrieval, sentence-embedding models make semantic matching more scalable. Sentence-BERT is designed to encode text into fixed-size embeddings that can be compared efficiently using cosine similarity, making it suitable for retrieval settings that require large-scale pairwise comparison [52]. Since the proposed framework ranks candidate precedents through cosine similarity over embeddings, Sentence-BERT provides a practical basis for context-aware precedent recommendation under realistic retrieval constraints.

Table 2. Adapted benchmark evidence supporting Sentence-BERT for efficient semantic similarity retrieval

Model / Source	Reported pattern from the study	Relevance to the proposed study
Average BERT embeddings [52]	Simple averaging of BERT embeddings performs weaker on semantic similarity tasks	Shows that plain contextual embeddings are not enough for similarity-based retrieval
Sentence-BERT (SBERT-NLI-base) [52]	Strong improvement over average BERT embeddings in cosine-based semantic similarity evaluation	Supports efficient semantic matching for precedent retrieval
Sentence-BERT (SBERT-NLI-large) [52]	One of the strongest SBERT variants in the reported STS benchmark results	Supports using Sentence-BERT as a scalable ranking model
Sentence-embedding family overall	Sentence-specific embedding architectures are more suitable than naive embeddings for cosine similarity search	Justifies embedding-based precedent recommendation

2.2.3. LLM-Assisted Relevance Judgment, Query Refinement, and Explanation

Recent studies position LLMs as controlled components for relevance judgment, query reformulation, or explanation rather than as fully autonomous legal search engines. For example, LLMs can label relevance for retrieved candidates using structured prompts, improving alignment with expert judgments but increasing cost and sensitivity to prompting [19]. Other work uses question generation or structured summaries to improve retrieval under context-length constraints [18].

2.2.4. Retrieval as Legal-Element Generation

Some approaches reframe retrieval as generating the legal elements expected in relevant cases (e.g., factors, issues, or doctrinal elements) and then retrieving cases containing those elements, sometimes constraining generation to sequences grounded in the corpus [33]. This direction reinforces the importance of structure and traceability for practitioner-facing retrieval.

2.2.5. Implications for Decision Support

Across retrieval studies, a recurring limitation is that technical ranking performance does not automatically translate to practitioner usefulness. Decision support requires transparent presentation and verification support so users can understand why a precedent is suggested and assess applicability [16], [17], [42], [43].

2.3. Legal Text Summarization for Judgment Understanding

2.3.1. Role of Summarization in Legal Review

Legal judgments are lengthy and structurally dense, making summarization a practical tool for reducing reading burden. Legal summarization must prioritize faithfulness: fluent summaries are not useful if they omit controlling facts, issues, reasoning, or the holding [7], [8].

2.3.2. Hybrid Extractive–Abstractive Summarization for Long Judgments

A common strategy in legal summarization is hybrid summarization, where the system first extracts the most important sentences for coverage and grounding, then rewrites or compresses them to improve readability. This design is repeatedly justified in legal settings because purely extractive summaries can be repetitive and fragmented, while purely abstractive summaries risk hallucination or losing legal details when judgments exceed model limits [31].

Ouyang et al. (2022) [31] represent this approach through a two-stage pipeline: an extractive component selects key sentences from a judgment, then an abstractive generator rewrites the selected content into a smoother summary. Their work explicitly targets long-document constraints by extending positional handling and using copying mechanisms to preserve important legal terms that may otherwise be lost or paraphrased incorrectly [31].

Legal summarization must balance readability with rigorous factual "faithfulness," as errors in controlling facts, issues, or rulings can compromise legal interpretation [7], [8]. Purely extractive methods effectively retain grounded facts but often lack the synthetic coherence preferred by practitioners; conversely, purely abstractive approaches improve fluency but risk hallucinating details when processing lengthy, structurally dense judgments that exceed model context windows [8].

Current research increasingly addresses these trade-offs through sequential "extract-then-abstract" architectures [7], [8], [31]. In this configuration, an extractive component, such as BERTSum, first isolates legally controlling "highlights," which are subsequently synthesized by a generative model [51]. By using extractive segments as a grounded input for the abstractive generator, this pipeline ensures the final summary remains anchored in traceable source text, satisfying the dual requirements of linguistic fluency and evidentiary precision [7], [31].

Table 3. Adapted literature support for the BERTSum → Command A sequential hybrid summarization design

Model / Study	Original setting	Reported strength	Relevance to the proposed study
BERTSUMEXT [51]	CNN/DailyMail summarization benchmark	Strong extractive sentence selection using pretrained encoders	Supports BERTSum for traceable extractive highlights
BERTSUMEXTABS [51]	CNN/DailyMail summarization benchmark	Combines extractive grounding with abstractive generation	Supports the logic of a hybrid summarization pipeline
Instruction-Tuned LLMs (e.g., Command A) / LLM-based Summarization [7]	Legal document summarization / Long-context reasoning	High structural adherence and fluent synthesis when grounded on extracted context	Supports using an instruction-tuned LLM for the final abstractive generation phase
BERT-Ext [8]	Long legal case summarization on IN-Ext	Strong extractive baseline in legal summarization evaluation	Supports retaining an extractive verification layer alongside abstraction
Main synthesis	Extractive pre-filtering combined with LLM-based abstraction offers optimal faithfulness and fluency	Strong extractive baseline in legal summarization evaluation	-

To show how extractive and abstractive methods compare directly in a legal summarization setting, Table 4 reproduces the document-wide evaluation reported by Shukla et al. (2022) [8].

Table 4. Document-wide ROUGE and BERTScore results on the IN-Ext dataset

Algorithm	**ROUGE Scores**			**BERTScore**
	R-1	R-2	R-L	
Extractive Methods				
Unsupervised, Domain Independent				
Luhn	0.568	0.373	**0.422**	**0.882**
Pacsum_bert	**0.59**	**0.41**	0.335	0.879
Unsupervised, Legal Domain Specific				
MMR	0.563	0.318	0.262	0.833
KMM	0.532	0.302	0.28	0.836
LetSum	**0.591**	**0.401**	**0.391**	**0.875**
Supervised, Domain Independent				
SummaRunner	0.532	0.334	0.269	0.829
BERT-Ext	**0.589**	**0.398**	**0.292**	**0.85**
Supervised, Legal Domain Specific				
Gist	0.555	0.335	0.391	0.864
Abstractive Methods				
Pretrained				
BART	0.475	0.221	0.271	0.833
BERT-BART	**0.488**	**0.236**	**0.279**	0.836
Legal-Pegasus	0.465	0.211	**0.279**	**0.842**
Legal-LED	0.175	0.036	0.12	0.799
Finetuned				
BART_MCS	0.557	0.322	0.404	0.868
BART_MCS_RR	0.574	0.345	0.402	0.864
BERT_BART_MCS	0.553	0.316	0.403	**0.869**
Legal-Pegasus_MCS	**0.575**	**0.351**	**0.419**	0.864
Legal-LED	0.471	0.26	0.341	0.863

Domain-focused hybrid designs also appear in specialized legal areas. For Chinese maritime judgments, Zhang et al. (2025) [30] propose an extractive–abstractive pipeline enhanced with a maritime lexicon, reporting that domain vocabulary support improves summary quality and term consistency in that setting [30]. More recent work by Patel et al. (2026) [37] similarly compares a hybrid engineered pipeline against a fine-tuned LLM approach and reports that the hybrid method performs better for preserving relevant content under their evaluation, reinforcing the view that structured hybrid pipelines remain competitive for long legal documents [37].

2.3.3. Structure-Guided and Knowledge-Guided Summarization

Structure-guided methods steer selection and generation using expected legal components (background/facts, issues, reasoning, disposition). Knowledge-guided methods inject domain signals or constraints so outputs better reflect legal logic. Recent work reports improved summarization when logical structure and legal knowledge are used to guide content and organization [22]. Dataset studies likewise suggest that high-quality expert summaries implicitly follow legal structure, capturing salient content across different parts of a decision [41].

2.3.4. Reasoning-Guided Summarization Using Judicial Logic

While structure-guided summarization organizes content by component, reasoning-guided methods aim to capture the logical progression of the judicial decision, often modeled through the concept of judicial syllogism, the inference path from the major premise (law), the minor premise (facts), and to the conclusion (holding). Recent literature demonstrates that modern LLMs can be effectively steered to perform this "chain-of-thought" reasoning, allowing the model to summarize not just what happened in a case, but why the court ruled as it did [23].

This reasoning-centered approach addresses a known gap where summaries that are structurally accurate may still fail to convey the court’s rationale [7], [22]. By framing summarization as a process of latent reasoning reconstruction, models are better able to preserve the essential holding of a decision. Dataset studies indicate that expert-authored summaries inherently prioritize this logical flow, identifying the "ratio decidendi" (the reason for the decision) as the core information unit [41]. Consequently, guiding a model to prioritize the judicial syllogism ensures that the final summary is not merely a condensation of facts, but a faithful reflection of the court's analytical process.

2.4. Benchmarking and Evaluation of Legal NLP Systems

Evaluation is a central concern in legal NLP because legal outputs are not judged only by readability. In legal settings, quality depends on whether the output preserves legally controlling facts, issues, reasoning, and outcomes. A summary or retrieved precedent can sound fluent but still be legally wrong. This matters because legal work is high-stakes and errors can mislead analysis even when the text appears credible [3, 7, 38]. As a result, legal NLP evaluation must account for (1) correctness, (2) reasoning preservation, and (3) the ability of users to verify claims, not only surface similarity.

2.4.1. Why Legal NLP Evaluation Is Harder Than General NLP Evaluation

Legal text has constraints that make evaluation stricter than in common NLP tasks. First, legal meaning is often carried by specific elements, such as dates, party roles, procedural posture, applicable legal rules, and narrow holdings. Dropping or changing one element can change the legal interpretation even if the output stays coherent [3, 7]. Second, legal reasoning requires alignment between facts, issues, rules, and conclusions. An output that accurately summarizes case facts but distorts the court's rationale (the ratio decidendi) fails as a legal summary, highlighting that "faithfulness" is a more critical dimension of quality than surface-level fluency [7]. Third, law is inherently context-dependent. Relevance and correctness depend heavily on jurisdiction, prevailing legal doctrine, and procedural history, complicating the definition of "ground truth" and making standard benchmarks difficult to generalize [3], [38]. These challenges necessitate evaluation frameworks that move beyond linguistic overlap to assess the preservation of judicial logic and legal truth.

2.4.2. Evaluation of Legal Retrieval and Precedent Recommendation

Retrieval is commonly evaluated with ranking metrics such as Precision@k, Recall@k, MAP, and NDCG [11]. However, legal relevance can be interpretive and multi-dimensional (issue match vs. fact-pattern match), making labels sensitive to definition and requiring expert judgment for defensibility [11], [19].

2.4.3. Evaluation of Legal Summarization and the Limits of Overlap Metrics

Overlap metrics (e.g., ROUGE, BLEU) remain the industry standard for general text summarization due to their computational efficiency. However, in the legal domain, they are increasingly viewed as insufficient metrics for quality. While these tools quantify surface-level lexical similarity, they fail to measure "legal faithfulness", the degree to which a summary accurately preserves the court's holding, reasoning, and dispositive outcomes [7].

Recent empirical evaluations emphasize that model scale is not the sole determinant of quality in legal summarization. Medina-Ramírez et al. [50] demonstrate that smaller, domain-optimized language models can achieve performance comparable to larger general-purpose models when evaluated using a rigorous, multi-faceted framework. This study further validates the necessity of moving beyond generic metrics like ROUGE, arguing that for legislative and judicial texts, evaluation must prioritize factual adherence and logical consistency over surface-level lexical overlap [50].

As demonstrated in figure 2, while standard overlap metrics such as ROUGE remain prevalent, they often fail to account for the structural and logical complexity of legal judgments. The summary of existing literature highlights that diverse datasets, ranging from Indian legal corpora to European case law, frequently rely on ROUGE to assess performance, yet this metric lacks the nuance required to evaluate legal faithfulness or the preservation of controlling holdings [7].

Work	Year	Method	Description	Dataset	Metric	HumEval
[32]	2024	Ensemble different summarization algorithms	Three ensemble methods are proposed: voting-based, ranked-list ensemble (using Borda count and Reciprocal Ranking), and graph-based (leveraging sentence similarity graphs to select identical sentences from connected components) for better summarization	IN-Ext, IN-abs	ROUGE, METEOR	No
[24]	2023	BSLT LPGN	A hybrid legal summarization approach combines the extractive model BSLT and the abstractive model LPGN based on Lawformer to generate the summary	CAIL2020	ROUGE	No
[88]	2023	Extractive then abstractive	A transfer learning approach combines extractive and abstractive techniques for legal summarization, addressing limited labeled data by selecting sentences and generating summaries using GPT-2	Australian Legal Case Report Dataset	ROUGE, FactCC	No
[44]	2023	Extractive then abstractive	Legal texts are normalized using dictionaries for abbreviations and article summaries, then processed with BART for extractive summarization and PEGASUS for abstractive summarization	SCI, IndianKanoon, Manupatra, ILDC	ROUGE	No
[43]	2023	Extractive then abstractive	Machine learning methods label sentences as important or not, followed by LSTMs for summarization, which are then compared to PEGASUS for performance evaluation	Opinion of the supreme court of Utah, Idaho, Arizona, New Mexico, Nevada, and Colorado	Recall, F1	No
[102]	2023	Extractive then abstractive	RoBERTa vectorizes document sentences, which are processed by a Dilated Gated CNN extractive model to select relevant sentences. The resulting corpus is then fed into T5 PEGASUS for abstractive summarization	CAIL	ROUGE	No
[121]	2022	Extractive then abstractive	Ripple Down Rules classify sentences into 13 rhetorical roles using C4.5 decision trees, Naive Bayes, SVM, Conditional Random Fields, and BiLSTM algorithms for improved sentence categorization	50 documents of Bombay High Court collected from Legal Search of Manupatra	ROUGE	Yes
[81]	2018	Ontology-based	An ontology-based approach extracts semantic knowledge from Chinese legal documents, summarizes it into knowledge blocks, computes block similarity, and classifies the documents into different categories	CTA, CDD	Accuracy	No

Figure 2. A Comprehensive Survey on Legal Summarization

This discrepancy between traditional automated scoring and the requirements of legal professionals underscores a major research gap: the need for evaluation frameworks that incorporate structural and knowledge-guided signals to ensure outputs are not only fluent but legally defensible.

As the field shifts toward more robust verification, recent literature has begun adopting multi-dimensional, expert-aligned evaluation frameworks. This modern standard incorporates not only lexical metrics but also semantic similarity (e.g., BERTScore) and, critically, LLM-as-a-Judge (G-Eval) methodologies to assess faithfulness, structural coherence, and legal correctness.

Model	R1	R2	RL
ORACLE	52.59	31.24	48.87
LEAD-3	40.42	17.62	36.67
Extractive			
SUMMARUNNER (Nallapati et al., 2017)	39.60	16.20	35.30
REFRESH (Narayan et al., 2018b)	40.00	18.20	36.60
LATENT (Zhang et al., 2018)	41.05	18.77	37.54
NEUSUM (Zhou et al., 2018)	41.59	19.01	37.98
SUMO (Liu et al., 2019)	41.00	18.40	37.20
TransformerEXT	40.90	18.02	37.17
Abstractive			
PTGEN (See et al., 2017)	36.44	15.66	33.42
PTGEN+COV (See et al., 2017)	39.53	17.28	36.38
DRM (Paulus et al., 2018)	39.87	15.82	36.90
BOTTOMUP (Gehrmann et al., 2018)	41.22	18.68	38.34
DCA (Celikyilmaz et al., 2018)	41.69	19.47	37.92
TransformerABS	40.21	17.76	37.09
BERT-based			
BERTSUMEXT	43.25	20.24	39.63
BERTSUMEXT w/o interval embeddings	43.20	20.22	39.59
BERTSUMEXT (large)	43.85	20.34	39.90
BERTSUMABS	41.72	19.39	38.76
BERTSUMEXTABS	42.13	19.60	39.18

Figure 3. Performance Comparison of Legislative Summarization Models

As illustrated in figure 3, recent empirical studies are now benchmarking language models using this comprehensive approach, demonstrating that model performance on legacy metrics like ROUGE does not always correlate with legal utility.

2.4.4. Benchmarks and Shared Tasks as Evaluation Infrastructure

Benchmarks function as the essential infrastructure for standardizing legal NLP, driving the community to move beyond simple classification tasks toward complex legal reasoning. LegalEval (SemEval 2023 Task 6) underscores the necessity of evaluation designs that strictly adhere to domain-specific constraints, such as rhetorical role labeling and legal explanation generation [39].

As the field pivots toward large-scale generative models, newer benchmarks, such as BriefMe [32] and LegalEval-Q [40], have emerged to address the specific challenges of LLM-generated legal text. Unlike static classification datasets, these frameworks prioritize the quality of generative outputs, focusing on metrics such as logical coherence, the quality of citations, and the mitigation of risks like hallucinations and unsupported claims [32], [40]. This evolution in benchmarking highlights a consensus in the field: as legal AI systems move from "understanding" (classification) to "assisting" (generation), evaluation protocols must evolve to simulate the rigorous verification standards required by legal practitioners.

2.4.5. Toward Practitioner-Aligned and Human-Centered Evaluation

For decision support, evaluation should measure workflow usefulness: whether summaries reduce reading time while preserving key elements, whether retrieved precedents come with traceable support, and whether outputs are stable enough for review and audit [3], [40], [42], [43].

2.5. Research Gaps and Synthesis

The reviewed literature shows clear progress in legal retrieval, legal summarization, and legal benchmarking. However, it also shows that these capabilities are still fragmented across tasks, datasets, and evaluation setups. In practice, legal analysis is workflow-based. A user must first understand the current case, then locate relevant precedents, then verify why those precedents matter. Many studies improve one step at a time, but fewer works connect the steps into a unified, human-centered decision-support process [3, 38, 43]. The gaps below consolidate what is consistently missing across the reviewed works and clarify the motivation for an integrated approach.

2.5.1. Fragmentation of Judgment Understanding and Precedent Identification

A dominant gap is that judgment understanding and precedent identification are often treated as separate problems. Workflow-level benchmarks that simulate multi-stage judicial reasoning also show that current systems struggle to perform reliably across stages, reinforcing that end-to-end integration remains immature [20]. Retrieval research improves ranking, relevance judgment, or element alignment [19, 26, 29, 33], while summarization research improves content compression, coherence, and structure preservation [22, 23, 30, 31, 37]. Even when a study uses summarization to support retrieval, the summary is often treated as a compression artifact for efficiency rather than as a structured understanding layer for users [21]. This separation matters because legal professionals typically need both outputs together: a grounded case understanding and a defensible set of recommended precedents, ideally with explanations that link the recommendation to legally controlling elements.

2.5.2. Long-Document Constraints and Cost–Quality Trade-offs Remain Unresolved

Long judgments continue to create practical limitations. Even when models and embeddings are strong, truncation, chunking, and token budgeting can drop legally controlling details [6, 7]. Several studies propose cost-efficient strategies (e.g., summarization-assisted retrieval, structured summaries, or split-and-aggregate document representations), but these often introduce trade-offs between efficiency, stability, and completeness [21, 36]. Related summarization pipelines for long legal documents also emphasize efficiency and readability, but typically evaluate summarization in isolation and do not test whether the summaries measurably improve downstream legal research tasks such as precedent retrieval and verification [34]. In applied settings, this creates a usability gap: systems must reduce reading burden without losing key facts, issues, and reasoning that determine legal relevance.

This gap is visible across jurisdictions and task types. Retrieval pipelines still struggle when relevance depends on specific legal factors rather than surface similarity [19, 33]. Summarization pipelines still risk omission or distortion when compressing long reasoning chains [22, 23, 31, 37]. This suggests a need for long-document handling that is both structure-aware and user-verifiable, instead of only token-efficient.

2.5.3. Evaluation Gaps: Legal Faithfulness, Stability, and Reproducibility

Current benchmarking literature reveals a critical consensus: legal correctness cannot be inferred from linguistic fluency or lexical overlap metrics alone [7], [39], [40]. Standard retrieval metrics, such as MAP and NDCG, effectively measure ranking behavior but fail to capture whether a retrieved precedent is legally defensible, particularly when relevance is highly interpretive [11], [19]. Similarly, traditional summarization metrics, such as ROUGE and embedding-based similarity, often reward surface-level coverage of non-essential content while missing whether the summary faithfully preserves the court's holding and rationale [7], [41]. As noted in recent quality-focused benchmarks, the field is beginning to reject this surface-level evaluation in favor of protocols that assess generated text on dimensions beyond mere readability, such as factual consistency and adherence to legal logic [40].

There is a recurring gap regarding the stability and reproducibility of legal AI systems. Professional legal work demands consistency as practitioners require evidence paths that are stable across varying prompts and retrieval configurations. However, many current studies lack comprehensive replication materials or fail to address the sensitivity of models to hyper-parameter changes, weakening the confidence that reported performance gains will translate into reliable, practitioner-facing tools. This underscores the need for evaluation frameworks that prioritize not only "peak" model performance but also the reliability and auditability of outputs in a professional legal environment.

2.5.4. Practitioner Alignment Gap: Systems vs. Workflows

Several works move toward realistic legal tasks (e.g., brief-oriented summarization and retrieval) and highlight that difficult tasks remain weak even for strong models [32]. At the same time, system-oriented platforms often emphasize document management, ingestion, and workflow efficiency, but stop short of deeper judgment understanding paired with context-aware precedent recommendation inside one unified interface [44]. Even scalable judgment recommendation systems based on semantic similarity and embeddings can improve retrieval efficiency, but they often provide limited support for judgment summarization and user-facing justification that helps practitioners evaluate why a recommended case matters [24]. Prototype legal assistance platforms show feasibility for RAG-style grounding, but they do not fully resolve integrated judgment summarization + precedent recommendation + verification as a single workflow [35].

In parallel, prediction-focused studies integrate retrieval or legal knowledge to improve outcome prediction, but retrieval is often used internally for model accuracy rather than exposed as transparent precedent support for users [25, 27, 28]. This reinforces a central gap: many pipelines optimize model performance, but fewer designs optimize human use, where the system must reduce cognitive load while keeping the user in control through traceable justifications [42, 43].

2.5.5. Consolidated Research Gap and Relevance to the Proposed Study

Across the reviewed works, the consolidated gap is the lack of an integrated, human-centered legal decision-support workflow that combines:

1. judgment understanding (preferably structured and legally grounded),
2. context-aware precedent identification (beyond surface similarity), and
3. verification support (clear evidence links, stable outputs, and evaluation aligned with legal correctness).

The literature supports the need for this integration. Retrieval research shows improved semantic relevance and reasoning-based judgment of relevance, but still needs stronger practitioner-facing explanations and stability [19, 33]. Summarization research shows that structure and legal logic improve quality, but often remains isolated from retrieval and user verification [22, 23, 31, 37]. Benchmarking research shows that evaluation must reflect legal constraints and real workflow demands, not only generic NLP scoring [32, 39, 40]. Finally, decision-support and AI-in-law literature emphasizes that AI should assist, not replace, legal judgment, which implies transparency and human oversight by design [42, 43].

This set of gaps directly motivates a unified approach where summarization and retrieval are treated as connected components within one decision-support pipeline, with outputs designed for traceability and human review rather than autonomous legal conclusions.

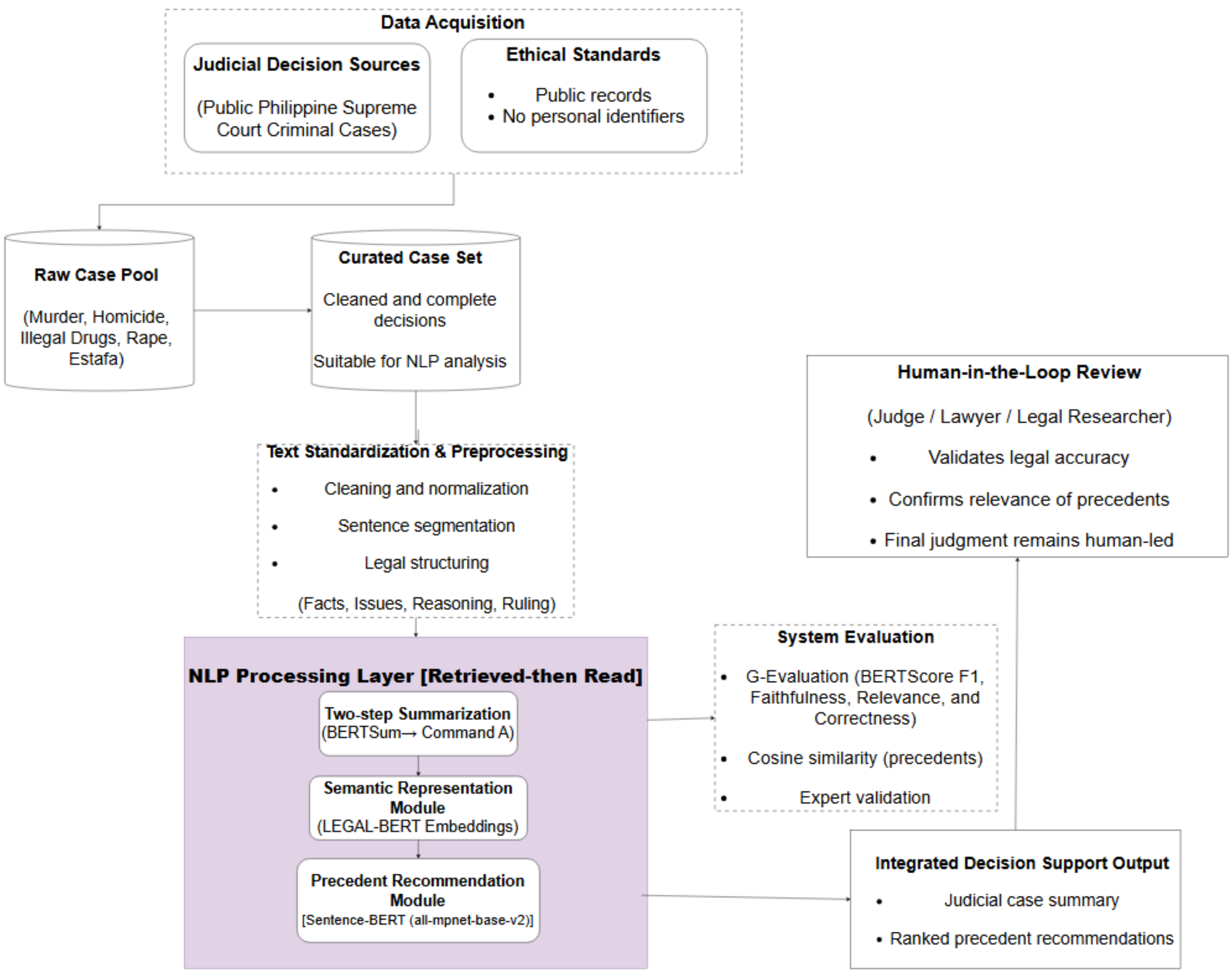

Figure 4. Conceptual Framework Diagram

The conceptual framework of the study is shown in figure 4. It demonstrates the process of collecting and curating Supreme Court criminal cases into a cleaned and complete case set, and then cleaning, normalizing, and segmenting them. The framework also focuses on the structure of decisions—Facts, Issues, Judicial Reasoning, and Ruling/Decision—ensuring that each task in the downstream NLP has a corresponding structure. The workflow after structuring is the NLP processing layer that uses a retrieve-then-read summarization strategy that is performed using BERTSum (extracting legally significant highlights) and Command A (synthesizing the extracted highlights). Simultaneously, semantic representation and precedent recommendation is done by using Sentence-BERT based similarity ranking by cosine similarity. The framework incorporates BERTSum-PH-v2 for extractive legal highlights, Command A for abstractive synthesis, and Sentence-BERT to efficiently rank the similarity of the precedents in a scalable manner. The framework also includes a human-in-the-loop review by legal professionals, judges, or researchers to ensure the legality of the responses, the relevance of the precedent, and to maintain human oversight over the final judgment. The framework concludes with a single integrated decision support output, the judicial case summary and list of ranked precedents, which is then evaluated by the system and expertly reviewed to ensure legal correctness and relevance.

Chapter 3

METHODOLOGY

This chapter explains the end-to-end methodology used to build and evaluate an NLP-driven decision support framework for Philippine Supreme Court criminal decisions. The workflow covers: (1) dataset acquisition and quality screening, (2) preprocessing and legal-section structuring, (3) model training for summarization and precedent recommendation, and (4) evaluation using automatic metrics and expert review.

3.1. Data Gathering

This subsection describes the procedure used to collect Supreme Court criminal case decisions and to organize them into a structured dataset for model training and evaluation. The data gathering process focuses on acquiring publicly accessible judicial decisions, ensuring category balance across selected offenses, and applying quality screening to retain only cases that are complete and technically usable for text extraction and analysis.

3.1.1 Data Sources

The primary source of judicial decisions is LawPhil, a publicly accessible legal repository that publishes Supreme Court decisions as HTML webpages containing the full text of the decision. The study restricts the scope to Supreme Court decisions only to ensure consistency in legal authority and document structure. Decisions may come from any promulgation year, subject to the inclusion and exclusion criteria defined in this subsection.

3.1.2. Case Categories and Dataset Size

The dataset comprises five criminal case categories: murder, homicide, illegal drugs, rape, and estafa. To ensure a balanced distribution across categories, the study targeted between 600 and 1,900 decisions per category, resulting in a total of 6,431 cases across the complete dataset.

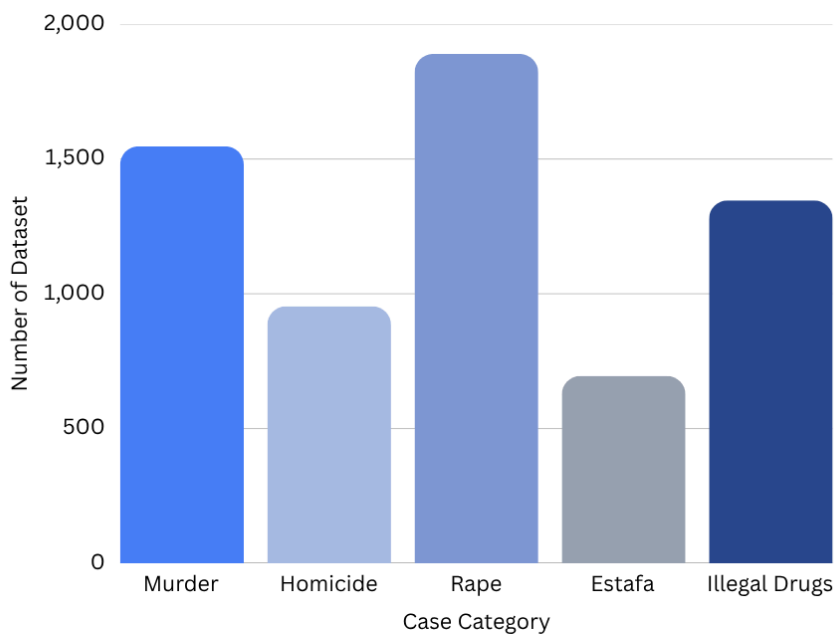

Figure 5. Dataset Distribution

Figure 5 presents the distribution of the collected dataset across the five target criminal case categories: murder, homicide, rape, estafa, and illegal drugs. The chart shows that rape cases form the largest portion of the dataset, followed by murder and illegal drugs cases. Homicide and estafa have smaller case counts, with estafa having the lowest representation among the five categories. Although the dataset is not perfectly equal across categories, each category remains within the target collection range and provides sufficient representation for training, validation, testing, and evaluation. This distribution supports the study's focus on Philippine Supreme Court criminal decisions while maintaining coverage across multiple offense types.

3.1.3. Inclusion Criteria

A case decision is included in the dataset if it satisfies the following conditions:

1. The document is a Philippine Supreme Court decision available within the designated jurisprudence repository.
2. The decision is published in HTML format, allowing its core body text to be successfully parsed and extracted for subsequent preprocessing steps, including cleaning and segmentation.
3. The extracted text contains specific keyword matrices that positively match one of the predefined target legal categories.

3.1.4. Exclusion Criteria and Quality Screening

To improve dataset reliability and to avoid technical failures during preprocessing, the following are excluded:

1. Documents hosted purely as PDF files are explicitly blocked by the scraper to ensure uniform and reliable text extraction via HTML parsing.
2. Decisions that have already been scraped and stored are bypassed, identified programmatically through exact source URL tracking to prevent redundancy.
3. HTML pages with empty, broken, or inaccessible body text that fail the initial text extraction phase are immediately discarded.

4. Decisions that, upon deep cleaning, do not contain the predefined terminology matrices required for the target legal categories. These are automatically flagged as "Uncategorized" by the system and discarded without being saved to the dataset.

3.1.5. Data Collection Procedure

Collection uses an automated retrieval support for link gathering/download. Each retained decision is stored in a standardized folder structure and documented in a metadata table.

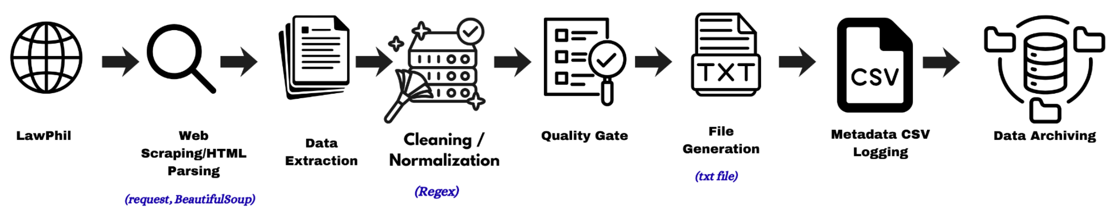

Figure 6. Data Collection and Annotation

Figure 6 presents the end-to-end data collection pipeline of the study. Starting from LawPhil, documents are retrieved through web scraping and HTML parsing using requests and BeautifulSoup, then subjected to data extraction, cleaning and normalization via Regex, and filtered through a quality gate. while their corresponding case information is recorded through metadata CSV logging. Finally, the generated dataset files are stored in Google Drive for organized storage, transfer, and dataset management.

3.1.5.1. Automated Case Retrieval and Web Scraping

The study employs an automated web scraping approach to systematically collect case decisions from the LawPhil repository. A custom Python script using requests, urllib3, and BeautifulSoup crawls LawPhil's monthly index pages, identifies candidate case links, and excludes PDF files to maintain a uniform HTML-based extraction process. Candidate case pages are retrieved as HTML content and parsed in memory. The raw HTML files are not permanently stored; instead, the relevant decision text is extracted, cleaned, categorized, and saved as a plain text file for later preprocessing and modeling. The automated retrieval procedure follows these steps:

1. Crawl LawPhil monthly index pages to collect candidate Supreme Court decision links.
2. Filter candidate URLs and exclude PDF files or inaccessible pages.
3. Retrieve each candidate decision in HTML format and parse the content using BeautifulSoup.
4. Extract the case title, G.R. number, decision text, source URL, and other available metadata.
5. Clean and normalize the extracted text using regular expressions.
6. Apply category matching through predefined keyword matrices for the target criminal categories.
7. Discard duplicate, empty, broken, or uncategorized cases through the quality gate.

8. Save approved cases as .txt files inside their corresponding category folders.
9. Record the corresponding metadata in a CSV file for traceability and dataset documentation.

3.1.5.2. File Naming, Storage, and Organization

All collected decisions are programmatically saved as plain text (.txt) files using a standardized naming convention: Category_CaseID_Year_Source.txt. The raw HTML is not stored permanently. Instead, the extracted text is written directly to the file system. Files are organized into specific category directories within a mounted cloud storage environment. Upon completion of the scraping process, the directories are compiled into their respective folder to facilitate efficient distribution, preprocessing, and model training.

3.1.5.3. Automated Data Extraction and Text Preprocessing

Data extraction is performed entirely in-memory using the BeautifulSoup library. The script isolates the main body of the HTML document and programmatically strips away navigation elements and scripts to capture the core decision text. Once extracted, the raw text undergoes an immediate automated preprocessing step utilizing Python's Regular Expressions module. This cleaning engine automatically removes website watermarks, optical character recognition artifacts, and stray pagination numbers while correcting broken sentence structures. This programmatic approach eliminates the need for manual text correction.

3.1.5.4. Automated Quality Gate and Data Validation

To preserve dataset integrity and prevent the inclusion of noisy data, the pipeline incorporates strict automated quality gates. As the script processes each URL, it enforces three primary checks. First, a duplicate filter verifies the URL to skip already scraped links. Second, an extraction gate discards the case if the parsed text body returns empty. Third, a categorization validation gate evaluates the cleaned text. If the text fails to align with any of the target legal categories, it is labeled as "Uncategorized" and immediately discarded. Cases that fail these automated checks are skipped entirely without manual intervention.

3.1.6. Dataset Documentation and Metadata Fields

Each approved case is documented through a metadata CSV file generated during scraping. The metadata includes identifying and processing details such as case title, G.R. number, source URL, category, local filename, and extraction status when available.

The metadata CSV supports traceability by linking each saved .txt file to its LawPhil source. It also helps verify dataset composition, support duplicate checking, and improve reproducibility.

3.1.7. Data Handling and Storage

All execution and data processing occur within Google Colab environment with the resulting text files and compressed archives stored directly in Google Drive. Since the scraped legal decisions are public records from an open repository, no private personal data collection is performed.

3.2. Data Pre-processing

After data collection, the study uses the cleaned .txt files generated by the scraping pipeline as input for preprocessing and legal section structuring. Since the scraping stage already performs HTML parsing, text extraction, cleaning, category filtering, and file generation, this stage focuses on converting the collected text files into structured legal records.

Each case is read from its category folder and processed through regex-based legal section detection. The system organizes each decision into standardized fields, namely Facts, Issues, Judicial Reasoning, and Ruling or Decision. Cases without a detectable ruling section are excluded because the ruling serves as the main reference for summarization and evaluation. The output of this stage is a structured JSON dataset used for model training, summarization, precedent retrieval, and evaluation.

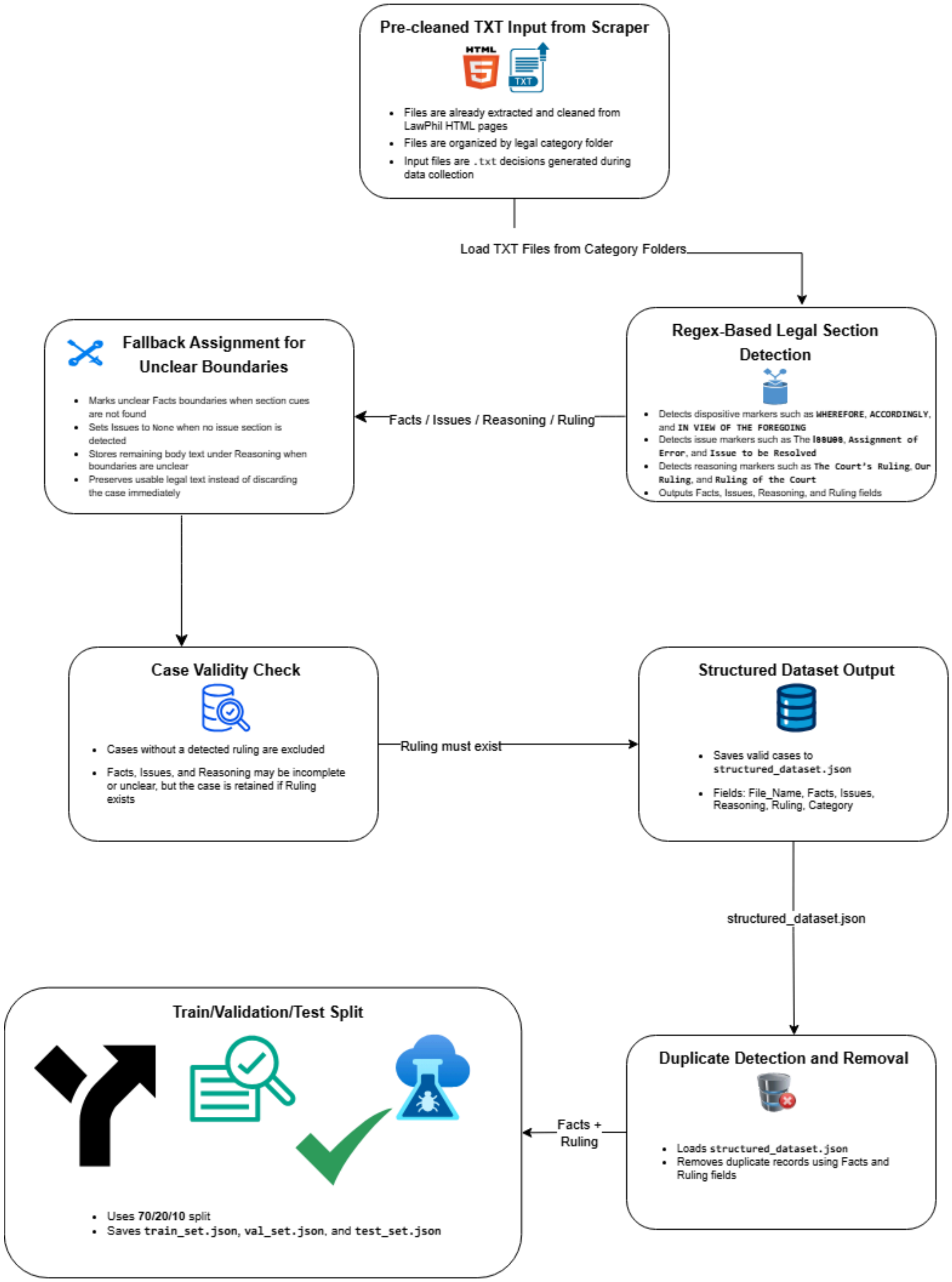

Figure 7. Preprocessing Architecture

Figure 7 presents the preprocessing architecture applied after the web scraping and data collection stage. The workflow begins with pre-cleaned .txt case files stored in category folders. Each file is loaded and processed through regex-based legal section detection to identify the Facts, Issues, Judicial Reasoning, and Ruling or Decision sections. When section boundaries are unclear, the parser assigns available text to the closest applicable field while preserving the case content. A validity check is then applied, where cases without a detectable Ruling or Decision section are excluded. Valid cases are assigned their category labels and saved as structured_dataset.json. The structured dataset is then checked for duplicate records using the Facts and Ruling fields before being divided into training, validation, and testing sets using a stratified 70/20/10 split.

3.2.1. Preprocessing Environment and Tools

Preprocessing is implemented in Google Colab using Python. The pipeline uses os for directory traversal, pandas for dataset construction, re for regex-based section detection, NLTK for later sentence-level processing, and scikit-learn for dataset splitting. The preprocessing stage operates on the .txt files produced by the scraping pipeline. The primary preprocessing method is rule-based legal boundary detection using regular expressions.

3.2.2. Cleaning, Segmentation, and Legal Section Structuring

Preprocessing removes extraction noise while preserving legal wording. Decisions are then segmented and structured into four standardized components: Facts, Issues, Judicial Reasoning, and Ruling/Decision.

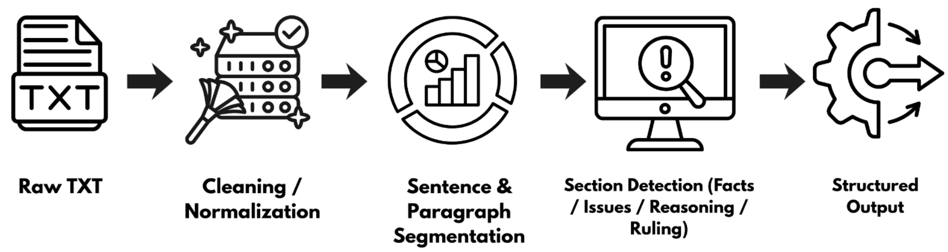

Figure 8. Preprocessing and Section Structuring Pipeline

Figure 8 shows the legal section structuring pipeline and preprocessing undertaken on every decision. This begins with raw text files, which are cleaned and normalized, broken into sentences and paragraphs, and identified using a set of rules: Facts, Issues, Judicial Reasoning and Ruling/Decision are key legal components. The structured output guarantees similarity in the presentation in a variety of cases allowing downstream NLP units to work on well-defined legal elements. This is an important structured segmentation that is the keystone in preserving legal meaning and enhancing the summarization fidelity and semantic retrieval accuracy.

3.2.3. Section Completeness and Case Validity Rules

- Ruling/Decision is mandatory for inclusion.
- If Facts/Issues/Reasoning cannot be reliably recovered through rule-based detection plus minimal boundary correction, the case is excluded to preserve dataset uniformity.
- Duplicate cases are removed prior to dataset splitting.

3.2.4. Dataset Splitting for Training and Evaluation

After structuring and filtering, the dataset is loaded from structured_dataset.json. Duplicate records are removed using the Facts and Ruling fields. The remaining cases are divided into training, validation, and testing sets using a 70/20/10 split.

Stratified splitting is applied using the case category to preserve category balance. The resulting files are saved as train_set.json, val_set.json, and test_set.json.

3.2.5. Output of Preprocessing Stage

The preprocessing stage produces structured training, validation, and testing files. Each record contains the filename, category, Facts, Issues, Judicial Reasoning, and Ruling or Decision.

These structured files serve as inputs for LEGAL-BERT domain adaptation, BERTSum-PH extractive training, Command-A summarization, Sentence-BERT precedent retrieval, and evaluation.

3.3. NLP Model Training and Fine-tuning

This subsection presents the procedures used to train and fine-tune the NLP components that support semantic representation, judicial judgment summarization, and precedent recommendation. Model development is conducted using the preprocessed and structured dataset to ensure that each component operates on consistent legal sections. The training procedure is designed to support decision assistance by producing interpretable outputs that remain aligned with the court's ruling and legal reasoning, while preserving the role of human experts in validating system results.

3.3.1. Semantic Representation via Sentence-BERT

To facilitate effective downstream similarity comparisons and precedent recommendation, semantic representation is executed using Sentence-BERT (SBERT). Utilizing the pre-trained all-mpnet-base-v2 architecture, dense vector embeddings are generated from the combined case narratives. These high-dimensional document vectors are captured and stored in a localized tensor matrix, serving as the baseline vector index for similarity matching.

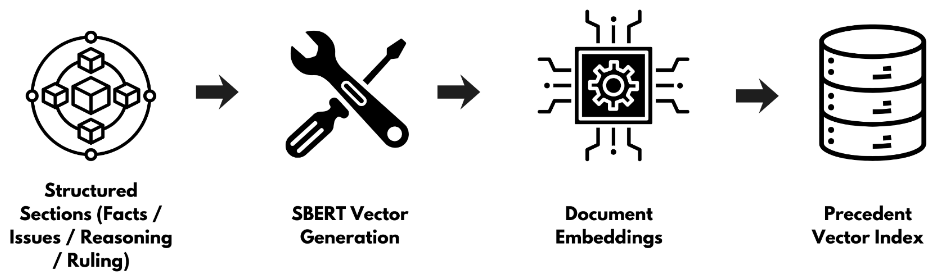

Figure 9. Sentence-BERT Vector Generation and Embedding Export Workflow

The workflow for establishing the semantic vector space can be observed in Figure 9. Structured textual segments from the historical precedent database are processed through the Sentence-BERT model to yield dense, pooled document-level embeddings. These vectors effectively capture contextual legal concepts, case nuances, and factual structures. The resulting mathematical representations are exported and compiled into a persistent tensor matrix to facilitate rapid, resource-efficient cosine similarity computation during live query executions. This specialized retrieval mechanism ensures that the support framework maps incoming query cases against relevant historical jurisprudence based on deep semantic meaning rather than shallow keyword overlapping.

3.3.2. Retrieve-then-Read Judicial Judgment Summarization (BERTSum-PH to Command-A)

The study utilizes a retrieve-then-read summarization strategy to process lengthy judicial documents. First, BERTSum-PH functions as the retrieval layer by extracting legally significant sentences directly from the judicial reasoning section. Second, Cohere Command-A acts as the reading layer, generating a structured abstractive summary by synthesizing the extracted highlights alongside the Facts and the final dispositive ruling.

This dual-stage approach significantly improves traceability. Because the generative model is strictly grounded on the algorithmically selected legal highlights rather than summarizing the entire unstructured decision directly, the framework preserves factual fidelity and ensures that the final output remains anchored to the court's original logic.

3.3.2.1. Reference Summary Definition

For training consistency, the Ruling/Decision section serves as the reference target, since it captures the legally determinative outcome.

3.3.2.2. Extractive Retrieval via BERTSum

BERTSum functions as the retrieval layer by producing extractive highlights, which are sentences copied directly from the decision, to improve traceability and verification. Sentence-level supervision is produced by selecting "oracle" sentences that best cover the reference content.

3.3.2.2.1. *Oracle Sentence Label Construction for BERTSum Fine-tuning*

To enable fine-tuning, the study constructs sentence-level labels using a decision-section-based reference. The Ruling/Decision section serves as the reference text. Sentences from the source decision are compared against the reference using overlap-based matching, and an oracle extractive set is formed by selecting sentences that best cover the reference content. Selected sentences are labeled as yi = 1, while non-selected sentences are labeled as yi = 0. This procedure produces extractive supervision while preserving the ruling-centered alignment of the summarization task

3.3.2.2.2. BERTSum Fine-tuning Procedure

Using the oracle sentence labels, BERTSum is fine-tuned to learn sentence importance scoring. During training, the model minimizes a binary classification objective over sentences. During inference, the fine-tuned model ranks sentences within a decision and returns the top-ranked sentences as extractive highlights, subject to a length constraint to maintain a concise output.

3.3.2.3. Abstractive Synthesis via Command A (Cohere)

The configuration of Command A during the abstractive summarization stage relies on structured prompting. The model input is formulated using the extracted BERTSum highlights, the Facts, and the Ruling/Decision section. This arrangement ensures that the generated narrative is legally determinative and strictly grounded in the source text, adding weight to the focus of the framework on structured and defensible summarization.

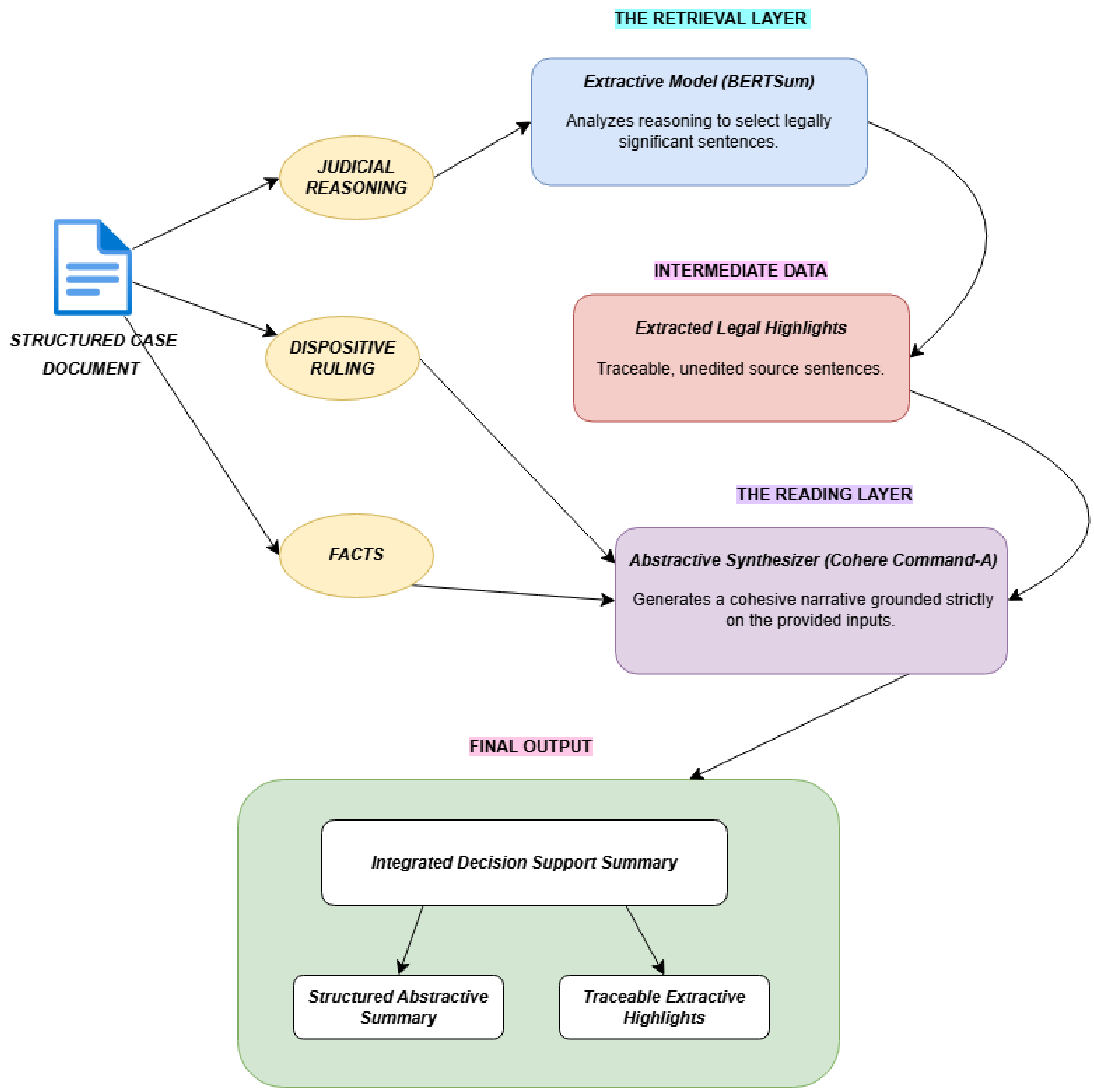

Figure 10. Retrieve-then-Read Summarization Workflow (BERTSum to Command-A)

Figure 10 illustrates the end-to-end architecture of the retrieve-then-read summarization strategy. The process begins with a parsed structured case document. To prevent the generative model from hallucinating or processing excessive tokens, only the

Judicial Reasoning section is routed through the Retrieval Layer. BERTSum acts as an extractive filter, isolating legally significant sentences to produce a set of traceable, unedited legal highlights. In the subsequent Reading Layer, the Cohere Command-A abstractive synthesizer receives three distinct, heavily grounded inputs: the extracted legal highlights, the original Facts, and the Dispositive Ruling. By forcing the synthesizer to generate a cohesive narrative strictly from these verified inputs, the framework produces an Integrated Decision Support Summary that successfully pairs a highly readable abstractive overview with mathematically traceable extractive highlights.

3.3.2.4. Integrated Summary Output

The final summarization output consists of the Command A abstract summary accompanied by BERTSum extractive highlights. This structure is intended to present a concise explanation while also providing traceable supporting statements for review.

3.3.3. Precedent Recommendation via Sentence-BERT

Each case is encoded into a dense embedding. For a query case, embeddings are compared against the dataset using cosine similarity, and the system returns the Top‑10 most similar cases with identifying metadata (case title, G.R. number, date, source, link, similarity score). Sentence-BERT may be fine-tuned using labeled case pairs (similar vs. dissimilar) derived from category and baseline similarity signals.

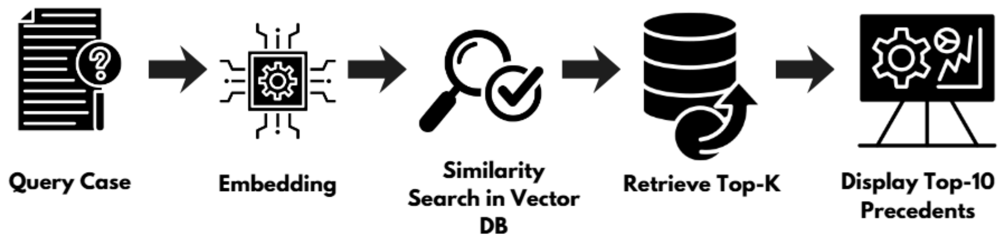

Figure 11. Precedent Retrieval and Ranking Workflow

Figure 11 shows the workflow on previous cases of retrieving precedents in a query case. The query is represented as an embedding and a comparison against the stored case embeddings is done with cosine similarity. Cases that are similar to the candidate are ranked in descending order and the top 10 similar cases are returned with identifying metadata and a similarity score. This flow of work illustrates the implementation of context-based recommendation (according to the legal reasoning patterns) with the help of the semantic similarity online search instead of key-word searches.

3.3.3.1. Similarity Pair Construction for Sentence-BERT Fine-tuning

To improve domain alignment, Sentence-BERT is fine-tuned using labeled case pairs derived from the dataset. Positive pairs are constructed from cases expected to share similar legal context, such as cases within the same category and with strong similarity under an initial embedding baseline. Negative pairs are constructed from cases expected to be dissimilar, such as cases across different categories or with low baseline similarity. The resulting pairs are assigned target similarity labels $y \in \{0,1\}$, where $y = 1$ indicates a similar pair and $y = 0$

indicates a dissimilar pair. These labels are used to train the model to bring semantically similar cases closer in embedding space while separating dissimilar cases.

3.3.4. Objective Functions and Loss Formulations

This subsection defines the objective functions used during training and fine-tuning. Formalizing these objectives supports methodological clarity and strengthens the replicability of the model development process.

Table 5. Objective Functions Used in Each NLP Component

Component	Task Type	Objective Function Used
LEGAL-BERT	Base Domain Adaptation (for BERTSum)	Masked Language Modeling (MLM) Loss
Command A	Abstractive Summarization	Negative Log-Likelihood (Sequence-to-Sequence)
BERTSum	Extractive Summarization	Binary Cross-Entropy Loss
Sentence-BERT	Precedent Similarity Ranking	Cosine Similarity + Supervised Cosine Regression Loss

Table 5 shows the objective functions that will be employed in training and fine-tuning of each component in the framework related to NLP. It demonstrates that LEGAL-BERT is undergoes domain adaptation via masked language modeling loss to serve as the base encoder for the extractive summarizer, Command A is abstractive summarizing with negative log-likelihood loss, BERTSum is extractive sentence selection with binary cross-entropy loss, and Sentence-BERT is learning similarity of precedent with cosine similarity loss and supervised regression loss.

3.3.4.1. LEGAL-BERT Masked Language Modeling Loss

LEGAL-BERT fine-tuning follows a masked language modeling objective, where a subset of tokens is masked, and the model learns to predict the original tokens based on surrounding context. Let M denote the set of masked token positions. The masked language modeling loss is defined as:

$$L_{MLM} = -\frac{1}{|M|}\sum_{i \in M} logP_{\theta}(x_i | x_{\setminus M}) \quad (1)$$

where x_i is the original token at position i, $x\setminus M$ is the input sequence with masked positions, and $P\theta$ is the predicted probability under parameters θ.

3.3.4.2. Command A Abstractive Summarization Loss

Command A is fine-tuned as a sequence-to-sequence model that generates a summary sequence given an input decision text. Let the target summary be $y = (y_1, y_2, \ldots, y_T)$. The training objective is the negative log-likelihood:

$$L_{ABS} = -\frac{1}{T}\sum_{t=1}^{T} logP_{\theta}(y_t | y_{<t}, x) \quad (2)$$

where x is the input text, and y < t are the previously generated tokens.

3.3.4.3. BERTSum Extractive Sentence Selection Loss

BERTSum performs extractive summarization by scoring sentences and selecting the most salient ones. Let N be the number of sentences, $y_i \in \{0,1\}$

indicate whether sentence i is selected, and p_i be the predicted probability. The binary cross-entropy loss is defined as:

$$L_{EXT} =- \frac{1}{N}\sum_{i=1}^{N}[y_i\, log(P_i) + (1 - y_i)\, log(1 - p_i)] \quad (3)$$

This objective supports learning a sentence-level importance scoring function for extractive refinement.

3.3.4.4. Sentence-BERT Similarity and Supervised Similarity Loss

Sentence-BERT similarity is computed using cosine similarity. For embeddings u and v, cosine similarity is:

$$s(u, v) = cos(u, v) = \frac{u \cdot v}{||u||\ ||v||} \quad (4)$$

For supervised fine-tuning using pair labels $y \in \{0,1\}$, a cosine regression objective is defined as:

$$L_{SIM} = (s(u, v) - y)^2 \quad (5)$$

This loss encourages predicted similarity values to approximate the target similarity labels used for legal case comparison.

3.3.5. Implementation Settings and Training Documentation

Model training and fine-tuning are conducted in Google Colab using Python. To strengthen reproducibility, the study records the configuration used for each trained component, including the model checkpoint name, maximum input length, batch size, number of epochs, learning rate, optimizer configuration, and validation-based selection criteria. Random seeds are fixed for dataset splitting and training runs where applicable.

The final reported configurations correspond to the settings that yield the best validation performance under the defined evaluation metrics.

3.4. Testing and Evaluation Procedures

This subsection describes how the summarization and precedent recommendation components are assessed using quantitative metrics and structured review. Evaluation is conducted on the held-out test set to measure summarization quality, recommendation usefulness, and basic computational feasibility under the defined dataset setting.

3.4.1. Summarization Evaluation

Summarization quality is evaluated using G-Evaluation by comparing the generated summary against the reference summary derived from the ruling or decision section. Faithfulness, Answer Relevance, and Answer Correctness, alongside BERTScore F1, are computed for test cases to quantify the semantic and factual alignment between generated and reference texts. In addition, extractive highlights produced by BERTSum are reviewed for alignment with the original decision text and for coverage of legally significant statements.

3.4.2. Precedent Recommendation Evaluation

Precedent recommendation performance is evaluated based on the relevance and usefulness of the ranked precedents produced for each query case in the test set. Similarity ranking is computed using cosine similarity between the query embedding and candidate embeddings stored from the dataset. The system returns the Top 10 most similar cases as recommended precedents. The evaluation emphasizes whether the returned precedents demonstrate meaningful similarity in terms of case context and legal reasoning, consistent with the decision-support objective of the system.

To support quantitative information retrieval evaluation, relevance judgments are obtained through human evaluation by two lawyers. For each query case, an evaluation pool of the Top 30 retrieved precedents is formed. Each precedent in the Top 30 pool is labeled as Relevant (1) or Not Relevant (0). Precision@10 is computed using the Top K results, while Recall@10 is computed using the total number of relevant precedents identified within the Top 30 pool as the denominator. For the main reporting, K is set to 10 to match the system's Top 10 recommendation output.

3.4.3. Computational Efficiency Measures

To assess feasibility for practical decision-support settings, computational measures are recorded during inference. These include the time required to preprocess a case, generate the Command A summary, compute BERTSum extractive highlights, and rank candidate precedents using similarity scoring. Resource usage may also be documented through memory and runtime indicators available in the execution environment.

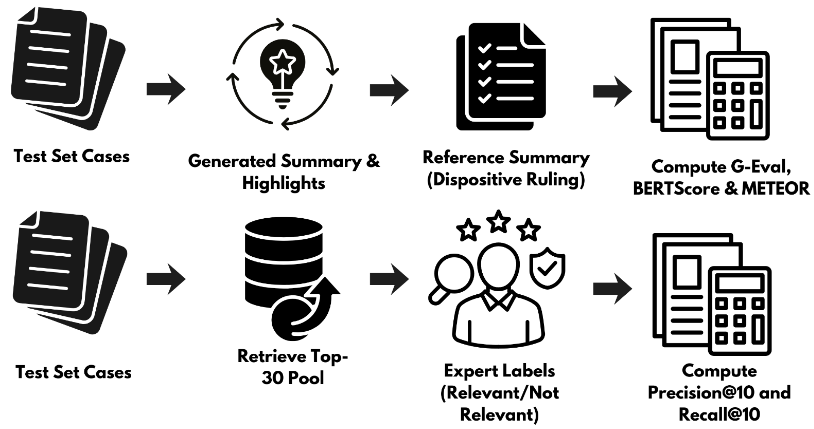

Figure 12. Evaluation Workflow

Figure 12 illustrates the parallel evaluation procedures for both the summarization and precedent recommendation modules. In the top track, the generated summaries and extractive highlights are evaluated against the ground-truth Reference Summaries (the actual Dispositive Rulings) to compute G-Evaluation metrics, BERTScore F1, and METEOR. In the bottom track, the system retrieves a Top-30 candidate pool for each test case query, which domain experts then manually label for relevance to calculate the final Precision@10 and Recall@10 performance. This dual-track workflow ensures that both automated algorithmic scoring and human-centered validation are comprehensively utilized.

3.4.4. Evaluation Metrics and Similarity Scoring Formulas

Table 6. Evaluation Metrics and Their Purpose

Metric	Component Evaluated	What It Measures
Faithfulness	Summarization	Factual consistency of the summary relative to the retrieved source context

Answer Relevance	Summarization	The degree to which the generated summary addresses the specific legal query
Answer Correctness	Summarization	The accuracy of the generated summary compared against the ground truth reference
BERTScore F1	Summarization	Semantic similarity between generated and reference summaries using contextual embeddings
METEOR	Summarization	Syntactic fluency and lexical overlap, accounting for stemming, synonyms, and word order fragmentation.
Cosine Similarity	Precedent Recommendation	Semantic similarity between case embeddings
Precision@K	Precedent Recommendation	Proportion of relevant precedents in Top K
Recall@K	Precedent Recommendation	Proportion of relevant precedents retrieved within Top K

Table 6 provides a summary of the evaluation metrics employed in the study, explicitly mapped against the specific system components they assess. It demonstrates that summarization quality is evaluated using a hybrid approach: LLM-as-a-Judge G-Evaluation metrics (Faithfulness, Answer Relevance, and Answer Correctness) assess legal truth and alignment, while algorithmic metrics (BERTScore F1 and METEOR) measure underlying semantic similarity and syntactic fluency. Conversely, the precedent recommendation module is evaluated using cosine similarity for mathematical vector ranking, followed by Precision@K and Recall@K to measure the practical accuracy and relevant-case coverage of the retrieval engine.

Table 7. Interpretation of Summarization Metric Score Ranges

Metric	Score Range	Interpretation	Implication in the Study
Faithfulness	0.00–0.40	Low Hallucination Control	The summary contains significant claims not supported by the retrieved legal context.
	0.41–0.70	Moderate Hallucination Control	The summary is generally grounded, though some minor legal nuances may lack direct support.
	0.71–1.00	High Hallucination Control	The summary is strictly derived from the retrieved text, ensuring high factual integrity.
Answer Relevance	0.00–0.40	Low Query Alignment	The summary fails to address the specific legal question or focuses on irrelevant case details.
	0.41–0.70	Moderate Query Alignment	The summary addresses the main query but includes redundant or tangential information.
	0.71–1.00	High Query Alignment	The summary is highly concise and directly addresses the user's legal inquiry.
Answer	0.00–0.40	Low Factual Accuracy	The summary contradicts the ground truth reference or

Correctness			misses the core ruling entirely.
	0.41–0.70	Moderate Factual Accuracy	The summary captures the general outcome but lacks precision in legal reasoning or citations.
	0.71–1.00	High Factual Accuracy	The summary closely aligns with the reference ruling in both outcome and legal justification.
BERTScore F1	0.00–0.40	Low Semantic Similarity	The summary uses language that is semantically distant from the reference legal text.
	0.41–0.70	Moderate Semantic Similarity	The summary conveys the general meaning of the case using different but relevant terminology.
	0.71–1.00	High Semantic Similarity	The summary demonstrates near-perfect semantic alignment with the expert-written reference.
METEOR	0.00–0.20	Low Syntactic Overlap	Indicates minimal exact or stemmed word matches, often resulting in fragmented grammar or unrelated vocabulary.
	0.21–0.40	Moderate Syntactic Overlap	Reflects typical performance for highly abstractive generation, where exact phrasing changes but meaning is retained via synonym usage.
	0.41–1.00	High Syntactic	Indicates heavy verbatim copying (extractive-heavy)

		Overlap	and strong syntactic alignment with the original reference.

Table 7 presents the descriptive interpretation of score ranges for the summarization metrics used in the study. Since summarization quality in the proposed framework is measured through G-Evaluation metrics (Faithfulness, Answer Relevance, and Answer Correctness) alongside algorithmic metrics (BERTScore F1 and METEOR), the table explains what low, moderate, and high values indicate in terms of factual consistency, query alignment, semantic similarity, and syntactic overlap between the generated summary and the reference summary. Higher values generally indicate stronger summarization fidelity; however, these scores should still be interpreted with caution because algorithmic metrics alone do not fully substitute for a comprehensive expert evaluation of complex legal reasoning or doctrinal nuances. This interpretation is consistent with the study's discussion that legal summarization evaluation must go beyond surface-level metrics and focus on the substantive accuracy of the legal output.

Table 8. Interpretation of Precedent Recommendation Metric Score Ranges

Metric	Score Range	Interpretation	Implication in the Study
	-1.00 to -0.01	Negative similarity	The query case and candidate precedent are semantically dissimilar and are unlikely to reflect related legal context.
	0.00	No similarity	No meaningful semantic relationship is indicated between the query case and candidate precedent.
	0.01-0.30	Weak similarity	The candidate precedent shares

Cosine Similarity			minimal semantic relation with the query case and may have limited usefulness as support.
	0.31–0.60	Moderate similarity	The candidate precedent shows some semantic relation and may be potentially relevant for expert review.
	0.61–0.80	Strong similarity	The candidate precedent is strongly related in semantic content and is more likely to be relevant to the query case.
	0.81–1.00	Very strong similarity	The candidate precedent is highly semantically aligned with the query case and is highly likely to rank as a strong recommendation.
Precision@10	0.00–0.20	Very low precision	Only a small portion of the Top-10 recommended precedents are relevant, indicating weak recommendation accuracy.
	0.21–0.40	Low precision	Some relevant precedents appear in the Top-10 results, but irrelevant cases still dominate the recommendations.
	0.41–0.60	Moderate precision	A fair portion of the Top-10 recommendations are relevant, indicating acceptable recommendation accuracy.
	0.61–0.80	High precision	Most of the Top-10 recommendations are relevant, indicating strong retrieval precision.
	0.81–1.00	Very high precision	Nearly all of the Top-10 recommended precedents are relevant, indicating excellent recommendation accuracy.
	0.00–0.20	Very low recall	The system retrieves only a

Recall@10			small fraction of all relevant precedents in the evaluation pool.
	0.21–0.40	Low recall	The system captures some relevant precedents, but many relevant cases are still missed.
	0.41–0.60	Moderate recall	The system retrieves a fair portion of the relevant precedents available in the evaluation pool.
	0.61–0.80	High recall	The system successfully retrieves most relevant precedents in the evaluation pool.
	0.81–1.00	Very high recall	The system captures nearly all relevant precedents in the evaluation pool, indicating very strong retrieval coverage.

Table 8 presents the descriptive interpretation of score ranges for the precedent recommendation metrics used in the study. The table explains how cosine similarity reflects the semantic closeness between the query case and candidate precedents, while Precision@10 and Recall@10 indicate the effectiveness of the Top-10 precedent recommendations produced by the system. Higher cosine similarity values suggest stronger semantic relatedness and increase the likelihood that a candidate case will rank highly in retrieval. Likewise, higher Precision@10 and Recall@10 values indicate stronger retrieval effectiveness in terms of recommendation accuracy and relevant-case coverage. Since the study evaluates recommended precedents using a Top-30 pool labeled by two lawyers, these score ranges help clarify the practical meaning of the retrieval results in the context of legal decision support.

3.4.4.1. Cosine Similarity for Precedent Ranking

For a query case embedding q and a candidate precedent embedding d_j, cosine similarity is computed as:

$$Sim(q, d_j) = cos(q, d_j) = \frac{q \cdot d_j}{||q|| \, ||d_j||} \tag{6}$$

Candidate cases are ranked in descending order Sim(q,d_j), and the system returns the Top 10 most similar precedents.

3.4.4.2. G-Evaluation Metrics (Faithfulness)

G-Evaluation assesses Faithfulness by utilizing an LLM-as-a-Judge (Llama 3.1 70B). Instead of strict lexical overlap, the LLM is provided with a strict evaluation rubric and the source text. It evaluates whether every claim in the generated summary can be directly inferred from the retrieved judicial context, outputting a normalized score from 1 to 5.

3.4.4.3. G-Evaluation Metrics (Answer Relevance)

Answer Relevance is evaluated via the LLM-as-a-Judge to measure how well the generated summary addresses the implicit legal query. The model assesses whether the summary successfully filters out tangential legal discussions and procedural noise, ensuring the output focuses entirely on the core legal issues and factual matrices of the case.

3.4.4.4. G-Evaluation Metrics (Answer Correctness)

Answer Correctness evaluates the ultimate legal truth of the output. The LLM-as-a-Judge compares the generated abstractive summary against the ground-truth reference (the court's actual Dispositive Ruling). Utilizing a chain-of-thought prompt, the model scores the semantic equivalence and legal alignment of the generated holding on a 1 to 5 continuous scale, ensuring the framework successfully synthesizes the court's rationale without altering the legally determinative outcome.

3.4.4.5. BERTScore F1

BERTScore leverages contextual embeddings to evaluate semantic similarity. Let (x) be the reference summary and $(\hat{x})$ be the generated summary. Precision (P_{BERT}) and Recall (R_{BERT}) are computed using cosine similarity matches between token embeddings:

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \in x} \sum_{\hat{x}_j \in \hat{x}}^{max} (x_i^T \hat{x}_j),\ P_{BERT} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_i \in \hat{x}} \sum_{x_j \in x}^{max} (x_i^T \hat{x}_j) \qquad (7)$$

The BERTScore F1 is the harmonic mean of these values:

$$F_{BERT} = 2 * \frac{P_{BERT} * R_{BERT}}{P_{BERT} + R_{BERT}} \qquad (8)$$

3.4.4.6. METEOR Score

The Metric for Evaluation of Translation with Explicit ORdering (METEOR) evaluates the abstractive generation capabilities of the pipeline. Unlike basic lexical overlap metrics that strictly require exact word matches, METEOR incorporates stemming and synonymy, making it highly effective for evaluating generative models that may use different terminology to convey the same legal meaning.The metric is calculated by first computing unigram precision P and unigram recall R based on the mapped unigrams m between the generated summary (with total unigrams t) and the reference summary (with total unigrams r):

$$P = \frac{m}{t} \tag{9}$$

$$R = \frac{m}{r} \tag{10}$$

METEOR places a stronger emphasis on recall by computing a parameterized harmonic mean $Fmean$, where the hyperparameter α is typically set to 0.9:

$$Fmean = \frac{P \cdot R}{\alpha p + (1 - \alpha)\, R} \tag{11}$$

To account for word order and fluency, METEOR applies a fragmentation penalty. It counts the number of contiguous matching unigram chunks C. Using hyperparameters γ and β, the penalty is calculated as:

$$Penalty = \gamma \left(\frac{C}{M}\right)^{\beta} \tag{12}$$

The final METEOR score is then computed by applying this penalty to the harmonic mean:

$$METEOR = F_{mean} \cdot (1 - Penalty) \tag{13}$$

By utilizing this metric alongside BERTScore and G-Eval, the framework ensures that the generated summaries are penalized for grammatical fragmentation while being rewarded for semantic fluency and accurate legal synonym usage.

3.4.4.7. Precision@K

Precision@K measures the proportion of relevant precedents among the Top K recommended cases. It is defined as:

$$Precision@K = \frac{\#\{relevant\ precedents\ in\ Top{-}K\}}{K} \tag{14}$$

where P denotes the evaluation pool size and P > K .

3.4.4.8. Recall@K

Recall@K measures the proportion of relevant precedents retrieved within the Top K results relative to the total relevant precedents identified in the evaluation pool. It is defined as:

$$Recall@K = \frac{\#\{relevant\ precedents\ in\ Top{-}K\}}{\#\{relevant\ precedents\ in\ Top{-}30\ pool\}} \tag{15}$$

3.5. Human-in-the-Loop Validation

Human-in-the-loop validation is implemented as a validation measure to ensure that the generated summaries and recommended precedents remain aligned with legal reasoning and professional judgment. Due to the scale of the test set under the expanded dataset, expert review is conducted on a stratified sample of test cases per category to maintain feasibility while preserving balanced evaluation coverage. Outputs are reviewed by two qualified evaluators, namely two lawyers, to confirm legal accuracy, interpretability, and precedent relevance, while maintaining the system's role as decision support rather than decision-making.

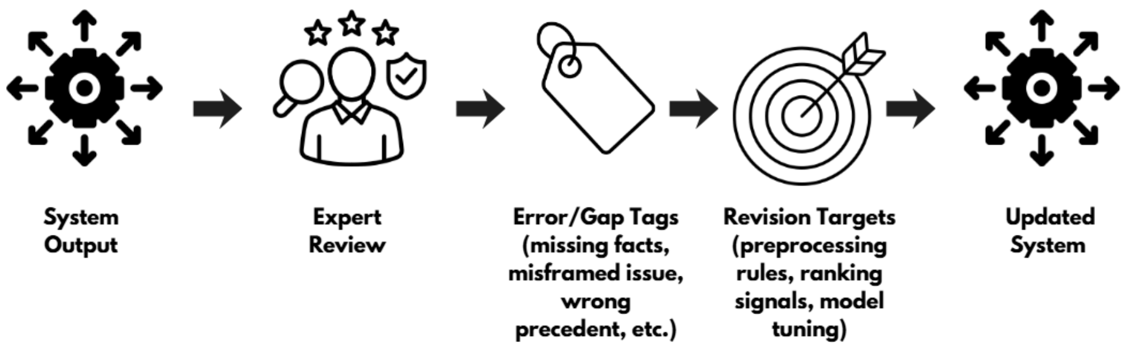

Figure 13. Human Review Loop

Figure 13 demonstrates the human in the loop validation process. Domain experts examine system outputs such as summaries and suggested precedents to find faults, discrepancies or lack of legal components. Feedback is summarized into areas of improvement, e.g. improving rules, tuning models, or changes of training signals. This is the loop that makes the framework strictly act as a decision support component in order to retain human control and enhance legal reliability.

3.5.1 Expert Task and Review Procedure

Expert evaluation is conducted on a stratified sample of test cases to ensure representation across the five case categories. For each sampled query case, the system produces: (a) a Command A abstractive summary, (b) BERTSum extractive highlights, and (c) a Top-10 ranked list of recommended precedents.

Table 9. Domain Expert Task

Scope of Responsibilities	What the Expert Reviews	How the Expert Performs It	Output
Reflecting on judicial summaries and judicial precedents generated by the system	(1) Command A abstractive summary, (2) BERTSum extractive highlights, (3) Top-10 recommended precedents (and Top-30 evaluation pool if used for scoring)	Reviews each sampled query case output and assesses whether the summary and recommended precedents make sense as decision-support outputs for that case	Notes per query case on strengths, issues, and missing elements
Evaluation of legality, applicability, and consistency of arguments	Summary content and precedent relevance for each query case	Checks whether the summary and recommended precedents are legally consistent with the case context, and whether the precedents are applicable (issue alignment, reasoning alignment, and defensible support)	Written comments explaining why outputs are legally consistent / inconsistent
Expert advice to make sure there is adherence to the set legal principles and judicial standards	Overall output quality across reviewed samples	Flags outputs that deviate from expected legal reasoning presentation or judicial standards (e.g., misrepresenting the ruling, weak basis for precedent support, unclear rationale)	Consolidated expert feedback points (common violations, recurring issues)
Determining limitations or concerns in the outputs and giving recommendations on ways to improve them	Error patterns across summaries and precedent recommendations	Identifies limitations (e.g., missing controlling facts, overgeneralized reasoning, irrelevant precedent matches) and proposes improvement)	List of limitations/concerns and recommended improvements

Table 9 outlines the roles and processes of evaluation to domain experts when human validation is in force. It provides the scope of review, which is appraisal of abstractive summaries, extractive highlights, and the recommended precedents. Another aspect that is described in the table is how the experts judge legal consistency, applicability and reasoning alignment, as well as how feedback should be recorded. This professional participation is organized so that assessment is not done based on an algorithmic approach but also professional legal opinion.

3.5.2 Likert-Scale Expert Evaluation

To complement the automatic evaluation metrics, the study uses a 5-point Likert scale for expert assessment of the generated judicial summaries and recommended precedents. This allows domain experts to rate the outputs in terms of legal quality, relevance, reasoning alignment, and practical usefulness for legal analysis. The Likert-scale results are used together with the automatic metrics to provide a more complete evaluation of the system.

Table 10. Legal Summarization Evaluation Criteria (5-Point Likert Scale)

Evaluation Criterion	**Description**	**1**	**2**	**3**	**4**	**5**
Accuracy of Legal Content	The summary correctly reflects the facts, issues, and ruling of the decision.	Very Poor	Poor	Acceptable	Good	Excellent
Completeness of Legal Elements	The summary includes key legal elements such as facts, issues, reasoning, and ruling.	Very Poor	Poor	Acceptable	Good	Excellent

Clarity and Readability	The summary is clearly written and easy to understand for legal review.	Very Poor	Poor	Acceptable	Good	Excellent
Preservation of Judicial Reasoning	The summary maintains the logical reasoning used by the court in reaching the decision.	Very Poor	Poor	Acceptable	Good	Excellent
Usefulness for Legal Research	The summary helps legal practitioners quickly understand the case.	Very Poor	Poor	Acceptable	Good	Excellent

This table presents the 5-point Likert scale used by domain experts to evaluate the generated judicial summaries. It assesses important aspects of summary quality such as accuracy, completeness, clarity, reasoning preservation, and usefulness for legal review. The scale complements the automatic evaluation metrics by adding expert judgment on the legal quality and practical value of the summaries

Table 11. Legal Precedent Recommendation Evaluation Criteria (5-Point Likert Scale)

Evaluation Criterion	**Description**	**1**	**2**	**3**	**4**	**5**
Relevance to the Query Case	The recommended precedent is relevant to the legal context of the query case.	Very Poor	Poor	Acceptable	Good	Excellent
Alignment of Legal Issues	The recommended precedent addresses legal issues similar or closely related to those in the query case.	Very Poor	Poor	Acceptable	Good	Excellent

Alignment of Judicial Reasoning	The recommended precedent reflects reasoning patterns or doctrinal logic consistent with the query case.	Very Poor	Poor	Acceptable	Good	Excellent
Defensibility as Precedent Support	The recommended precedent can be reasonably defended as supporting or informing the legal analysis of the query case.	Very Poor	Poor	Acceptable	Good	Excellent
Usefulness for Legal Research	The recommended precedent is useful in helping a legal practitioner verify, compare, or deepen understanding of the query case.	Very Poor	Poor	Acceptable	Good	Excellent

Table 11 presents the 5-point Likert scale used to evaluate the quality of the recommended precedents generated by the system. Since the framework does not only summarize judicial decisions but also returns a Top-10 ranked list of semantically similar cases, expert review is needed to determine whether the suggested precedents are legally relevant, issue-aligned, reasoning-consistent, defensible as support, and useful for legal research. The scale allows the domain experts to provide a structured assessment of the recommendation output beyond the binary relevance labels used for Precision@10 and Recall@10, thereby strengthening the human-centered validation component of the study.

3.5.3 Survey Dataset Description

In order to aid the human inspection process, a survey based evidence created a dataset of humanally rated legal experts that read the products created by the framework.

Table 12. Survey Dataset Characteristics for Expert Evaluation

Component	**Description**
Respondents	Licensed legal professionals
Number of Experts	2 domain experts
Sampling Method	Purposive sampling
Data Collection Instrument	Structured questionnaire
Evaluation Scope	Generated summaries, extractive highlights, and precedent recommendations
Rating Method	5-point Likert scale
Additional Feedback	Written comments and improvement suggestions

Table 12 presents the survey data features such as characteristics of the respondents, sampling strategy, data collection tool, area of evaluation and rating tool that was applied during expert evaluation.

3.5.4 Data Privacy and Ethical Considerations

The proposed framework is designed and evaluated as a legal decision-support tool rather than an autonomous legal decision-making system. Although the case decisions used in this study are publicly available judicial records, responsible use remains necessary because the framework transforms large volumes of legal text into generated summaries, extractive highlights, and recommended precedents that may influence how users interpret a case. For this reason, all system outputs must be treated as assistive materials subject to human review and verification, and not as final legal conclusions, legal advice, or substitutes for judicial or professional legal judgment. Users are expected to examine the original decision text and the cited or recommended precedents before relying on any generated output. Ethical use of the framework therefore requires transparency regarding the source of the data, clarity about the intended decision-support role of the system, and continued human oversight to reduce the risk of overreliance, unsupported conclusions, or misapplication of generated results. In any future web deployment, the framework should be accompanied by clear terms and conditions, a privacy or data-use notice, and an explicit disclaimer stating that the platform uses publicly available case decisions and is intended only to support legal research and review.

Chapter 4

RESULTS AND DISCUSSION

This chapter outlines the findings and assessment of the proposed NLP-based legal decision-support system for criminal decisions made by the Supreme Court of the Philippines. It provides an analysis of the performance of the hybrid retrieve-then-read architecture and evaluates the effectiveness of the system through the use of a variety of evaluation metrics and comparative model analysis.

4.1. Experimental Configuration

In this study, we assessed a variety of summarisation pipelines, retrieval configurations, and LLM-as-a-Judge evaluation setups in multiple iterative experiments. The same structured preprocessing pipeline was applied to each configuration and the metrics G-Evaluation, BERTScore F1, METEOR, Precision@10, and Recall@10 were used to evaluate each configuration.

Table 13. Experimental Configurations Used in the Study

Experiment ID	Configuration	Pipeline Description	Evaluation / Modification
E1	PEGASUS	Original summarization pipeline	Baseline evaluation
E2	FLAN-T5	Original summarization pipeline	Baseline evaluation
E3	BART	Original summarization pipeline	Baseline evaluation

E4	Dual (BERTSum + BART)	Hybrid extractive–abstractive pipeline	Original configuration
E5	Dual Extractive (Regex + BERTSum)	Rule-based + neural extractive pipeline	Original configuration
E6	Gemini 2.5	Hybrid Extractive Highlights	Original configuration
E7	Qwen 2.5	Extractive highlighting pipeline for key legal contexts	Baseline evaluation
E8	Qwen 2.5	Original pipeline with BERTScore and METEOR integration	Added semantic evaluation metrics
E9	Command R	Original summarization pipeline	Baseline evaluation
E10	Llama 3.3 70B	Hybrid Extractive Highlights	Original configuration
E11	Llama 3.3 70B	Original pipeline with BERTScore and METEOR integration	Added semantic evaluation metrics
E12	Command A	Hybrid Extractive Highlights	Original configuration
E13	Command A	Original retrieve-then-read pipeline	Baseline configuration
E14	Command A	Modified evaluation metrics	Added metric refinements
E15	Command A + Command R Plus	LLM-as-a-Judge Integration with G-Evaluation	API-based evaluation
E16	Command A + OpenAI Judge	LLM-as-a-Judge integration	OpenAI evaluator

E17	Command A + Llama 3.3 Judge (Grok API)	Judge replacement	Low-parameter inference setup
E18	Command A + Gemini 2.5 Judge	Judge replacement	API-based evaluation
E19	Command A + Llama 3.1 8B Judge (Ollama API)	Judge replacement	Local API inference
E20	Command A + Qwen 2.5 7B Judge	Local evaluation pipeline	Local deployment
E21	Command A + Qwen 2.5 7B Judge	Added preprocessing Phase 1 & 2	Extended preprocessing
E22	Command A + Qwen 2.5 14B Judge (Ollama)	Judge replacement	Larger local model
E23	Command A + Qwen 2.5 14B Judge	Token and parameter tuning	Increased token allocation
E24	Command A + Qwen 3.1 14B Judge	Judge replacement	Updated Qwen variant
E25	Command A + Llama 3.1 32B Judge (Ollama)	Judge replacement	Larger local inference
E26	Command A + Qwen 2.5 7B Judge	Retrieval-first workflow	Retrieve-then-read modification
E27	Command A + Qwen 2.6 27B Judge	Judge replacement	Expanded parameter configuration
E28	Command A + Llama 3.1 70B Judge	Local evaluation pipeline	High-capacity judge
E29	Command A + Llama 3.1 70B Judge	Added helper functions	Pipeline optimization
E30	Command A + Llama 3.1 70B Judge	Increased token allowance	Expanded context window

E31	Command A + Llama 3.1 70B Judge	Removed output limitations	Full-context evaluation
E32	Command A + Llama 3.1 70B Judge	Migrated to NVIDIA A100 GPU	Hardware optimization
E33	Command A + Llama 3.1 70B Judge	Added evaluation dimensions	Correctness, Faithfulness, Relevancy
E34	Command A + Llama 3.1 70B Judge	Expanded evaluation context to 4k words	Final evaluation setup

The experimental progression shows how the integrated legal NLP decision-support framework was developed by an iterative optimization process. The first experiments were conducted with benchmarking standalone transformer summarization architectures (PEGASUS, FLAN-T5, BART, Gemini 2.5, Qwen 2.5, Command R, Llama 3.3 70B). Later experiments moved towards hybrid architectures in which the retrieving and reading were merged, namely the BERTSum extractive retrieval with the Command A abstractive synthesis. Later stages focused on optimizing the LLM-as-a-Judge evaluation framework, including several changes in evaluators, preprocessing improvements, restructuring the retrieval, expanding the tokens, optimizing for hardware (NVIDIA A100 GPU), and introducing multidimensional legal evaluation metrics, such as correctness, faithfulness and relevance. Final configuration: Command A + Llama 3.1 evaluation model with an expanded 4k-word evaluation context.

4.2. Experimental Results

The study explored 34 different experimental architectures (Table 14) in an iterative process to identify the optimal architecture for legal decision support.

The first baseline models (Configurations E1–E12) were abstractive models (PEGASUS, FLAN-T5, BART, and Command A, respectively) running alone.

One regular problem observed in these initial baseline tests was that standalone abstractive models were unable to cope with the very long-length of the tokens in Philippine Supreme Court decisions. These models struggled to handle full documents, had too much aggression in information compression, failed to capture supporting legal arguments, and had an elevated risk of hallucination. On the other hand, purely extractive (BERTSum standalone) baseline tests resulted in high lexical overlap and factual fidelity, but the summaries produced were cognitively challenging for legal practitioners to read and were disjointed and fragmented.

This tight coupling of readability and factual fidelity was overcome by switching the study to the integrated "Retrieve-then-Read" architecture (Configurations E13 to E34). The system was successfully used to retrieve legally-relevant salient sentences using only BERTSum-PH to retrieve such sentences in order to take the risk of hallucination down, and Command A was used as a synthesizer solely based on the salient sentences retrieved by BERTSum-PH to produce a coherent narrative. The final pipeline selection was made based on the most mathematically sound and normalized evaluation footprint, as evaluated by a Llama 3.1 70B evaluator, on the pipeline's extractions.

4.2.1. PEGASUS Baseline Results

The PEGASUS model was one of the main baseline models chosen for the study because of its good performance on abstractive summarization tasks. The model was tested with the default pipeline setting, without any extra retrieval augmentation or context optimization. In experimental tests, however, the model proved to have a great many shortcomings with respect to long legal documents and longer case settings. PEGASUS was not very efficient with long-context input leading to repetitive sentence generation, duplication of phrases, and decreased coherence of the summary. The impact of these repetitions was negative on the generated summaries in terms of the overall readability, faithfulness and relevance. Therefore, it was decided not to use the model in this study since it is not suited for the summarization task of large-context legal texts and so the possibility of using other retrieve-then-read and hybrid architectures was explored.

4.2.2. BART with Dual Regex Extractive Results

Table 14. BART with Dual Regex Extractive Results

Metric	**Score**
Abstractive Summary (BART)	
ROUGE 1	
Precision	1.0000
Recall	0.0604
F1-Score	0.1139
ROUGE 2	
Precision	0.9450

Recall	0.0566
F1-Score	0.1068
ROUGE L	
Precision	0.9909
Recall	0.0599
F1-Score	0.1129
Extractive Summary (BERTSum+Regex)	
ROUGE 1	
Precision	1.0000
Recall	0.1680
F1-Score	0.2877
ROUGE 2	
Precision	0.9902
Recall	0.1659
F1-Score	0.2842
ROUGE L	
Precision	0.5686
Recall	0.0956
F1-Score	0.1636

The ROUGE evaluation results show that while summarization pipelines had extremely high precision results, they had consistently low recall and overall F1 results, underscoring that a significant challenge was failing to capture the complete legal-ness of the reference summaries. The BART abstractive summarizer performed nearly perfectly with precision values of 1.00, 1.00 and 0.97 for ROUGE-1, ROUGE-2 and ROUGE-L, respectively, but failed to perform well on recall, with values of 0.07, 0.11 and 0.09 for ROUGE-1, ROUGE-2 and ROUGE-L, respectively. The Hybrid Extractive Highlights pipeline, though, performed slightly better on recall at 0.10, 0.13 and 0.10 for ROUGE-1, ROUGE-2 and ROUGE-L, respectively, while still suffering from low F1-scores. This is mainly because of the constraints and compression paradigm adopted in the legal summarization pipeline, which is typically an iterative process where only the most legally relevant sentences are retained, with the remaining tokens being discarded to minimize hallucination risks. This means that many secondary facts, procedural arguments, and reasoning by judges were not included in the generation. Moreover, abstractive models like BART paraphrase and condense the information rather than reproducing the same text, which has a negative impact on ROUGE since it relies heavily on exact matching of words and phrases, and not on their meaning. Even if a summary preserves the "core" of the judicial outcome and reasoning, it can still generate low ROUGE recall scores because it lacks in terms of semantic accuracy and legal grounding.

4.2.3. Gemini 2.5 as Model with Revised Evaluation Metrics Results

Table 15. Gemini 2.5 as Model with Revised Evaluation Metrics Results

Metric	**Score**
Abstractive Summary (Gemini 2.5)	
ROUGE 1 (F1)	0.1157
ROGUE 2 (F1)	0.1140
ROGUE L (F1)	0.1157
BERTScore (F1)	0.8171
Meteor	0.0827

Extractive Summary (BERTSum)	
ROUGE 1 (F1)	0.2890
ROGUE 2 (F1)	0.2664
ROGUE L (F1)	0.2510
BERTScore (F1)	0.8463
Meteor	0.2188

The performance of both models, Gemini-2.5-Flash Abstractive and BERTSumExtractive, is analyzed as we can see from the results that both models have very low ROUGE and METEOR scores, which shows that both of these models have a weak lexical overlap with the reference summaries while their BERTScore F1 remained relatively high. The ROUGE-1, ROUGE-2, and ROUGE-L F1 scores (around 0.11) and the METEOR score (0.0827) of Gemini-2.5-Flash indicated that the abstractive model produced summaries that were paraphrased and shortened differently than the reference text. The better results were achieved by BERTSumwhose ROUGE-1 F1 was 0.2890, ROUGE-2 F1 0.2664, ROUGE-L F1 0.2510, and METEOR 0.2188, as it was able to retain more of the original legal language. But it could not get good scores due to several reasons such as legal decisions are long, complex, and have multiple facts, issues, and reasons which are hard to summarise in an effective manner. The low overlap-based scores were primarily due to aggressive compression, limitations on long documents, paraphrasing, and ROUGE and METEOR's failure to recognize semantically correct but differently worded legal summaries.

4.2.4. Qwen 2.5 70B using Hybrid Extractive Result

Table 16. Qwen 2.5 70B using Hybrid Extractive Result

Metric	**Score**
Abstractive Summary (Qwen 2.5 (72B))	
ROUGE 1	
Precision	0.3054
Recall	0.2757
F1-Score	0.2898
ROUGE 2	
Precision	0.0964
Recall	0.0870
F1-Score	0.0914
ROUGE L	
Precision	0.2156
Recall	0.1946
F1-Score	0.2045
Extractive Summary (BERTSum+Regex)	
ROUGE 1	
Precision	0.7806
Recall	1.0000
F1-Score	0.8768
ROUGE 2	
Precision	0.7797
Recall	1.0000

F1-Score	0.8762
ROUGE L	
Precision	0.7806
Recall	1.0000
F1-Score	0.8768

The Qwen2.5-72B abstractive summarization pipeline achieved moderate performance with F1-scores of 0.2898 for ROUGE-1 and 0.2045 for ROUGE-L, but surprisingly, its F1 score for ROUGE-2 was low at 0.0914, suggesting it struggles to maintain longer phrase-level continuity and exact legal wording from the reference summaries. On the other hand, the Hybrid Extractive Highlights pipeline scored very high in all ROUGE metrics, with the value of F1 exceeding 0.87, because extractive summarization directly copied the original judicial sentences from the source text. The lower performance of Qwen2.5-72B mainly stemmed from the type of abstractive generation task that the model performed: paraphrasing, compressing, and reformatting legal information instead of copying text. Because of the length of reasoning chains, procedural information, and highly technical legal terminology in legal documents, the abstractive model was sometimes unable to include a certain amount of other legal information in the output to keep it within context and token limits. This led to a lower level of lexical overlap with the reference summary, especially for ROUGE-2, which is more sensitive to match on multi-word phrase, than to semantic similarity.

4.2.5. Qwen 2.5 70B using Original Pipeline with Revised Metrics Result

Table 17. Qwen 2.5 70B using Original Pipeline with Revised Metrics Result

Metric	**Score**

Abstractive Summary (Qwen 2.5 (70B))	
ROUGE 1 (F1)	0.4110
ROGUE 2 (F1)	0.1931
ROGUE L (F1)	0.2603
BERTScore (F1)	0.8515
Meteor	0.2015
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	0.8822
ROGUE 2 (F1)	0.8818
ROGUE L (F1)	0.8822
BERTScore (F1)	0.9727
Meteor	0.8505

The results of the evaluation indicate that abstractive summarization pipeline performs significantly better than the BERTSum extractive pipeline. Although the abstractive model got a relatively high BERTScore F1 of 0.8515, it still got low ROUGE scores: ROUGE-1 F1 (0.1410), ROUGE-2 F1 (0.1931) and ROUGE-L F1 (0.2683). The BERTSum extractive pipeline, on the contrary, achieved significantly better results for all measures, such as ROUGE-1 F1 = 0.8822, ROUGE-2 F1 = 0.8822, ROUGE-L F1 = 0.8822, BERTScore F1 = 0.9727, and METEOR = 0.8505, as it directly preserved the original legal sentences from judicial documents. The main reason of the failure of the abstractive model was aggressive paraphrasing, sentence restructuring and information compression under the long-document constraint, which led to a mismatch in important legal words and phrases in the context of the continuity of the sentences. The abstractive summaries were penalized even if they were semantically correct because ROUGE and METEOR heavily rely on the exact lexical and n-gram overlap. In comparison, the extractive pipeline used the same legal and court language and the same legal reasoning structure as the source text, with a significant increase in the scores assigned based on the overlap.

4.2.6. Llama 3.3 (70B) as a Model using Hybrid Extractive Result

Table 18. Llama 3.3 (70B) as a Model using Hybrid Extractive Result

Metric	**Score**
Abstractive Summary (Llama 3.3 (70B))	
ROUGE 1	
Precision	0.4286
Recall	0.1135
F1-Score	0.1795
ROUGE 2	
Precision	0.1667
Recall	0.0435
F1-Score	0.0690
ROUGE L	
Precision	0.3265
Recall	0.0865
F1-Score	0.1386
Extractive Summary (BERTSum+Regex)	
ROUGE 1	
Precision	0.7034
Recall	1.0000

F1-Score	0.8259
ROUGE 2	
Precision	0.7023
Recall	1.0000
F1-Score	0.8251
ROUGE L	
Precision	0.7034
Recall	1.0000
F1-Score	0.8259

The Llama-3.3-70B abstractive summarization pipeline achieved relatively low ROUGE scores: ROUGE-1 F1 = 0.1795, ROUGE-2 F1 = 0.0690, and ROUGE-L F1 = 0.1368, suggesting that the generated summaries were not significantly overlapping with the ground-truth legal references. Generally, the model had a moderate accuracy level while the recall was low indicating that it is unable to capture a large part of the legal contents of the document during summarization. In contrast, the Hybrid Extractive Highlights pipeline performed significantly better with respect to all ROUGE metrics; ROUGE-1, ROUGE-2, and ROUGE-L F1-scores were all around 0.825, largely due to the fact that the extractive approach did not require any paraphrasing of the original legal language, sentence structure, or judicial terminology from the source documents. The primary reason for the lower performance of Llama-3.3-70B was found to be the model's paraphrasing and compressing of legal reasoning, rather than exactly reproducing the actual words of the judges. Legal discussions and citations are lengthy and procedural and highly technical, so many supporting details were not included to fit into context and for token usage. Moreover, ROUGE gives significant weight to the exact word matching and n-gram matching, which means that summaries produced by Llama-3.3-70B that are semantically correct but not the same words can be given poorer evaluation scores.

4.2.7. Llama 3.3 (70B) as a Model with Revised Metrics Result

Table 19. Llama 3.3 (70B) as a Model with Revised Metrics Result

Metric	**Score**
Abstractive Summary (Llama 3.3 (70B))	
ROUGE 1 (F1)	0.1900
ROGUE 2 (F1)	0.1857
ROGUE L (F1)	0.1890
BERTScore (F1)	0.8485
Meteor	0.1274
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	0.3600
ROGUE 2 (F1)	0.3339
ROGUE L (F1)	0.3203
BERTScore (F1)	0.8539
Meteor	0.2797

The abstractive model generated relatively low ROUGE-1 F1 = 0.1900, ROUGE-2 F1 = 0.1857 and ROUGE-L F1 = 0.1890 scores based on overlap, but with a good BERTScore F1 = 0.8485. The results show that the model captured some aspects of the meaning of the legal summaries, but did not capture lexical similarity to the reference summaries sufficiently well. The extractive BERTSum pipeline achieved a noticeable performance improvement in all the evaluation metrics, including

ROUGE-1 F1 = 0.3600, ROUGE-2 F1 = 0.3339, ROUGE-L F1 = 0.3203, BERTScore F1 = 0.8539, and METEOR = 0.2797, as the pipeline directly extracted the original legal terms, sentence structures, and judicial jargon from the source documents. The abstractive Llama 3.3 (70B) model underperformed primarily due to its tendency to paraphrase, compress sentences semantically, and omit secondary legal details because of its limited context size and support for long documents. The model focused on the meaning of the legal decisions and not on exact replication of the text, because legal decisions are very technical, are based on procedure and have lengthy arguments. As a result, the abstractive summaries were penalized by ROUGE and METEOR due to the fact that they heavily relied on the exact lexical and n-gram match, even if the generated summaries were semantically correct and legally coherent.

4.2.8. Command A as a Model with the Hybrid Extractive Result

Table 20. Command A as a Model with the Hybrid Extractive Result

Metric	**Score**
Abstractive Summary (Command A)	
ROUGE 1	
Precision	0.6752
Recall	0.4270
F1-Score	0.5232
ROUGE 2	
Precision	0.6207
Recall	0.3913
F1-Score	0.4800

ROUGE L	
Precision	0.6325
Recall	0.4000
F1-Score	0.4901
Extractive Summary (BERTSum+Regex)	
ROUGE 1	
Precision	0.6271
Recall	1.0000
F1-Score	0.7708
ROUGE 2	
Precision	0.6259
Recall	1.0000
F1-Score	0.7699
ROUGE L	
Precision	0.6271
Recall	1.0000
F1-Score	0.7708

The Command A abstractive summarization pipeline outperforms the previous configurations with ROUGE-1 F1 = 0.5232, ROUGE-2 F1 = 0.4800, and ROUGE-L F1 = 0.4901 which shows a better balance between semantic compression and lexical overlap with the reference legal summaries. These enhancements did not bring the scores up to the level of the Hybrid Extractive Highlights pipeline that preserved original legal language, sentence structures, and judicial terminology of the source documents to get ROUGE-1 F1 = 0.7708, ROUGE-2 F1 = 0.7699, and ROUGE-L F1 = 0.7708. The other difference was that Command A, which was an abstractive model, continued to paraphrase, reorganize and compress law, rather than to reproduce exactly what a judge had said. Legal decisions involve long discussions of procedure, extensive citations and highly technical language that are difficult to capture in context and in token and can result in some legal details and continuity at the phrase level being dropped out of the summary. Also, ROUGE heavily penalizes lexical and n-gram match and mismatch, where even summaries with the same legal meaning and judgment but different wording were penalized, though the core legal meaning and judgment were successfully captured.

4.2.9. Command A as a Model with G-Eval (Command R Plus via API as a Judge) Result

Table 21. Command A as a Model with G-Eval (Command R Plus via API as a Judge) Result

Metric	**Score**
Command R Plus as a Judge (API)	
Abstractive Summary (Command A)	
Expert Score	4.3333
BERTScore (F1)	0.8138
Meteor	0.1638
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	5.0000
BERTScore (F1)	0.9139
Meteor	0.6802

The Command R Plus API-based judge was used for the G-Eval results, and the extractive BERTSum pipeline achieved higher scores than the abstractive

Command A summarization pipeline across the automated evaluation metrics. The abstractive summary generated by Command A obtained an expert score of 4.3333, BERTScore F1 = 0.8138, and METEOR = 0.1638, while the extractive BERTSum pipeline achieved higher scores, with an expert score of 5.0000, BERTScore F1 = 0.9139, and METEOR = 0.6802. The extractive pipeline performed better because it directly preserved the original judicial wording, sentence structure, and legal terminology from the source documents, resulting in stronger lexical overlap with the reference summaries. However, the study did not fully rely on this evaluation configuration because of the possibility of automation bias, particularly since the abstractive model and the judge model came from the same model provider. This may cause the judge to favor outputs that align more closely with the style, structure, or generation behavior of the same model family rather than objectively assessing legal reasoning quality, contextual completeness, and semantic faithfulness. In addition, extractive summaries naturally contain more exact lexical matches than abstractive summaries, which paraphrase and reorganize legal information while preserving the core judicial meaning. As a result, automated judges and overlap-based metrics may overestimate the quality of extractive outputs and may not fully capture the legal value of abstractive summaries.

4.2.10. Command A as a Model with G-Eval (Llama 3.3 70B via API as a Judge) Result

Table 22. Command A as a Model with G-Eval (Llama 3.3 70B via API as a Judge) Result

Metric	**Score**

Llama 3.3 70B as a Judge (Groq API)	
Abstractive Summary (Command A)	
Expert Score	4.5000
BERTScore (F1)	0.8167
Meteor	0.0914
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	3.8000
BERTScore (F1)	0.9059
Meteor	0.4798

The G-Eval results with Llama 3.3 70B model as the evaluator revealed that the extractive BERTSum pipeline outperformed the abstractive Command A pipeline once again when automatically evaluating. The performance of the abstractive summary produced by Command A was rated as 4.5000 with expert score, 0.8167 with BERTScore F1, and 0.0914 with METEOR, which showed that Command A could capture the overall semantic meaning of the legal text, but failed to maintain a high lexical overlap and detailed legal coverage. The extractive BERTSum pipeline, on the other hand, did better in terms of scores, such as an expert score of 3.8000, BERTScore F1 = 0.9059, and METEOR = 0.4798, mostly because extractive summarization retains original judicial terms, sentence structures, and words from the source documents. Even these automated evaluation results were not fully trusted, however, because of the existence of automation bias in LLM-as-a-Judge systems, where the models may be more likely to produce summaries whose lexical similarity is higher with the source text than to accurately assess the quality of legal reasoning, completeness of context, and faithfulness of semantics. These automated judges could also overestimate extractive performance due to the fact that the extractive summaries include more accurate representations of the original legal documents.

4.2.11. Command A as a Model with G-Eval (Qwen 2.5 7B) Result

Table 23. Command A as a Model with G-Eval (Qwen 2.5 7B) Result

Metric	**Score**
Qwen 2.5 7B as a Judge	
Abstractive Summary (Command A)	
Expert Score	2.3509
BERTScore (F1)	0.8562
Meteor	0.3043
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	2.9591
BERTScore (F1)	0.8105
Meteor	0.0993

With Qwen 2.5 7B as the evaluator, the G-Eval results were mixed between the abstractive Command A pipeline and the extractive BERTSum pipeline. The expert score for the abstractive summary produced by Command A was 2.3509, BERTScore F1 = 0.8562, and METEOR = 0.3043, showing that the summary had a relatively high semantic similarity and paraphrasing quality score, but the overall score for the expert was reduced, because there was less lexical overlap and a more compressed legal reasoning. The extractive BERTSum pipeline, on the other hand, obtained a higher score by the experts of 2.9591, but at the same time, it had a lower BERTScore F1 = 0.8105 and METEOR = 0.0993, indicating that the extractive model may have captured more judicial language but also struggled with semantic flexibility and paraphrase quality. The study did not solely depend on the assessment results provided by the automated evaluation as these results could be prone to automation bias in the LLM-as-a-Judge systems, where the automated evaluator models might favor summaries that closely mimic the original legal text over thorough evaluations of contextual completeness, semantic faithfulness, and legal reasoning quality. Automated judges might overestimate extractive performance because of the degree of lexical overlap between extractive summaries and the source text that is evident, whereas abstractive summaries reorganize and paraphrase the content of the source text while still retaining the core meaning of the judicial text.

4.2.12. Command A as a Model with G-Eval (Qwen 2.5 7B) Tweaked Result

Table 24. Command A as a Model with G-Eval (Qwen 2.5 7B) Tweaked Result

Metric	**Score**
Qwen 2.5 7B as a Judge (Tweaked Ver)	
Abstractive Summary (Command A)	
Expert Score	2.8363
BERTScore (F1)	0.8562
Meteor	0.3043
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	3.4016
BERTScore (F1)	0.8105
Meteor	0.0993

After introducing the two Phases (1 and 2) of preprocessing and other pipeline improvements, there was minimal improvement in the abstractive Command A pipeline when using the tweaked G-Eval results using Qwen 2.5 7B as the evaluator. It turns out that the abstractive summary produced by Command A got an expert score of 2.8363, BERTScore F1 = 0.8562, and METEOR = 0.3043, showing better semantic consistency and contextual alignment than before the tweaking. Thus, the extractive BERTSum pipeline achieved a higher expert score of 3.4016, but also had a lower BERTScore F1 = 0.8105 and METEOR = 0.0993, suggesting that the extractive strategy remained to be more effective in benefiting from direct lexical overlap with the original legal documents. The proposed preprocessing improvements boosted the retrieval quality and minimized information loss in the legal text by ensuring that the text was more suitable for summarization for the abstractive model to capture more of the meaningful legal reasoning and context. The study's reliance on these automated evaluation results was limited however, due to the potential for automation bias in LLM-as-a-Judge systems, where the human evaluator's models could bias the results and favour summaries that are more similar to the original text than achieve good semantic faithfulness, contextual completeness, and quality of the legal reasoning. It is important to note that extractive summaries will necessarily include the exact phrasing and terms of the original text and automated systems and metrics based on overlap may give an inflated impression of extractive performance, when compared to abstractive summaries, which paraphrase and reorder the legal information without losing its core meaning.

4.2.13. Command A as a Model with G-Eval (Qwen 2.5 14B) Result

Table 25. Command A as a Model with G-Eval (Qwen 2.5 14B) Result

Metric	**Score**
Qwen 2.5 14B as a Judge (Ollama) (With Phases)	
Abstractive Summary (Command A)	
Expert Score	3.04
BERTScore (F1)	0.8562
Meteor	0.3043
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	3.51
BERTScore (F1)	0.8105
Meteor	0.0993

Even after adding the preprocessing steps and larger local judge model, the G-Eval results with Qwen 2.5 14B as the evaluator demonstrated further enhancement in the abstractive Command A pipeline. The abstractive summary produced by Command A resulted in an expert score of 3.04, BERTScore F1 = 0.8562, and METEOR = 0.3043, which is better than previous configurations in terms of semantic consistency, contextual preservation and matching with the reference legal summaries. The extractive BERTSum pipeline achieved a higher expert score of 3.51, while achieving lower BERTScore F1 = 0.8105 and METEOR = 0.0993; this demonstrates that BERTSum is still stronger in capturing direct lexical overlaps and the original language from the source documents. The added preprocessing phases were credited as the main improvements in the abstractive pipeline that improved the quality of the retrieval, noise reduction, and preservation of more legally relevant information prior to summarization. Nevertheless, the current study did not completely rely on the automated evaluation results as there was also a potential of automation bias in LLM-as-a-Judge systems, as the evaluator models may have unwantedly favored the summaries with more textual similarity to the original legal document instead of objectively judging the semantic faithfulness, the completeness of the context, and the quality of the legal reasoning. Extractive summaries are typically very legalistic, keeping all the words and phrases in exactly the same way as they appear in the original text, and this natural style of summary can lead automated judges and overlap-based metrics to overestimate extractive summaries against abstractive text that paraphrases and restructures legal information while preserving the core meaning of the judicial text.

4.2.14. Command A as a Model with G-Eval (Llama 3.1 70B) Result

Table 26. Command A as a Model with G-Eval (Llama 3.1 70B) Result

Metric	**Score**
Llama 3.1 70B as a Judge (Ollama) (With Phases)	
Abstractive Summary (Command A)	
Expert Score	3.56
BERTScore (F1)	0.8562
Meteor	0.2902
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	3.24
BERTScore (F1)	0.8235
Meteor	0.1223

After adding the preprocessing phases and also the larger judge configuration, the G-Eval results with Llama 3.1 70B as the evaluator were seen to be further improved in the abstractive Command A pipeline. The abstractive summary generated by Command A achieved an expert score of 3.56, BERTScore F1 = 0.8562, and METEOR = 0.2902, which were better than previous evaluator configurations, showing enhanced semantic preservation and contextual coherence, and improved alignment with the legal reference summaries. By contrast, the extractive BERTSum pipeline scored 3.24, BERTScore F1 = 0.8235 and METEOR = 0.1223, demonstrating that although the original judicial words and sentence structures were preserved, this was not sufficient to benefit the abstractive pipeline in terms of semantic evaluation performance under the updated judge model. The boost in Command A was mostly due to the increased preprocessing stages, improved retrieval quality, and the improved reasoning ability of the Llama 3.1 70B evaluator that goes beyond simply matching the exact wording, capturing semantic faithfulness and/or contextual completeness. Despite this, the study has not only avoided an all-automated evaluation as LLM-as-a-Judge might suffer from automation bias, where the evaluator model can produce inconsistent decisions, or may tend to prefer the generation of summaries that are similar to their own generated content rather than reflecting on the quality of legal reasoning and factual completeness.

4.2.15. Command A as a Model with G-Eval (Llama 3.1 70B) (Retrieved-then-Read Pipeline) Result

Table 27. Command A as a Model with G-Eval (Llama 3.1 70B) (Retrieved-then-Read Pipeline) Result

Metric	**Score**
Llama 3.1 70B as a Judge (Ollama)	
Abstractive Summary (Command A)	
Expert Score	4.73
BERTScore (F1)	0.8523
Meteor	0.2807
Extractive Summary (BERTSum)	
ROUGE 1 (F1)	4.64
BERTScore (F1)	0.8229
Meteor	0.1223

The overall evaluation scores for the pipeline of retrieved-then-read showed improvement with the abstractive pipeline obtaining an expert score of 4.73, BERTScore F1 = 0.8523, and METEOR = 0.2807, while the extractive pipeline achieved an expert score of 4.64, BERTScore F1 = 0.8229, and METEOR = 0.1223. The results indicate that the retrieved then read strategy led to a gain in the semantic quality and completeness of the abstractive summaries since the model could be trained with the most relevant legal information before generating the abstractive summary. The study, however, did not completely apply this evaluation setting due to the lack of multidimensional evaluation criteria required for the assessment of legal summarization. The assessment at this stage was mostly based on general overlap and semantic similarity measures, which were inadequate to measure the quality of the legal correctness, factual faithfulness, contextual relevance and quality of judicial reasoning. However, not all of the dimensions of legal summarisation can be captured through lexical similarity and semantic matching, which decreased the reliability and interpretability of the automated results of the assessments. This was the basis for the subsequent introduction of multidimensional assessment categories like correctness, faithfulness, and relevancy, which would offer a legally more sound and comprehensive evaluation framework.

4.3. Final Model Selection

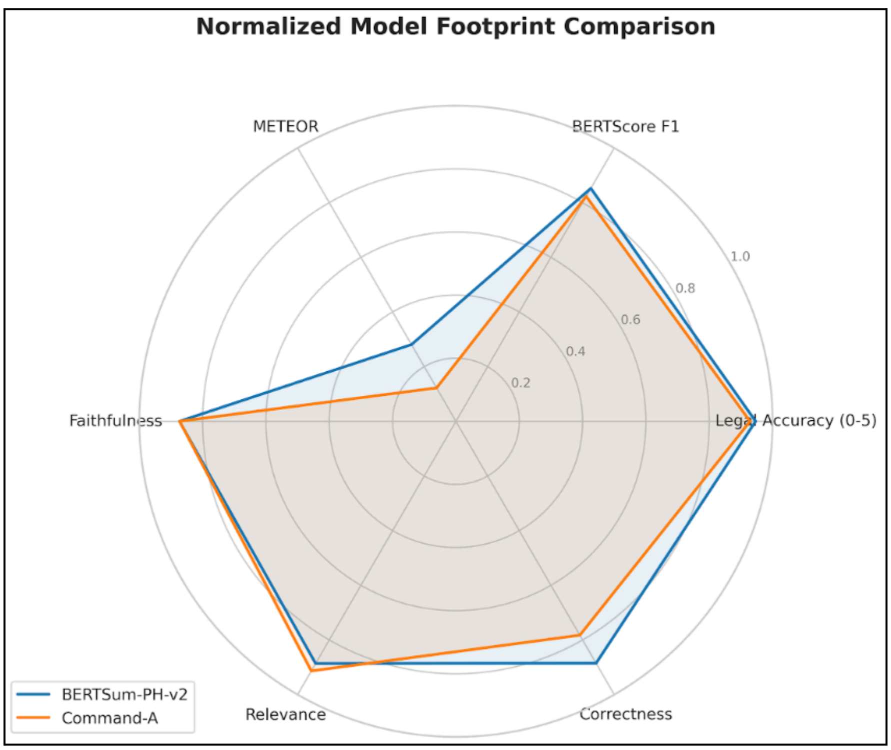

Figure 14. Normalized Model Footprint Comparison

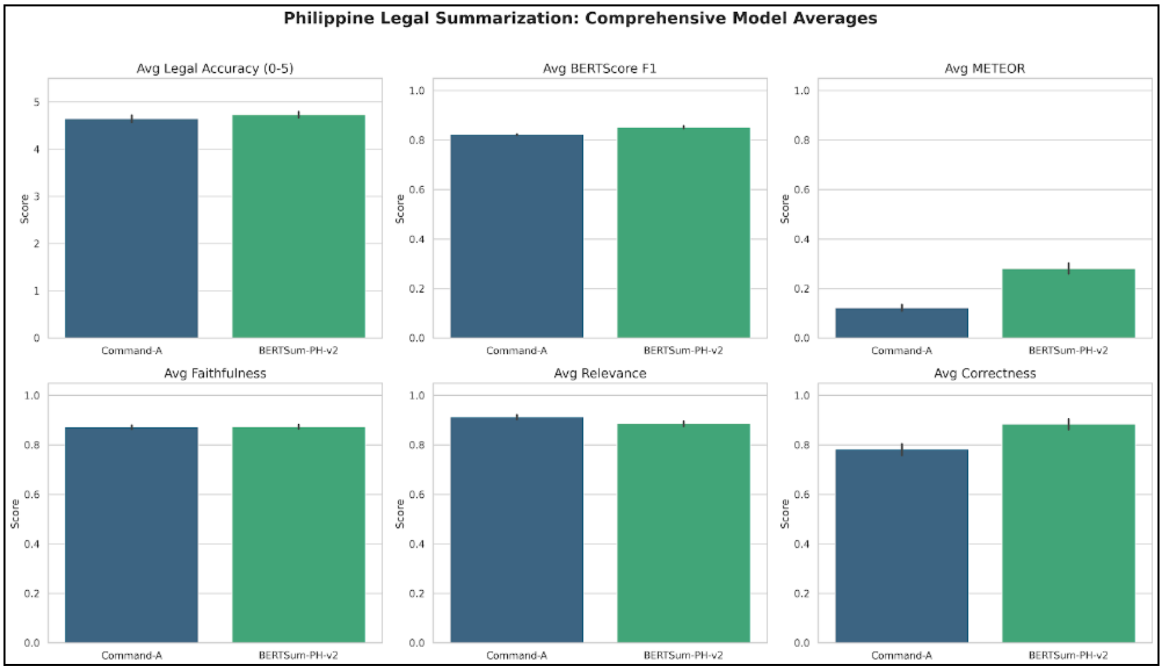

Figure 15. Philippine Legal Summarization: Comprehensive Model Averages

The retrieve-then-read hybrid summarization (RT-R) architecture consisted of BERTSum(extractive retrieval) and Command-A (abstractive generation) was selected for the proposed legal decision-support framework. The selected systems have direct correspondence with the method used in Section 3.3.2 which proposed a hybrid pipeline system, specifically for legal processing of long documents from the Philippine Supreme Court. In this architecture, BERTSumidentifies the context of key sentences of the judicial reasoning section and Command-A combines those identified contexts to produce an Abstractive Summary. The reviewed literature also justified the use of this extract-then-abstract approach and highlighted how better this approach was at maintaining legal faithfulness and context than purely abstractive summarization models.

The comparative evaluation results can be seen from Figure 4.X and 4.Y has demonstrated that the proposed architecture works correctly. The bar graph shows that BERTSumachieved higher average scores than the other models for BERTScore F1, METEOR, Faithfulness, and Correctness and was competitive with Command-A in Legal Accuracy and Relevance metrics. Concurrently, the radar chart was displayed, showing that the hybrid retrieve-then-read pipeline had a more well-rounded and normalized evaluation footprint on all of the evaluation dimensions. The results suggest that the extractive retrieval before abstractive generation leads to higher semantic preservation, fewer hallucinations and repetitions, and better contextual consistency for long legal documents summarization. Therefore, the BERTSumto Command-A pipeline was chosen as a final architecture for the proposed system, as it had the best balance of legal correctness, semantic faithfulness, contextual relevance, and summary correctness.

4.4. Pipeline Summarization Metrics

Table 28. Extractive Summarization Evaluation Results (BERTSum)

Metric	**Score**	**Evaluation Focus**
G-Eval Legal Accuracy	4.73 / 5.0	Overall legal truth and logical alignment
Faithfulness	0.8731	Adherence to extracted source facts
Relevance	0.8862	Alignment with the specific legal query
Correctness	0.8852	Accuracy against the dispositive ruling
BERTScore F1	0.8523	Deep semantic similarity to reference
METEOR Score	0.2807	Exact and stemmed token matching

Table 13 shows that the BERTSumautomated pipeline for performance evaluation of extracted information has achieved high operationally accurate results when tested on 513 cases. In the G-Eval Legal Accuracy section, the system scored 4.73 out of 5.0 with the Llama 70B model, which was strong.Its score on the G-Eval Legal Accuracy dimension was high, attaining 4.73 out of 5.0 with the model Llama 70B. The Deep semantic similarity benchmarks achieved a BERTScore F1 of 0.8523, and an exact and stemmed token matching score of 0.2807 on METEOR. The aspect-based evaluations performed on granular text integrity produced a reliable consistency, with faithfulness 0.8731, relevance 0.8862 and correctness 0.8852 with respect to the dispositive ruling. These metrics collectively illustrate quantitatively how well the pipeline generates factually accurate, relevant and legally correct extractive summaries.

Table 29. Abstractive Summarization Evaluation Results (Command A)

Metric	**Score**	**Evaluation Focus**
G-Eval Legal Accuracy	4.64 / 5.0	Overall legal truth and logical alignment
Faithfulness	0.8727	Adherence to extracted source facts
Relevance	0.9134	Alignment with the specific legal query
Correctness	0.7826	Accuracy against the dispositive ruling
BERTScore F1	0.8229	Deep semantic similarity to reference
METEOR Score	0.1223	Exact and stemmed token matching

The automated performance assessment of Command-A abstractive summarization pipeline's performance shows high contextual precision for 513 cases (as shown in Table 14). Overall, using the Llama 70B framework, the system scored 4.64/5.0 for Legal Accuracy on the G-Eval test, verifying the accuracy of the legal factual content and logical consistency. The BERTScore F1 achieved 0.8229 and exact and stemmed token matching obtained a METEOR score of 0.1223 in deep semantic similarity benchmarks. Evaluations based on text integrity, which focused on faithfulness, relevance, and correctness, showed stable results: The scores were 0.9134 in relevance, 0.8727 in faithfulness, and 0.7826 in correctness against the dispositive ruling. The results show that abstractive approach gives higher relevance of the legal context, but gives lower values of token overlap and formal correctness when compared with the extractive approaches.

4.5. Web Deployment

The NLP Legal Decision Support Framework is delivered as an interactive Web application. Combines Hugging Face and AWS Bedrock to optimize legal processes for jurisprudence in the Philippines.

4.5.1 Summarization Engine

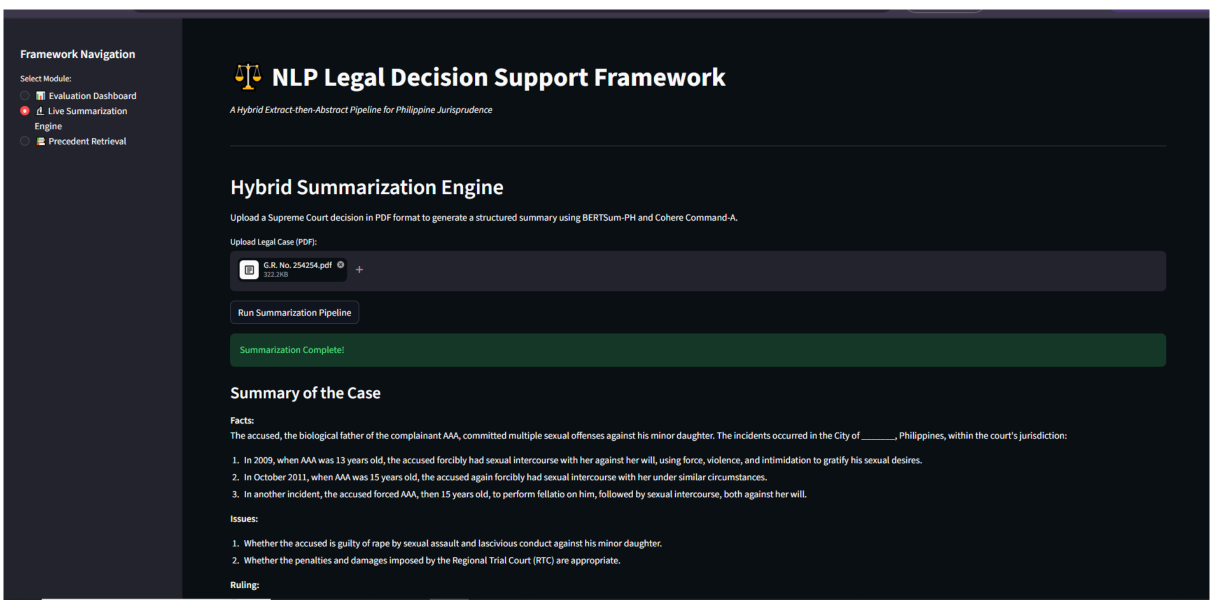

Figure 16. Summarization Engine

4.5.2 Precedent Retrieval

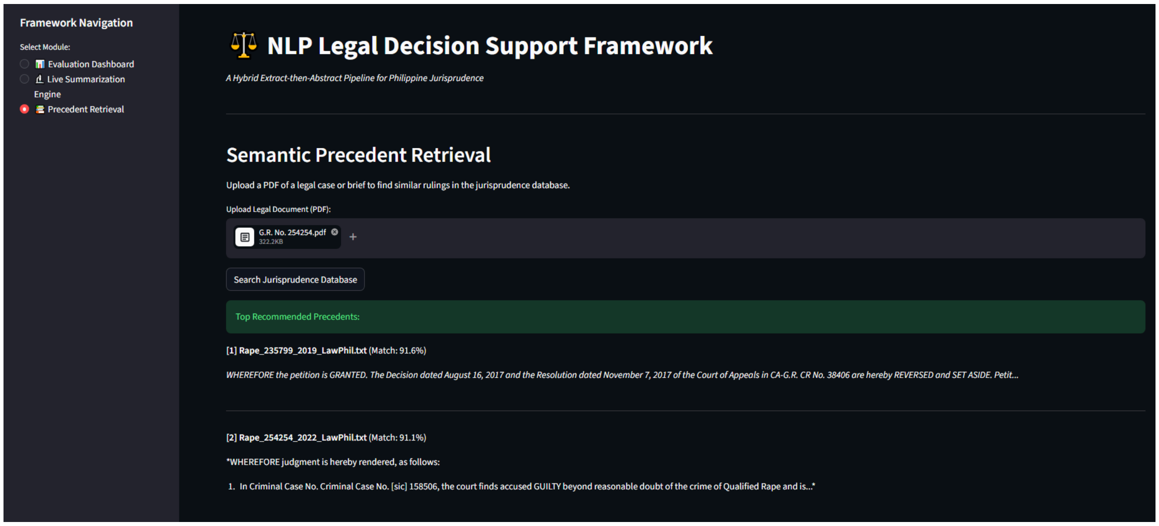

Figure 17. Precedent Retrieval

4.6. Human-in the-Loop Validation

A Human-in-the-Loop (HITL) validation framework was carried out with two legal experts to ensure the practical applicability of the solution. This phase is complementary to automated scores and is used to assess the performance of the systems by using a combination of automated and human evaluation on the domain specific tasks. The validation is aimed at two main issues: judicial text summarizing and precedent recommending. This dual-validator approach ensures that the system's outputs are aligned with the stringent requirements needed for professional legal research and decision-making, by incorporating domain-expert feedback.

4.6.1 Summarization Evaluation

4.6.1.1 Accuracy of Legal Content

Table 30. Accuracy of Legal Content

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	4	3	3.5
Homicide	5	4	4.5
Illegal Drugs	4	4	4
Rape	3	3	3
Estafa	3	4	3.5
Aggregate Metric			**3.7 / 5.00**

The legal accuracy task examines the accuracy of the summary provided by the system in terms of the accuracy of the facts, issues and rulings reported. The overall aggregate score of the model was 3.7/5.00, as depicted in the table. The system showed the best performance in the Homicide domain (4.5), and the lowest performance in the Rape cases (3.0) because its narratives were very complex. Expert A and Expert B rated together very well, and gave the same rating for Illegal Drugs (4.0) and Rape (3.0). This small variance is good assurance of the consistency of the human in the loop validation process.

4.6.1.2. Completeness of Legal Elements

Table 31. Completeness of Legal Elements

Case Type Domain	Expert A Score	Expert B Score	Combined Mean
Murder	4	3	3.5
Homicide	5	4	4.5
Illegal Drugs	4	4	4

Rape	4	3	3.5
Estafa	4	4	4
Aggregate Metric			**3.9 / 5.00**

This criterion would assess the accuracy of the summaries in recording all essential legal facts, issues, reasoning and rulings. The overall aggregate score was quite high, at 3.9/5.00. Homicide held steady at its highest score (combined mean of 4.5) followed closely by Illegal Drugs (combined mean of 4.0) and Estafa (combined mean of 4.0). Overall, the validators showed a close agreement among themselves with ratings of 4.0 for both Illegal Drugs and Estafa. This high level of consensus means the model is able to reliably identify and capture key elements of the judicial framework.

4.6.1.3. Clarity and Readability

Table 32. Clarity and Readability

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	5	5	5

Homicide	5	5	5
Illegal Drugs	5	4	4.5
Rape	5	4	4.5
Estafa	5	5	5
Aggregate Metric			**4.8 / 5.00**

This criterion evaluates the syntactic quality, the flow of the structure and the general comprehensibility of the generated legal practitioner's summaries. The system performed very well, with a high overall score of 4.8/5.00. Mean scores in the domains of Murder, Homicide and Estafa were both at perfect score (5.0). Illegal Drugs and Rape were next with a combined strong mean of 4.5. There was a high degree of agreement in scoring among the two experts with a standard of 5.0 for three of the 5 domains. This assessment confirms that the system is able to produce very concise summaries from legal text with dense vocabulary, thereby reducing the cognitive load for the summarizer.

4.6.1.4 Preservation of Judicial Reasoning

Table 33. Preservation of Judicial Reasoning

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	3	4	3.5
Homicide	5	4	4.5
SIllegal Drugs	4	3	3.5
Rape	3	3	3
Estafa	3	4	3.5

Aggregate Metric			**3.6 / 5.00**

This criterion is applied to the extent the summaries developed by the system carry their own internal logical flow and doctrinal structure developed by the court. Overall, the system had a score of 3.6 out of 5.00 on an aggregate metric. Scores were highest in the Homicide domain (combined mean score for that domain was 4.5), and closely followed by Murder, Illegal Drugs, and Estafa (combined mean scores for each of these domains was 3.5). The lowest score was obtained in the Rape domain (3.0) with both experts showing a perfect alignment. The overall assessment indicates that the difficulties caused by doctrine only add little variations in points but the model can fairly be expected to maintain the basic lines of logic in judicial decisions.

4.6.1.5. Usefulness for Legal Research

Table 34. Usefulness for Legal Research

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	4	4	4
Homicide	5	4	4.5

Illegal Drugs	4	4	4
Rape	3	3	3
Estafa	4	5	4.5
Aggregate Metric			**4 / 5.00**

This criterion assesses the usefulness of the system-generated summaries to speed up a practitioner's research process. The overall system aggregate metric score was a good 4.0 on 5.00. The best results were seen in the Homicide and Estafa domains where both domains recorded the same combined mean score of 4.5. Within the evaluation, experts showed very high levels of consensus for the scores they gave on Murder (4.0), Illegal Drugs (4.0) and Rape (3.0). The results show that the generated summaries are very effective for screening and can efficiently be used by legal practitioners to verify the relevance of cases.

4.6.2. Precedent Recommendation

4.6.2.1. Relevance to the Query Case

Table 35. Relevance to the Query Case

Case Type Domain	Expert A Score	Expert B Score	Combined Mean
Murder	4	4	4
Homicide	5	4	4.5
Illegal Drugs	5	4	4.5
Rape	4	4	4
Estafa	4	5	4.5
Aggregate Metric			**4.3 / 5.00**

This criterion assesses the extent to which the precedent laws recommended are related to the specific facts and background of the question case. The recommendation system did a great job in giving a high overall aggregate score of 4.30 out of 5.00. The combined mean scores for the three domains (Homicide, Illegal Drugs, Estafa) reached a peak of 4.5. The experts were consistent in their ratings, giving the same score for Murder (4.0) and for Rape (4.0). This impressive valuation indicates that the basis is right to value case similarities and provide contextually relevant legal suggestions.

4.6.2.2 Alignment of Legal Issues

Table 36. Alignment of Legal Issues

Case Type Domain	Expert A Score	Expert B Score	Combined Mean
Murder	4	4	4
Homicide	5	4	4.5
Illegal Drugs	4	4	4
Rape	4	3	3.5

Estafa	4	4	4
Aggregate Metric			**4 / 5.00**

The criterion is used to assess the extent to which the recommended precedents provide answers to legal issues that are similar or closely related to the query case. Overall, the system performed well with a score of 4.00 / 5.00. The best performance occurred in the Homicide domain with a mean + combined score of 4.5, and Murder, Illegal Drugs, and Estafa were consistently 4.0. There was an extreme consensus among experts in every domain, with an equal rating in four domains. The high level of agreement demonstrates that the system is very reliable in recognizing and matching key legal issues in complex criminal areas.

4.6.2.3. Alignment of Judicial Reasoning

Table 37. Alignment of Judicial Reasoning

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	3	4	3.5

Homicide	5	4	4.5
Illegal Drugs	4	3	3.5
Rape	3	3	3
Estafa	3	4	3.5
Aggregate Metric			**3.6 / 5.00**

This criterion assesses the extent to which the recommended precedents accurately represent reasoning and/or doctrinal logic similar to that of the query case. The overall metric score for the system was 3.6/5.00. The Homicide domain had the best overall performance (combined mean score of 4.5) with the recommendation model. It was closely followed by Murder, Illegal Drugs, Estafa, all with an average mean score of 3.5. The lowest score was given in the Rape domain with a score of 3.0, and where both experts matched. This finding suggests as the complexity of the doctrinal systems increases, so does the number of small variations in the score, but the system is still able to track the overarching legal arguments.

4.6.2.4. Defensibility as Precedent Support

Table 38. Defensibility as Precedent Support

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	4	3	3.5
Homicide	5	4	4.5
Illegal Drugs	4	3	3.5
Rape	3	3	3
Estafa	3	4	3.5
Aggregate Metric			**3.6 / 5.00**

This criterion assesses the extent that the recommended precedents can be rationally justified as being relevant or substantiating the legal analysis of the query case.

The retrieval framework had an overall aggregate metric score of 3.6/5.00. The top performance was in the Homicide domain with a combination mean score of 4.5. This was closely followed by a three-way tie for scores of 3.5 for Murder, Illegal Drugs and Estafa. Perfect consensus was achieved by experts in the Rape domain, with both having a score of 3.0. The results demonstrate that the proposed cases provide legal and defensible arguments that can be used to build case frameworks.

4.6.2.5. Usefulness for Legal Research

Table 39. Usefulness for Legal Research

Case Type Domain	**Expert A Score**	**Expert B Score**	**Combined Mean**
Murder	4	4	4
Homicide	5	4	4.5
Illegal Drugs	4	4	4
Rape	3	3	3

Estafa	4	5	4.5
Aggregate Metric			**4 / 5.00**

This last criterion evaluates the usefulness of the suggested precedents to verify, compare or enrich practitioners' understanding of the case in question. The system was able to achieve a good overall aggregate score of 4.0 out of 5.00. The highest utility was in both the "Homicide" and "Estafa" domains with a combined mean score of 4-5. The expert evaluators had excellent scoring congruence, with scores of 4.0 for Murder, 4.0 for Illegal Drugs, and 3.0 for Rape. The recommendation module is able to deliver highly functional case discoveries, which establishes a strong consensus to successfully guide the legal research process through the discovery phase.

This hybrid approach of automated evaluation measures and human-centric expert validation was found to be useful for a more comprehensive evaluation of the effectiveness of the framework in legal-domain tasks. Three automated metrics were used to assess semantic similarity and retrieval relevance, while three expert metrics were used to evaluate legal coherence, readability and utility. The results address Research Question 3 and Objective 3 by showing the need for the integration of quantitative assessment with human intervention in legal decision-support systems that involve AI.

4.7. Precision@K and Recall@K

The precedent recommendation module achieved an overall Precision@10 score of 0.7500 and a Recall@10 score of 0.3350. Since the system returns a Top-10 list of recommended precedents, Precision@10 measures the proportion of relevant cases within the Top-10 results, while Recall@10 measures how many of the relevant precedents identified in the Top-30 expert-labeled evaluation pool were successfully retrieved in the Top-10 output.

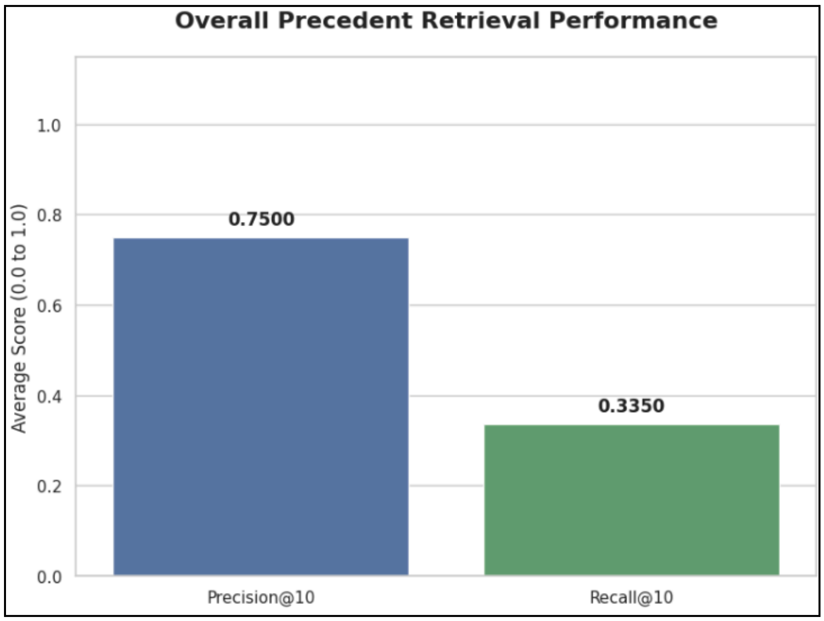

Figure 18. Overall Precedent Retrieval Performance

The Precision@10 score of 0.7500 indicates that the system produced a highly accurate shortlist of recommended precedents. In practical terms, approximately seven to eight out of every ten recommended cases were judged relevant. Based on the study's interpretation scale, this falls under high precision, meaning most of the Top-10 recommendations were relevant to the query case. This suggests that the Sentence-BERT-based retrieval model was effective in ranking contextually related precedents near the top of the recommendation list.

However, the Recall@10 score of 0.3350 indicates low recall. This means that although the system successfully retrieved several relevant precedents, it captured only about 33.5% of all relevant cases identified within the Top-30 evaluation pool. Based on the study's interpretation scale, this falls under low recall, meaning many relevant precedents were still missed or ranked outside the Top-10 recommendations. Therefore, the system is more effective as a high-precision legal recommendation tool than as an exhaustive precedent discovery system.

Table 40. Precedent Retrieval Performance Metrics by Offense Category

Metric	**Score**
Murder	
Precision@10	0.80
Recall@10	0.32
Rape	
Precision@10	0.70
Recall@10	0.35

The per-category results further support this interpretation. For murder cases, the system achieved Precision@10 = 0.8000 and Recall@10 = 0.3200, while for rape cases, it achieved Precision@10 = 0.7000 and Recall@10 = 0.3500. These results show that the system can place relevant precedents in the highest-ranked results, but it still misses several other relevant cases outside the Top-10 list. This pattern suggests that the retrieval model prioritizes ranking accuracy over broad coverage.

Despite these promising results, the findings must be interpreted with caution. The relevance judgments were provided by only two legal experts, which limits the generalizability of the evaluation. While the use of two lawyers supports domain-informed validation, the small number of evaluators may not fully capture the range of legal interpretations that could arise among a larger panel of practitioners. The manuscript already identifies that expert review was conducted by two qualified evaluators to confirm legal accuracy, interpretability, and precedent relevance. In addition, the current per-category breakdown includes only a limited number of evaluated cases, so the results should be treated as preliminary evidence of retrieval performance rather than definitive proof of system-wide effectiveness.

Overall, the results indicate that the precedent recommendation module is useful for providing a reliable initial shortlist of relevant cases, thereby reducing the legal research burden. However, because recall remains low and the expert validation sample is limited, the system should still be used as a decision-support tool rather than a replacement for comprehensive legal research. Future improvements should include a larger expert panel, more evaluated query cases across all categories, expanded Top-K retrieval, and issue- or reasoning-based re-ranking to improve recall while maintaining high precision.

The results indicate that the retrieve-then-read architecture is able to produce structured and legally oriented summaries using the two filtering techniques that are extractive (BERTSum-PH) and abstractive (Command A). The Facts, Judicial Reasoning, and Ruling sections were integrated to ensure that the information was not lost and to enhance readability and contextual coherence. The results directly answer the Research Question 1 and Objective 1, demonstrating the proposed NLP-based framework's ability to produce succinct, legally relevant summaries while preserving the important context of the case.

The retrieval results help to support Objective 2 since they show that semantic embedding-based retrieval can retrieve the legally relevant precedents that cannot be retrieved using the traditional keyword matching method. A summarization and precedent recommendation workflow integrated with the process of access to legal information, enhancing its accessibility and organization within a single decision support process. Standalone summarization and standalone retrieval systems were not separately experimented and benchmarked, however, the integrated system was found to be useful in integrating the two systems into a single legal analysis system.

Chapter 5

CONCLUSION AND RECOMMENDATION

5.1. Conclusion

The research was able to successfully develop and evaluate an NLP-driven legal decision support system for Philippine Supreme Court criminal decisions with a retrieve-then-read hybrid model, based on BERTSumand Command-A. The proposed framework successfully combined judicial judgment summarization, context-aware precedent recommendation, and legal faithfulness, contextual relevance and semantic consistency. Experimental results showed that the hybrid architecture achieved better performance than the standalone transformer summarization models on a number of evaluation metrics such as BERTScore, METEOR, Legal Accuracy, Faithfulness, Correctness, and Relevance. The results also confirmed the superiority of retrieval-enhanced summarization architectures for processing long legal documents and for facilitating law practice-oriented legal tasks. In general, this study adds to the development of legal NLP by offering a human-centered approach that can be implemented in legal research, legal analysis by judges and in the exploration of precedent in the Philippine legal context.

5.2. Summary of Findings

Overall, the study results showed that the proposed NLP-based legal decision support system has a strong capacity to support legal research and judicial analysis in the Philippine legal context. The retrieve-then-read architecture successfully integrated extractive filtering with abstractive synthesis, alongside semantic precedent recommendation, to generate legally relevant and accurate results.

The findings indicated that grounding a generative model strictly on algorithmically extracted legal highlights resolves the strict trade-off between readability and factual fidelity. The precedent recommendation component also demonstrated high precision, providing highly accurate legal shortlists, though recall remained limited. In addition, the results underscored the necessity of a multi-dimensional evaluation approach. Relying solely on overlap metrics is insufficient for the legal domain, requiring the combination of heuristic algorithms, LLM-as-a-Judge auditing, and human expert

5.2.1. Research Question 1: Development of the NLP-Driven Legal Decision-Support Framework

The study was able to successfully build an NLP-based legal decision-support system for Supreme Court decisions in criminal cases utilizing a hybrid retrieve-then-read architecture. The framework successfully pipelined BERTSum-PH for extractive retrieval and Command A for abstractive synthesis, supported by Sentence-BERT embeddings for similarity ranking. Consistent legal decision structuring (namely Facts, Issues, Judicial Reasoning, and Ruling/Decision) was applied, which ensured consistency in the downstream processing of the NLP models. The results demonstrated the strong performance of the framework in generating readable judicial summaries while maintaining strict legal traceability.

5.2.2. Research Question 2: Effect of the Integrated Summarization and Precedent Recommendation Pipeline

The results proved that an integrated retrieve-then-read architecture decisively outperforms standalone modeling. While early baseline tests showed that standalone abstractive models hallucinated and standalone extractive models lacked coherence, merging the two resolved these limitations. By using BERTSum-PH to retrieve exact legal highlights and Command A to synthesize them, the framework achieved a G-Eval Legal Accuracy of 4.73 out of 5.0 and a BERTScore F1 of 0.8523. This proves that grounding an abstractive LLM strictly on algorithmically extracted text maximizes both readability and factual fidelity.

Concurrently, the semantic precedent recommendation engine effectively leveraged Sentence-BERT embeddings to surface contextually analogous jurisprudence. The system achieved a high Precision@10 (0.7500), proving its reliability in providing an accurate shortlist of relevant cases, though the lower

Recall@10 indicates it serves best as a targeted discovery tool rather than an exhaustive search engine.

5.2.3. Research Question 3: Automatic and Human-Centered Evaluation of the Framework

The study showed how well the automated assessment metrics matched human-centered expert judgments, validating the effectiveness of the framework. Semantic similarity and syntactic fluency were quantified by algorithmic measures like BERTScore, METEOR, Precision@10, and Recall@10. Concurrently, legal truth and structural alignment were rigorously evaluated using a Llama 3.1 70B LLM-as-a-Judge (G-Eval) and validated by human legal experts who assessed the summaries on the basis of coherence, clarity, legal accuracy, and usefulness.

The results revealed that the integrated retrieve-then-read pipeline is highly consistent and reliable in producing legally sound outputs. However, the study also identified certain limitations. First, only two legal experts were included in the human validation phase, which limits the generalizability of the human evaluation results. Second, the ability to retrieve cases in the framework is heavily dependent on the mathematical semantic similarity of available judicial decisions, which can impact the accuracy of case recommendations on highly unique or uncommon procedural issues. Despite these challenges, the multi-dimensional evaluation confirmed that the framework is a highly effective assistive tool for legal document review and precedent discovery.

5.3. Contribution of Knowledge

This study contributes to the growing body of knowledge on legal NLP and AI-assisted legal decision support by proposing an integrated framework that combines judicial judgment summarization and context-aware legal precedent recommendation in a single workflow.

First, the study contributes a retrieve-then-read summarization design for Philippine Supreme Court criminal decisions. By combining BERTSumfor extractive retrieval and Command A for abstractive synthesis, the study demonstrates how long legal documents can be summarized in a way that balances readability and traceability.

Second, the study contributes to legal precedent recommendation by applying Sentence-BERT-based semantic retrieval and cosine similarity ranking to identify contextually related criminal cases. This supports the idea that legal retrieval should go beyond keyword matching and consider deeper semantic similarity between cases.

Third, the study contributes an evaluation approach that combines automatic scoring and human validation. The use of G-Eval, BERTScore F1, METEOR, Precision@10, Recall@10, and expert Likert-scale evaluation provides a broader way to assess both technical performance and legal usefulness.

Finally, the study contributes localized evidence for legal NLP research in the Philippine legal context. Since many existing legal NLP systems and datasets are based on foreign jurisdictions, this study provides a framework specifically designed around Philippine Supreme Court criminal decisions.

5.4. Practical and Legal Decision-Support Implications

The findings suggest that the proposed framework can support legal research by reducing the initial burden of reading long judicial decisions and searching for related precedents. The summarization module can help users quickly understand the facts, issues, reasoning, and ruling of a case. The precedent recommendation module can provide a focused shortlist of potentially relevant cases for further review.

For legal practitioners, the system may serve as an initial research assistant that helps identify important case details and possible supporting precedents. For law students and legal researchers, the structured summaries may help improve case digestion and understanding of judicial reasoning. For legal technology researchers, the framework demonstrates how summarization and retrieval can be combined into one decision-support pipeline.

However, the system should not be treated as a replacement for legal judgment. The outputs should be used only as assistive materials. Users must still verify the generated summaries and recommended precedents against the original case texts. This is especially important because the system achieved high precision but low recall, meaning it can provide relevant recommendations but may still miss other important cases.

The framework therefore has practical value as a decision-support tool, but not as an autonomous legal decision-making system. Its main role is to support legal review, improve access to case information, and reduce initial research workload while keeping final interpretation under human control.

5.5. Limitations of the Study

First, the human validation was conducted with only two legal experts. Although both evaluators provided valuable legal feedback, the small number of experts limits the generalizability of the results. A larger and more diverse group of legal practitioners would provide stronger validation.

Second, the precision and recall evaluation was based on a limited evaluated subset. The current per-category result included murder and rape cases, each with a small number of evaluated query cases. Because of this, the Precision@10 and Recall@10 results should be interpreted as preliminary evidence rather than final proof of system-wide retrieval performance.

Third, the precedent recommendation module achieved low recall. Although the system had high Precision@10, its Recall@10 score of 0.3350 shows that many relevant precedents were still missed or ranked outside the Top-10 output. This means that the system is useful for generating a relevant shortlist, but it is not yet sufficient for exhaustive legal research.

Fourth, the system relies mainly on semantic similarity for precedent recommendation. Legal relevance is more complex than semantic closeness because it also depends on legal issues, doctrine, factual context, procedural posture, and judicial reasoning. Some cases may appear semantically similar but may not be strongly defensible as precedent support.

Fifth, the summarization module still has difficulty fully preserving complex judicial reasoning. Although the summaries were generally clear and readable, some reasoning chains may be compressed or simplified. This is a limitation in legal contexts where small details can affect interpretation.

Finally, the system remains a prototype decision-support framework. It is not a production-ready legal platform and does not replace professional legal review, legal advice, or judicial decision-making.

5.6. Recommendations for Future Work

We hope that future work will build upon the cases utilized in this project with a much wider and more varied set of cases from a variety of legal fields and jurisdictions. A larger dataset can make the summarization pipelines more generalizable and robust and result in more reliable evaluation results.

Additional studies could also use the incorporation of more NLP/transformer models and evaluate their performance against the current extractive BERTSum and abstractive Command A models. Another alternative is to consider hybrid summarization methods which combine extractive and abstractive approaches to enhance coherence, context information and factual consistency in the generated text.

Future studies should also be conducted with greater numbers of legal experts and evaluators in order to further validate and ensure the reliability of human assessment results. This research was conducted with two expert evaluators, so more evaluators can be used to minimize any possible bias and gain a more complete picture of the quality of the summary.

Researchers can also look into other legal-domain specific metrics, such as measuring factual correctness, preserving legal reasoning, or interpretability. Moreover, future research could explore ways to reduce automation bias, especially when evaluation and generation models come from the same organization/ecosystem.

Finally, further development could involve the implementation of a user-friendly system of legal decision support in real time for legal practitioners, students and researchers. In practical legal applications, combining the explainable AI capabilities with the retrieval-augmented generation techniques could enhance transparency, trustworthiness, and usability even more.

5.7. Concluding Statement

This study successfully designed and tested a legal case summarization framework that utilizes extractive and abstractive summarization techniques in the field of legal decision support with the assistance of Artificial Intelligence. The results demonstrated that the extractive BERTSum pipeline performed better than the abstractive Command A pipeline on various automated evaluation metrics: BERTScore and METEOR, and higher expert evaluation scores.

The results show that extractive summarization is still more accurate in maintaining factual consistency and key information regarding legal documents (case) in the case document. Abstractive approach had the potential to create more natural summaries; however, it was noted that problems with coherence, context preservation and evaluation consistency were found.

While the size of the datasets, the number of evaluators, and the potential for automation bias were all factors, the study provides insightful lessons for the use of NLP technologies in the legal realm. The suggested framework emphasizes that AI-powered summarization systems can be valuable tools for legal professionals, making legal documents more accessible, faster to review, and more helpful for legal analysis.

In conclusion, the study provides a foundation for future research and development in AI-based legal summarization and decision-making tools, highlighting the critical need for accuracy, transparency, and human oversight in the application of AI within the legal field.

APPENDICES

February 16, 2026

Atty. Crisanto C. Carrillo, Jr.

Subject: Letter of Commitment – Domain Expert

Dear Atty. Crisanto C. Carrillo, Jr.,

This letter confirms your agreement to serve as the Domain Expert for our undergraduate thesis project.

In signing this document, you thereby indicate your readiness to offer your judicial assessment of selected system-generated products during **Thesis 1, 2 and 3**. Your work will be directed at the **assessment of the legal accuracy**, relevance of the context, soundness of the doctrine and general credibility of the created judicial summaries and legal precedents recommendations.

Project Title: An Integrated NLP-Driven Decision Support Framework for Automated Judicial Judgment Summarization and Context-Aware Legal Precedent Recommendation

Scope of Responsibilities:

- Reflecting on judicial summaries and judicial precedents generated by the system.
- Evaluation of legality, applicability and consistency of arguments.
- Expert advice to make sure there is adherence to the set legal principles and judicial standards.
- Determining the limitations or concerns in the outputs and giving recommendations on ways to improve them.

The experience you have in the judiciary and knowledge in law are priceless in maintaining a system that is not a substitute but a supplement to good judicial reasoning.

Your time and advice in making our research more academic and practical are greatly valued by us.

Sincerely,

Sophia Erin Nicole D. Castro | **Juan Paolo Angelo T. Dellomas** | **Maria Junella May E. Oalican**

Conforme:

Atty. Crisanto C. Carrillo, Jr.
Domain Expert

Date: 19 Feb. 2026

Appendix A. Letter of Commitment – Domain Expert A

The signed original copies of the Letters of Commitment are retained by the researchers and may be presented to the thesis panel upon request. Identifying information is withheld in this manuscript to preserve confidentiality.

February 16, 2026

Atty. Soledad Cabangis

Subject: Letter of Commitment – Domain Expert

Dear Atty. Cabangis,

This letter confirms your agreement to serve as the Domain Expert for our undergraduate thesis project.

In signing this document, you thereby indicate your readiness to offer your judicial assessment of selected system-generated products during **Thesis 1, 2 and 3**. Your work will be directed at the **assessment of the legal accuracy**, relevance of the context, soundness of the doctrine and general credibility of the created judicial summaries and legal precedents recommendations.

Project Title: An Integrated NLP-Driven Decision Support Framework for Automated Judicial Judgment Summarization and Context-Aware Legal Precedent Recommendation

Scope of Responsibilities:

- Reflecting on judicial summaries and judicial precedents generated by the system.
- Evaluation of legality, applicability and consistency of arguments.
- Expert advice to make sure there is adherence to the set legal principles and judicial standards.
- Determining the limitations or concerns in the outputs and giving recommendations on ways to improve them.

The experience you have in the judiciary and knowledge in law are priceless in maintaining a system that is not a substitute but a supplement to good judicial reasoning.

Your time and advice in making our research more academic and practical are greatly valued by us.

Sincerely,

Sophia Erin Nicole D. Castro	**Juan Paolo Angelo T. Dellomas**	**Maria Junella May E. Oalican**

Conforme:

Soledad T. de la Vega Cabangis,
Atty. Soledad Cabangis
Domain Expert

Date: 17/02/2026

Appendix B. Letter of Commitment – Domain Expert B

The signed original copies of the Letters of Commitment are retained by the researchers and may be presented to the thesis panel upon request. Identifying information is withheld in this manuscript to preserve confidentiality.

Appendix C. Dataset Metadata Schema and Sample Entry

Table 41. Dataset Metadata Schema

Field	Entry
Case Title	People of the Philippines v. Rossano Samson y Tiongco
G.R. No.	262579
Decision Date	February 28, 2024
Case Category	Murder
Source Repository	Supreme Court Website
Source Link	https://sc.judiciary.gov.ph/262579-people-of-the-philippines-vs-rossano-samson-y-tiongco/
Local File Name	Murder 262579 2024 SC.html
File Format	HTML
Extraction Status	(To be filled after extraction: Full / Partial / Failed)
Included / Excluded	(To be determined after quality screening)
Exclusion Reason	N/A (if included)
Notes	Initial sample entry for dataset documentation

Each collected case is documented using the metadata schema shown above. The metadata record supports dataset traceability, duplicate detection, extraction quality verification, and transparent reporting of inclusion and exclusion decisions. This structured documentation strengthens reproducibility and methodological clarity for subsequent preprocessing, training, and evaluation stage

Appendix D. Domain Expert Profile

Each expert is described in the study using the following profile details: title (lawyer/judge), current affiliations, years of experience, area of practice, and a brief description of their relevant role. This profiling supports transparency of the evaluation context without disclosing personally identifying information.

Table 42. Domain Expert Profile A

Identifier	Expert A
Professional Role	Licensed Attorney
Years of Experience	More than 15 years of institutional and professional legal experience
Area of Practice	Legal administration, professional legal education regulation, compliance monitoring, and evaluation of legal programs
Relevant Role Description	A senior legal professional with extensive experience in legal administration and regulatory oversight functions. Has supervised accreditation, program monitoring, compliance evaluation, and policy implementation within a national-level legal institution. Provides expertise in assessing legal accuracy, procedural consistency, and adherence to established judicial and professional standards.

Table 43. Domain Expert Profile B

Identifier	Expert B
Professional Role	Licensed Attorney
Years of Experience	More than 12 years of institutional and professional legal experience, with documented court appearances as counsel since at least 2013 and continuing leadership/legal roles through 2024.
Area of Practice	Corporate legal services, regulatory compliance, governance oversight, and tax-related litigation support
Relevant Role Description	A senior legal professional with extensive experience in corporate legal advisory work and regulatory compliance. Has served in managerial and governance roles within nationally operating institutions and cooperative-sector organizations. Provides expertise in legal compliance monitoring, policy oversight, institutional governance, and legal risk assessment. Experienced in handling tax-related legal matters and supporting regulatory proceedings.

REFERENCES

[1] Goal 16: Peace, justice and strong institutions. United Nations Development Programme. https://www.undp.org/sustainable-development-goals/peace-justice-and-strong-institutions.

[2] Sansone, C. and Sperlí, G. 2022. Legal Information Retrieval systems: State-of-the-art and open issues. Information Systems. 106 (2022), 101967. https://doi.org/10.1016/j.is.2021.101967.

[3] Ariai, S. et al. 2025. Natural Language Processing for the Legal Domain: A Survey of Tasks, Datasets, Models, and Challenges. ACM Computing Surveys. 58, 6 (December 2025), Article 163, 36 pages. https://doi.org/10.1145/3777009.

[4] Chalkidis, I. et al. 2020. LEGAL-BERT: The Muppets straight out of Law School. In Findings of the Association for Computational Linguistics: EMNLP 2020 (Online, 2020), 2898–2904. Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.findings-emnlp.261.

[5] Chalkidis, I. et al. 2022. LexGLUE: A Benchmark Dataset for Legal Language Understanding in English. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (Dublin, Ireland, 2022), 4310–4330. Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.297.

[6] Beltagy, I., Peters, M. E., and Cohan, A. 2020. Longformer: The Long-Document Transformer. arXiv preprint arXiv:2004.05150. https://arxiv.org/abs/2004.05150.

[7] Akter, M., Çano, E., Weber, E., Dobler, D., and Habernal, I. 2025. A Comprehensive Survey on Legal Summarization: Challenges and Future Directions. arXiv preprint arXiv:2501.17830. https://arxiv.org/abs/2501.17830.

[8] Shukla, A. et al. 2022. Legal Case Document Summarization: Extractive and Abstractive Methods and their Evaluation. In Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (Online only, 2022), 1048–1064. Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.aacl-main.77.

[9] Dan, H. et al. 2025. Enhancing legal judgment summarization using an integrated semantic and structural information-based approach. Artificial Intelligence and Law. 33 (2025), 271–292. https://doi.org/10.1007/s10506-023-09381-8.

[10] Jain, N. et al. 2024. A sentence is known by the company it keeps: Improving legal document summarization using deep clustering. Artificial Intelligence and Law. 32 (2024), 165–200. https://doi.org/10.1007/s10506-023-09345-y.

[11] Feng, Y., Li, C., and Ng, V. 2024. Legal Case Retrieval: A Survey of the State of the Art. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (Bangkok, Thailand, 2024), 6472–6485. Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.350.

[12] Ma, Y. et al. 2023. Incorporating Structural Information into Legal Case Retrieval. ACM Transactions on Information Systems. 42, 2 (November 2023), Article 40, 28 pages. https://doi.org/10.1145/3609796.

[13] Li, H. et al. 2023. SAILER: Structure-aware Pre-trained Language Model for Legal Case Retrieval. In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (Taipei, Taiwan, 2023), 2595–2599. ACM, New York, NY. https://doi.org/10.1145/3539618.3591761.

[14] Althammer, S. et al. 2021. DoSSIER@COLIEE 2021: Leveraging Dense Retrieval and Summarization-Based Re-ranking for Case Law Retrieval. In Proceedings of COLIEE 2021 Workshop (2021), 7 pages. https://arxiv.org/abs/2108.03937.

[15] Mentzingen, J., Habernal, I., and Gurevych, I. 2024. Textual similarity for legal precedents discovery: Assessing the performance. International Journal of Information Management Data Insights. 4 (2024), 100247. https://doi.org/10.1016/j.jjimei.2024.100247.

[16] Richmond, K. M. et al. 2024. Explainable AI and Law: An Evidential Survey. Digital Society. 3, 1 (2024), Article 1, 33 pages. https://doi.org/10.1007/s44206-023-00081-z.

[17] Collenette, L., Atkinson, K., and Bench-Capon, T. 2023. Explainable AI tools for legal reasoning about cases: A study on the European Court of Human Rights. Artificial Intelligence. 317 (2023), 103861. https://doi.org/10.1016/j.artint.2023.103861.

[18] Vishnuprabha V., Daleesha M. Viswanathan, Rajesh R., and Aneesh V. Pillai. 2025. Augmented Question-guided Retrieval (AQgR) of Indian Case Law with LLM, RAG, and Structured Summaries. https://doi.org/10.48550/arXiv.2508.04710.

[19] Shengjie Ma, Qi Chu, Jiaxin Mao, Xuhui Jiang, Haozhe Duan, and Chong Chen. 2024. Leveraging Large Language Models for Relevance Judgments in Legal Case Retrieval. arXiv preprint arXiv:2403.18405. https://doi.org/10.48550/arXiv.2403.18405.

[20] Zhitao He, Pengfei Cao, Chenhao Wang, Zhuoran Jin, Yubo Chen, Jiexin Xu, Huaijun Li, Xiaojian Jiang, Kang Liu, and Jun Zhao. 2024. AgentsCourt: Building

Judicial Decision-Making Agents with Court Debate Simulation and Legal Knowledge Augmentation. https://doi.org/10.48550/arXiv.2403.02959.

[21] Hugo Mentzingen, Nuno António, and Fernando Bacao. 2025. Effectiveness in retrieving legal precedents: exploring text summarization and cutting-edge language models toward a cost-efficient approach. Artificial Intelligence and Law (2025). https://doi.org/10.1007/s10506-025-09440-2.

[22] Wei Gao, Shuai Yu, Yongbin Qin, Caiwei Yang, Ruizhang Huang, Yanping Chen, and Chuan Lin. 2025. LSDK-LegalSum: Improving legal judgment summarization using logical structure and domain knowledge. Journal of King Saud University Computer and Information Sciences 37 (2025), Article 3, 15 pages. https://doi.org/10.1007/s44443-025-00022-5.

[23] Yumei Song, Yongbin Qin, Ruizhang Huang, Yanping Chen, and Chuan Lin. 2025. Legal text summarization via judicial syllogism with large language models. Journal of King Saud University Computer and Information Sciences 37 (2025), Article 111, 13 pages. https://doi.org/10.1007/s44443-025-00113-3.

[24] Jenish Dhanani, Rupa Mehta, and Dipti Rana. 2022. Effective and scalable legal judgment recommendation using pre-learned word embedding. Complex & Intelligent Systems 8 (2022), 3199–3213. https://doi.org/10.1007/s40747-022-00673-1.

[25] Yifei Zhang, Yanling Li, and Fengpei Ge. 2025. LAS-SCR: knowledge-driven enhanced legal judgment prediction via the semantic and similar case retrieval of law articles. Journal of King Saud University – Computer and Information Sciences 37 (2025), Article 351, 17 pages. https://doi.org/10.1007/s44443-025-00383-x.

[26] Mi-Young Kim, Juliano Rabelo, Housam Khalifa Bashier Babiker, Md Abed Rahman, and Randy Goebel. 2024. Legal Information Retrieval and Entailment Using Transformer-based Approaches. The Review of Socionetwork Strategies 18 (2024), 101–121. https://doi.org/10.1007/s12626-023-00153-z.

[27] Qihui Zhao. 2025. Legal judgment prediction via legal knowledge extraction and fusion. Journal of King Saud University – Computer and Information Sciences 37 (2025), Article 31, 16 pages. https://doi.org/10.1007/s44443-025-00019-0.

[28] Sanjay Kumar Sahoo, Avinash Samantra, Chinmaya Kumar Mohapatra, Bishnu Prasad Swain, Zulfiqar Ali, and Ghulam Muhammad. 2025. LABDT (Lexi AdaBoost Decision Tree): An Approach for Legal Outcome Prediction Fusing Lexical, Semantic, and Similarity-based Features. International Journal of Computational Intelligence Systems 18, 279 (2025). https://doi.org/10.1007/s44196-025-01003-2.

[29] Mi-Young Kim, Juliano Rabelo, Kingsley Okeke, and Randy Goebel. 2022. Legal Information Retrieval and Entailment Based on BM25, Transformer and Semantic

Thesaurus Methods. The Review of Socionetwork Strategies 16 (2022), 157–174. https://doi.org/10.1007/s12626-022-00103-1.

[30] Lin Zhang, Yanan Li, and Hongyu Zhang. 2025. Deep Learning-Based Automatic Summarization of Chinese Maritime Judgment Documents. Applied Sciences 15, 10 (2025), 5434. https://doi.org/10.3390/app15105434.

[31] Jianquan Ouyang, Wenpeng Huang, and Tianming Liu. 2022. A Hybrid Automatic Text Summarization Model for Judgment Documents. In Proceedings of the International Conference on Intelligent Computing (ICIC 2022). Springer, 1–14. https://doi.org/10.1007/978-3-031-06767-9_31

[32] Jesse Woo, Fateme Hashemi Chaleshtori, Ana Marasović, and Kenneth Marino. 2025. BriefMe: A Legal NLP Benchmark for Assisting with Legal Briefs. In Findings of the Association for Computational Linguistics: ACL 2025, 13139–13190. Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.findings-acl.681.

[33] Chaeeun Kim, Jinu Lee, and Wonseok Hwang. 2025. *LegalSearchLM: Rethinking Legal Case Retrieval as Legal Elements Generation*. https://doi.org/10.48550/arXiv.2505.23832

[34] Sai Srinivas Vellela, Singu Kavya, Kakarla Lakshmi Sravani, Tadikonda Poojitha, and Sattenapalli Siva Nageswara Rao. 2025. NLP-Driven Summarization: Efficient Extraction of Key Information from Legal and Financial Documents. International Journal for Modern Trends in Science and Technology 11, 03 (2025), 341–349. https://doi.org/10.2139/ssrn.5250908

[35] Jenith Selva Melba M. and Savitha R. 2025. AI Based Legal Assistance Platform Using Natural Language Processing. International Research Journal on Advanced Engineering Hub (IRJAEH) 2, 12 (December 2025), 4164–4178. https://doi.org/10.47392/IRJAEH.2025.0611

[36] Yasin Ortakci and Burak Borhan. 2025. Optimizing SBERT for long text clustering: two novel approaches with empirical insights. The Journal of Supercomputing 81 (2025), Article 950, 41 pages. https://doi.org/10.1007/s11227-025-07414-4

[37] Dharil Patel, Shruti Patil, Deepali Vora, and Aniket K. Shahade. 2026. Legal text summarization with optimized hybrid models and fine-tuned LLaMA-2. Discover Computing 29, 31 (2026). https://doi.org/10.1007/s10791-026-09916-y

[38] Daniel Martin Katz, Dirk Hartung, Lauritz Gerlach, Abhik Jana, and Michael J. Bommarito. 2023. Natural Language Processing in the Legal Domain. https://doi.org/10.48550/arXiv.2302.12039.

[39] Ashutosh Modi, Prathamesh Kalamkar, Saurabh Karn, Aman Tiwari, Abhinav Joshi, Sai Kiran Tanikella, Shouvik Kumar Guha, Sachin Malhan, and Vivek Raghavan. 2023.

SemEval 2023 Task 6: LegalEval - Understanding Legal Texts. https://doi.org/10.48550/arXiv.2304.09548.

[40] Yunhan Li and Gengshen Wu. 2025. LegalEval-Q: A New Benchmark for the Quality Evaluation of LLM-Generated Legal Text. https://doi.org/10.48550/arXiv.2505.24826.

[41] Luca Ragazzi, Gianluca Moro, Stefano Guidi, and Giacomo Frisoni. 2025. LAWSUIT: a LArge expert-Written SUmmarization dataset of ITalian constitutional court verdicts. Artificial Intelligence and Law 33 (2025), 1151–1187. https://doi.org/10.1007/s10506-024-09414-w.

[42] Rachid Ejjami. 2024. AI-Driven Justice: Evaluating the Impact of Artificial Intelligence on Legal Systems. International Journal for Multidisciplinary Research (IJFMR) 6, 3 (May–June 2024), Article IJFMR240323969, 29 pages. https://doi.org/10.36948/ijfmr.2024.v06i03.23969.

[43] Mohsen Soori, Fooad Karimi Ghaleh Jough, Roza Dastres, and Behrooz Arezoo. 2026. AI-based decision support systems in Industry 4.0, a review. Journal of Economy and Technology 4 (2026), 206–225. https://doi.org/10.1016/j.ject.2024.08.005.

[44] Valerio Bellandi, Silvana Castano, Stefano Montanelli, and Stefano Siccardi. 2025. Streamlining Legal Document Management: A Knowledge-Driven Service Platform. SN Computer Science 6 (2025), Article 166. https://doi.org/10.1007/s42979-025-03694-y.

[45] Rabee Al-Qaesm, Mohannad Hendi, and Banan Tantour. 2025. Alkafi-llama3: fine-tuning LLMs for precise legal understanding in Palestine. Discover Artificial Intelligence 5, 107 (2025). https://doi.org/10.1007/s44163-025-00313-w.

[46] Supreme Court E-Library. 1987 Constitution of the Republic of the Philippines, Article VIII, Sec. 14–15. https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/45/11448

[47] Supreme Court E-Library. Republic Act No. 386 (Civil Code), Article 8. https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/53360

[48] Supreme Court of the Philippines. Decisions. https://sc.judiciary.gov.ph/decisions/

[49] Supreme Court E-Library. Supreme Court E-Library. https://elibrary.judiciary.gov.ph/

[50] Supreme Court of the Philippines. eCourt PH. https://sc.judiciary.gov.ph/ecourt-ph/

[51] Yang Liu and Mirella Lapata. 2019. Text Summarization with Pretrained Encoders. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing

(EMNLP-IJCNLP). Association for Computational Linguistics, Hong Kong, China, 3730–3740. DOI:https://doi.org/10.18653/v1/D19-1387

[51] Reimers, Nils, and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Association for Computational Linguistics, Hong Kong, China, 3982–3992. https://doi.org/10.18653/v1/D19-1410